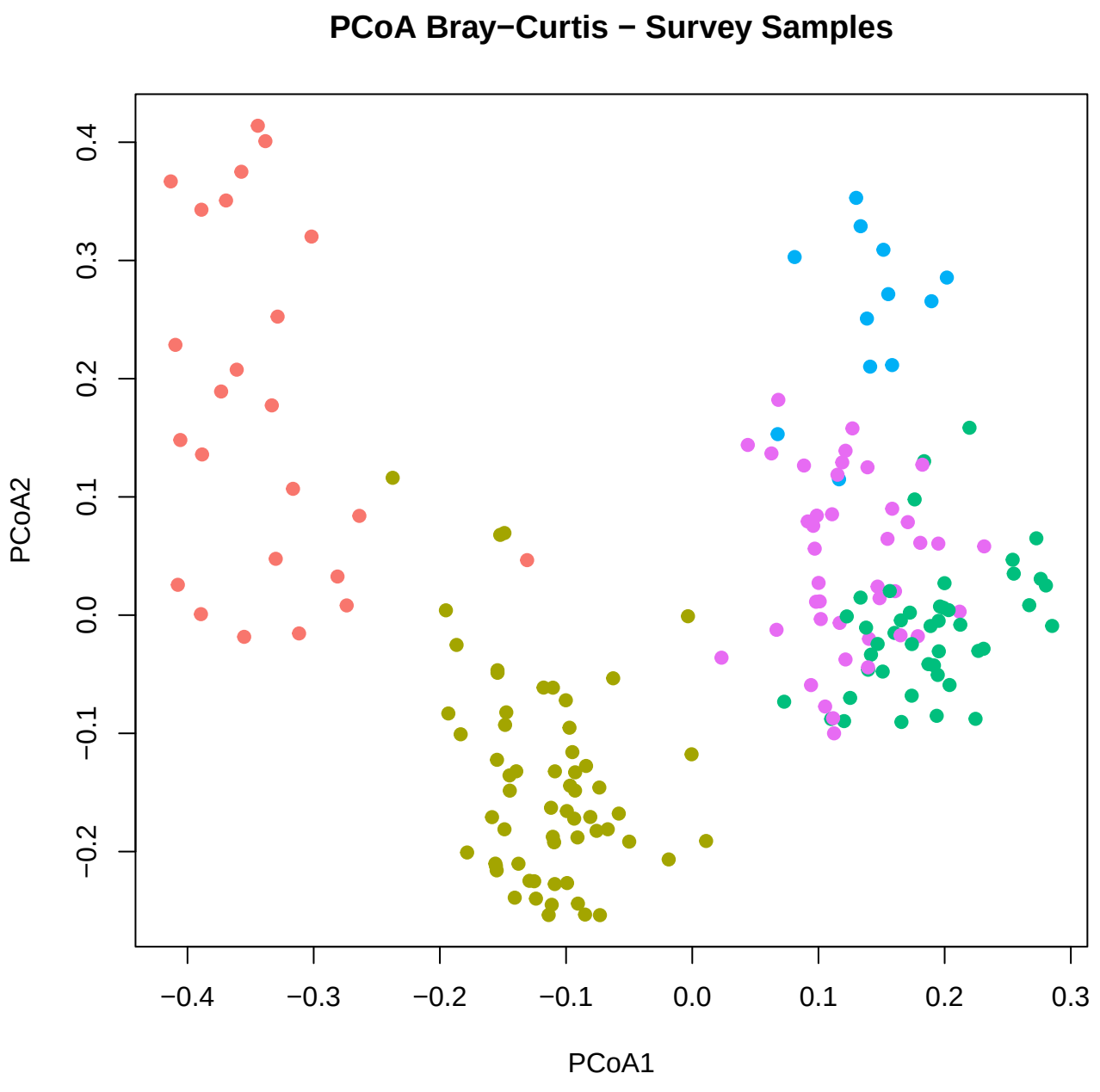
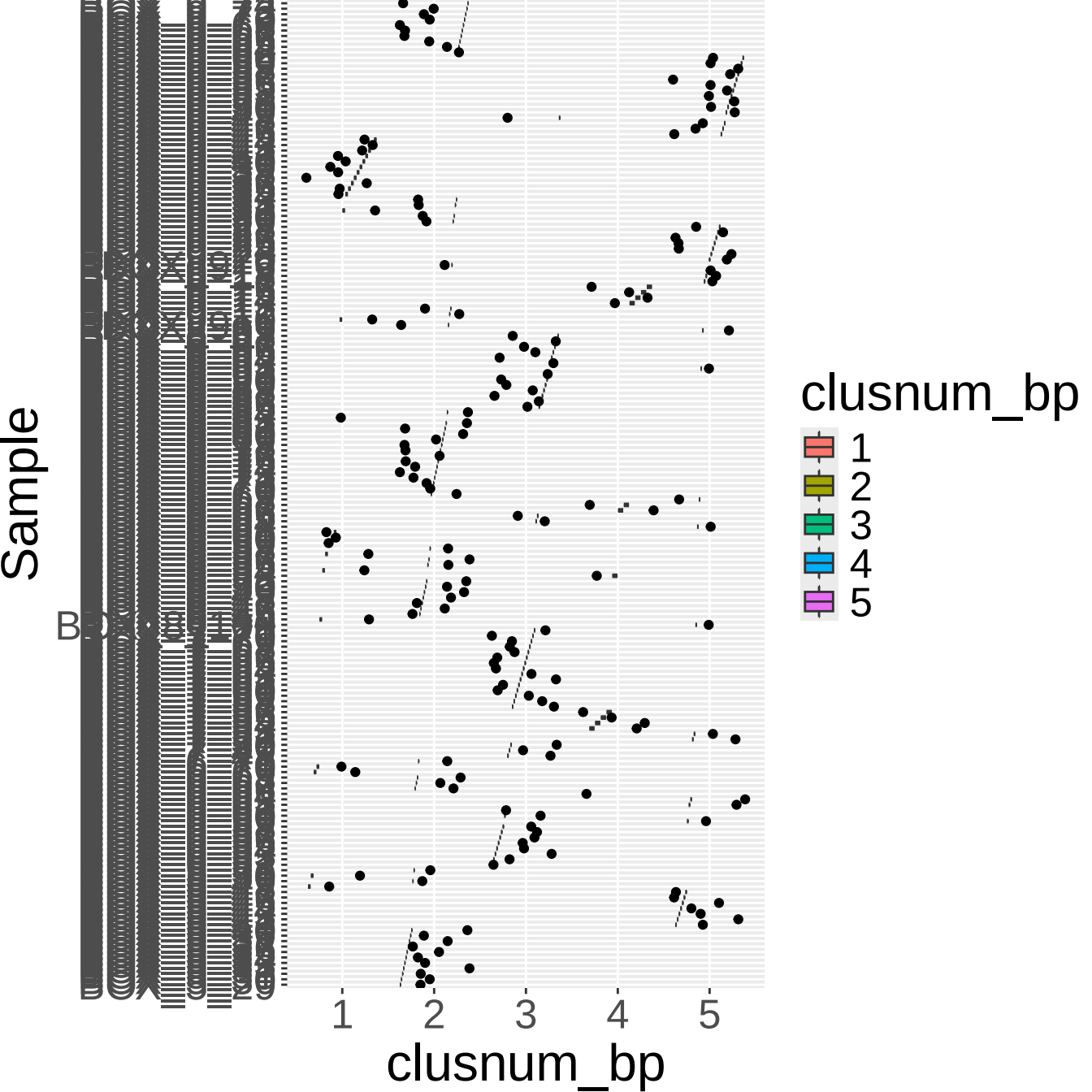
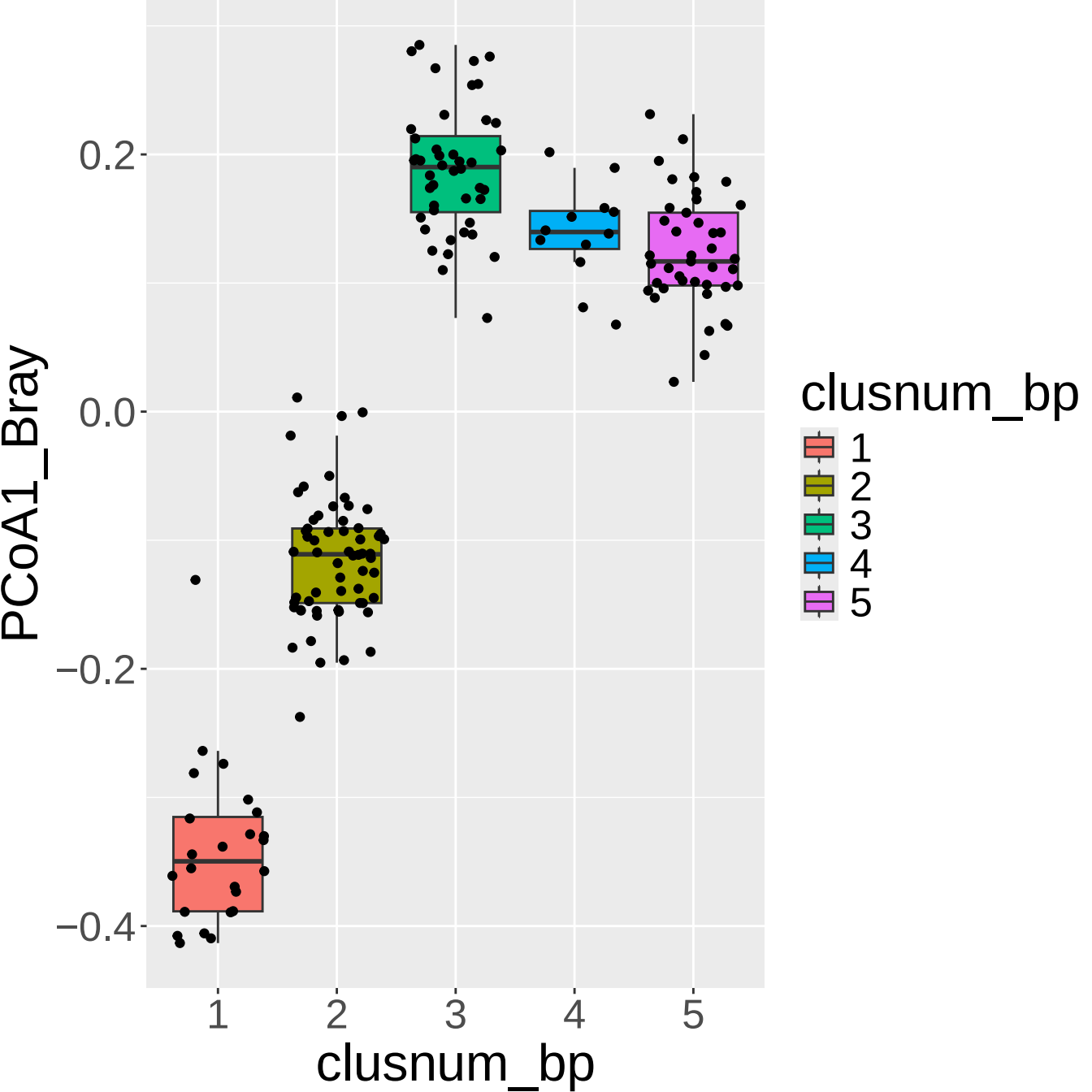
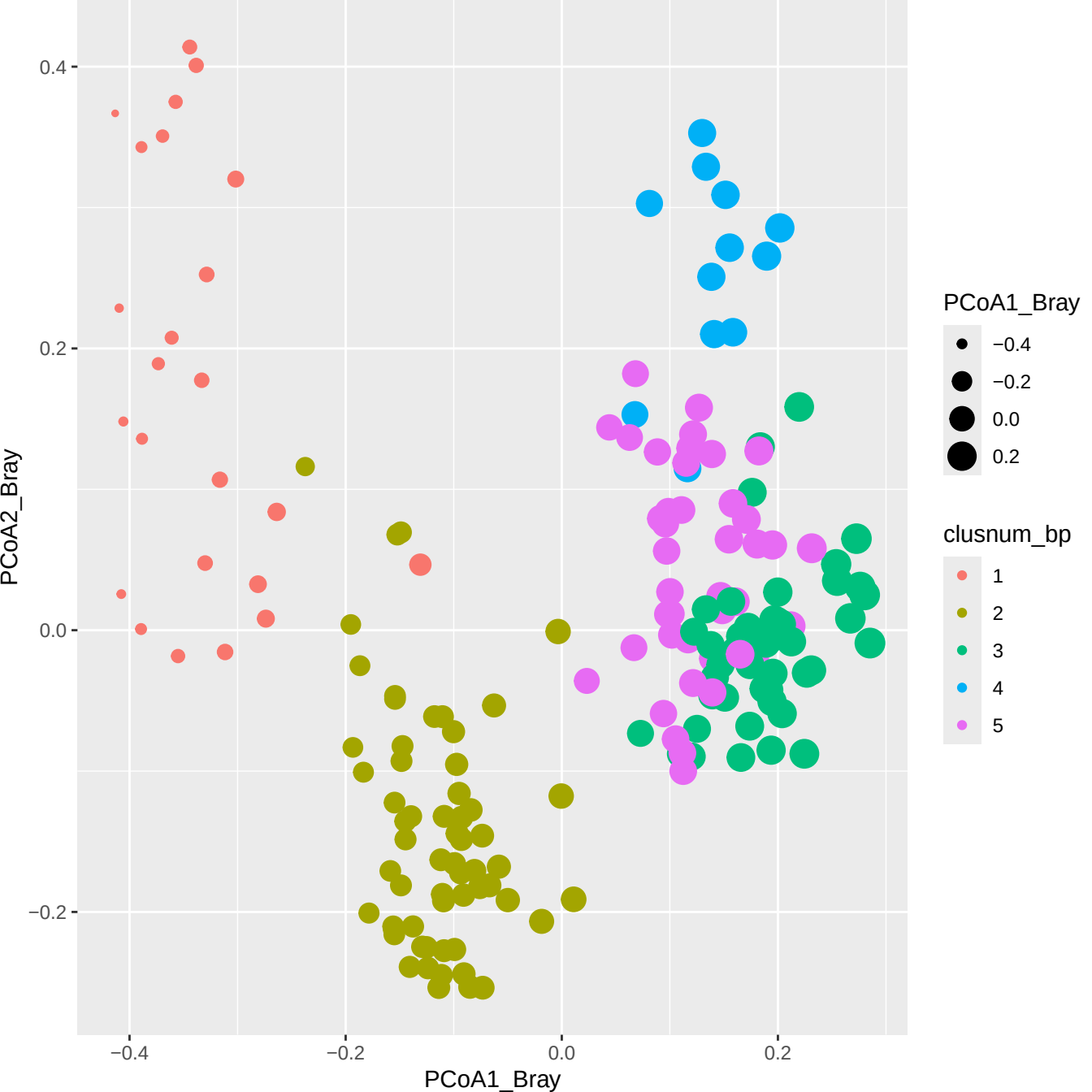
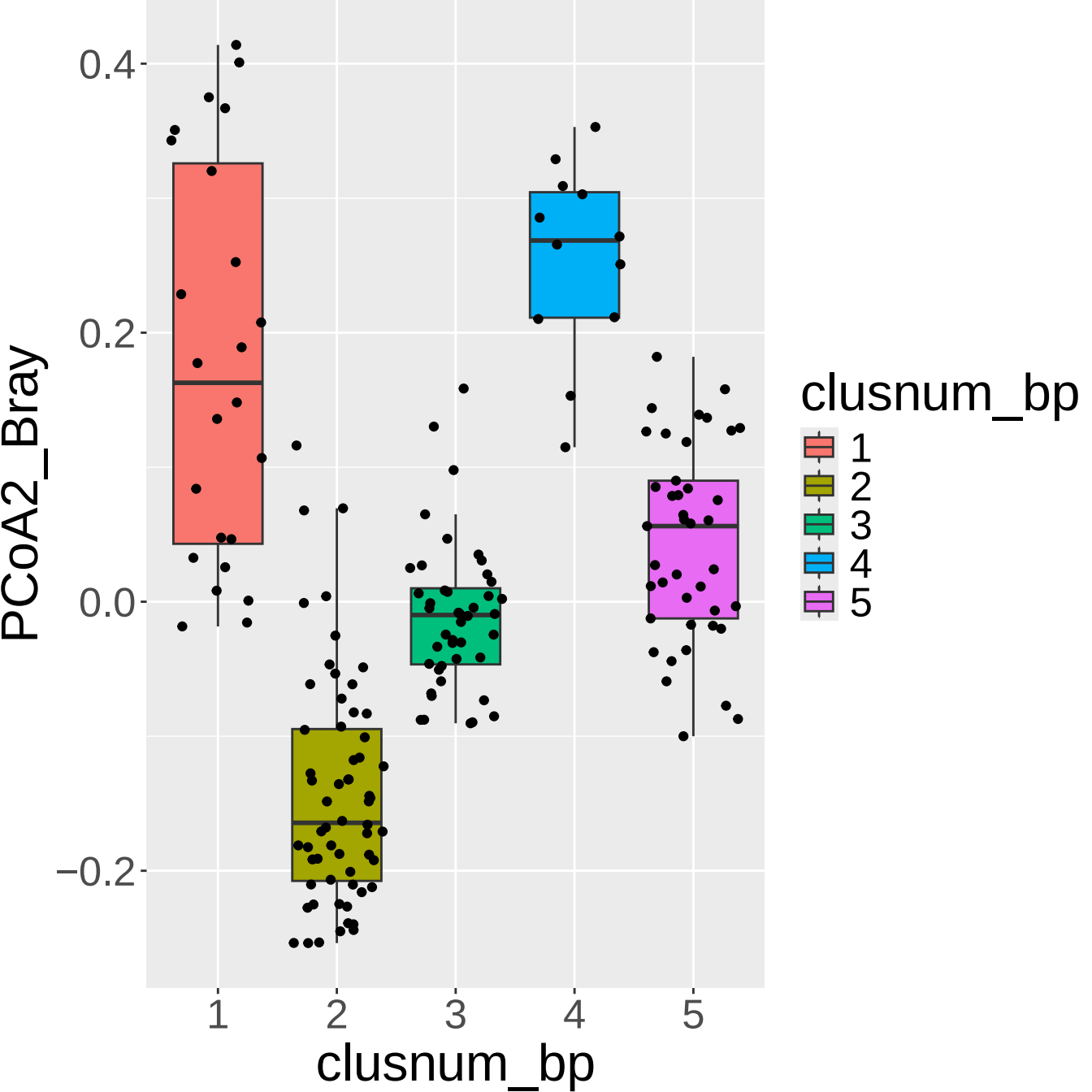

PCoA Bray−Curtis − Survey Samples
PCoA2
PCoA1
0.4
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.4
−0.3
−0.2
−0.1
0.0
0.1
0.2
0.3

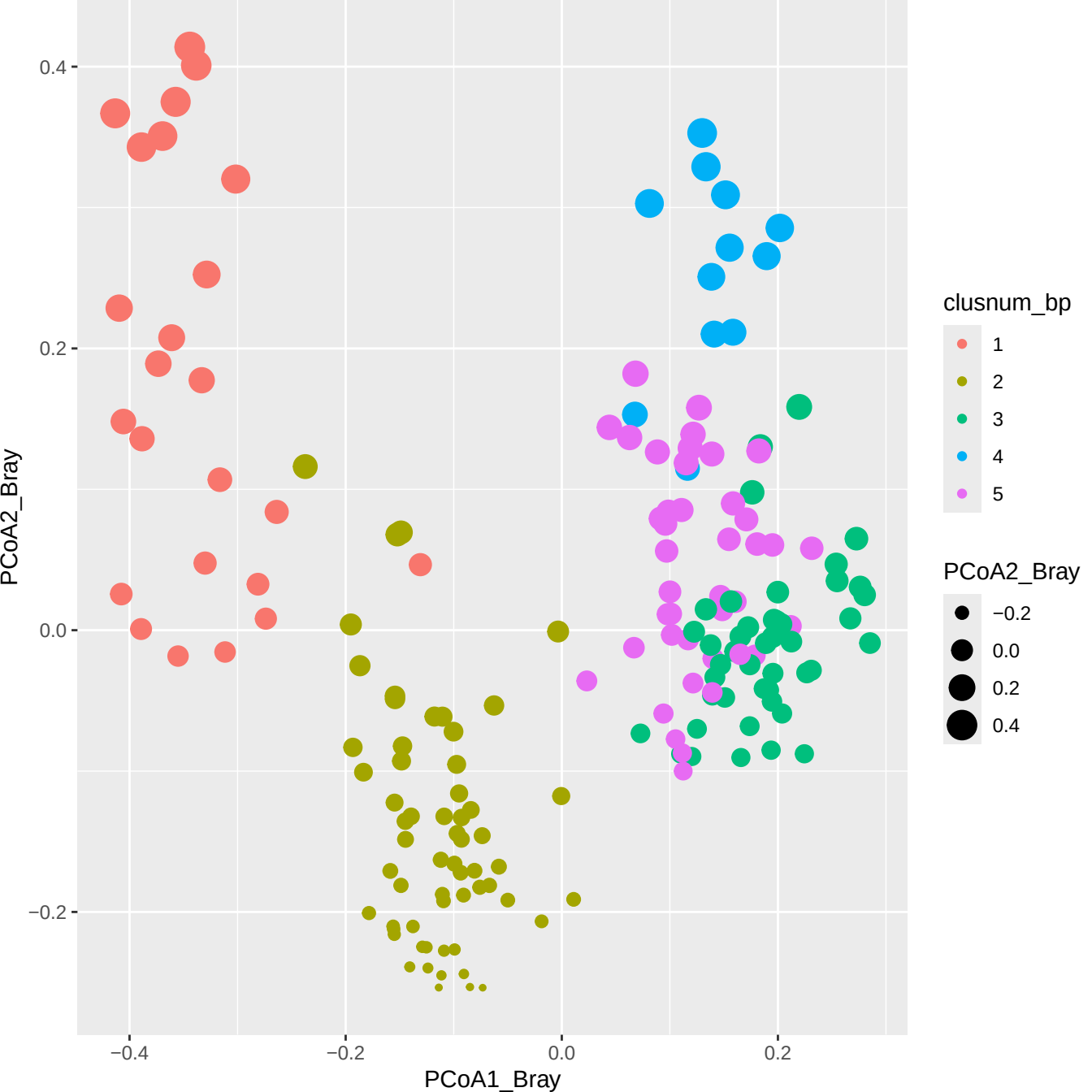

Sample
clusnum_bp
1
2
3
4
5
clusnum_bp
1
2
3
4
5

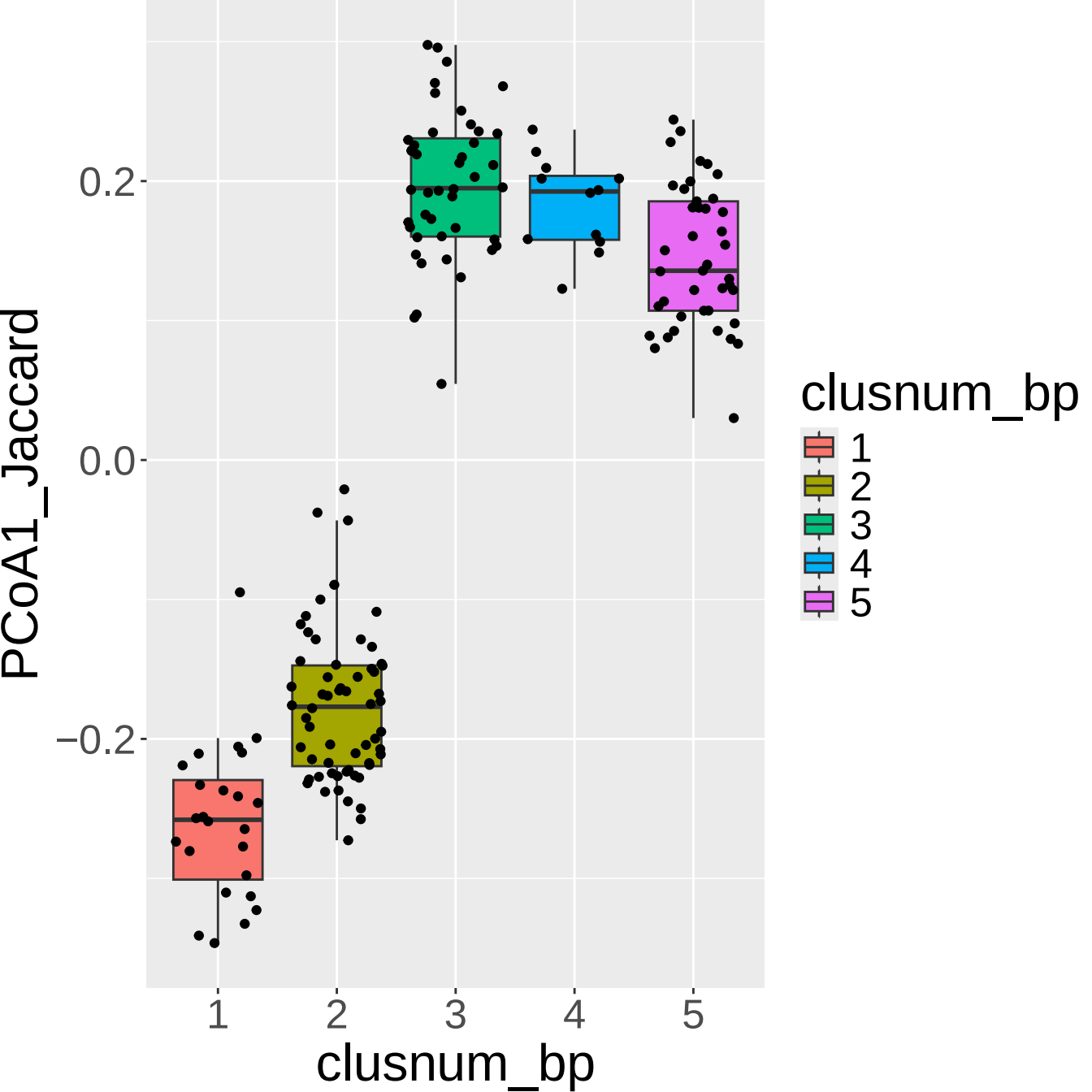
●	BOX_5_49	●	BOX_6_68	●	BOX_7_63	●	BOX_8_54	●	BOX_8_74	●	BOX_8_93	●	BOX_9_20	●
●	BOX_5_50	●	BOX_6_69	●	BOX_7_64	●	BOX_8_55	●	BOX_8_75	●	BOX_8_94	●	BOX_9_23	●
●	BOX_5_51	●	BOX_6_70	●	BOX_7_65	●	BOX_8_56	●	BOX_8_76	●	BOX_8_95	●	BOX_9_24	●
●	BOX_6_51	●	BOX_6_71	●	BOX_7_66	●	BOX_8_57	●	BOX_8_77	●	BOX_8_96	●	BOX_9_25	●
●	BOX_6_52	●	BOX_7_48	●	BOX_7_67	●	BOX_8_58	●	BOX_8_78	●	BOX_8_97	●	BOX_9_26	●
●	BOX_6_53	●	BOX_7_49	●	BOX_7_68	●	BOX_8_59	●	BOX_8_79	●	BOX_8_98	●	BOX_9_27	●
●	BOX_6_54	●	BOX_7_50	●	BOX_7_69	●	BOX_8_60	●	BOX_8_80	●	BOX_8_99	●	BOX_9_28	●
●	BOX_6_55	●	BOX_7_51	●	BOX_7_70	●	BOX_8_61	●	BOX_8_81	●	BOX_9_1	●	BOX_9_29	●
●	BOX_6_56	●	BOX_7_52	●	BOX_7_71	●	BOX_8_62	●	BOX_8_82	●	BOX_9_10	●	BOX_9_30	●
●	BOX_6_57	●	BOX_7_53	●	BOX_8_100	●	BOX_8_64	●	BOX_8_83	●	BOX_9_11	●	BOX_9_31	●
●	BOX_6_58	●	BOX_7_54	●	BOX_8_44	●	BOX_8_65	●	BOX_8_84	●	BOX_9_12	●	BOX_9_32	●
●	BOX_6_59	●	BOX_7_55	●	BOX_8_45	●	BOX_8_66	●	BOX_8_85	●	BOX_9_13	●	BOX_9_33	●
●	BOX_6_60	●	BOX_7_56	●	BOX_8_46	●	BOX_8_67	●	BOX_8_86	●	BOX_9_14	●	BOX_9_34	●
●	BOX_6_61	●	BOX_7_57	●	BOX_8_47	●	BOX_8_68	●	BOX_8_87	●	BOX_9_15	●	BOX_9_35	●
●	BOX_6_62	●	BOX_7_58	●	BOX_8_48	●	BOX_8_69	●	BOX_8_88	●	BOX_9_16	●	BOX_9_36	●
●	BOX_6_63	●	BOX_7_59	●	BOX_8_49	●	BOX_8_70	●	BOX_8_89	●	BOX_9_17	●	BOX_9_37	●
●	BOX_6_64	●	BOX_7_60	●	BOX_8_50	●	BOX_8_71	●	BOX_8_90	●	BOX_9_18	●	BOX_9_38	●
●	BOX_6_66	●	BOX_7_61	●	BOX_8_52	●	BOX_8_72	●	BOX_8_91	●	BOX_9_19	●	BOX_9_39	●
●	BOX_6_67	●	BOX_7_62	●	BOX_8_53	●	BOX_8_73	●	BOX_8_92	●	BOX_9_2	●	BOX_9_40	●

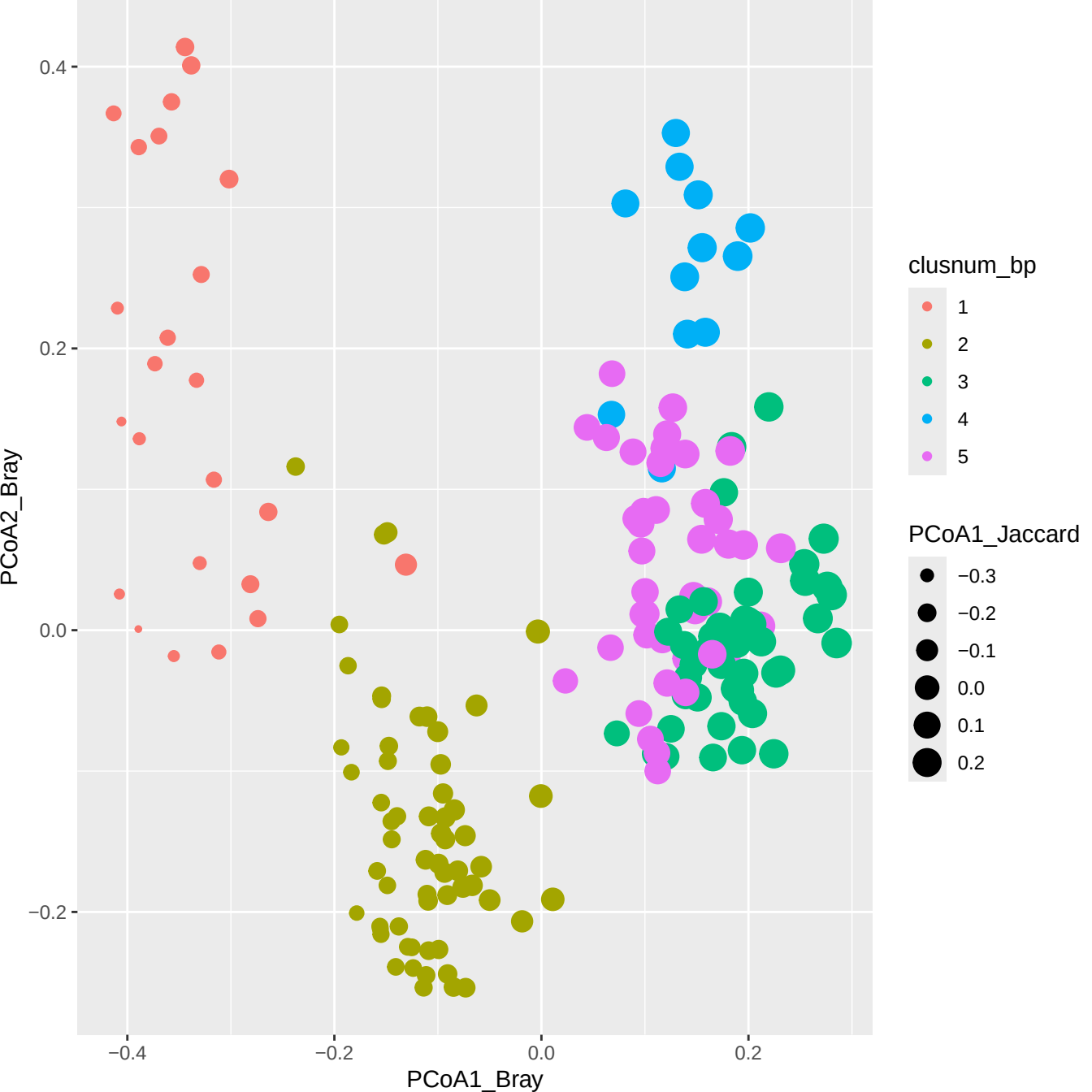

PCoA1_Bray
0.2
0.0
−0.2
−0.4
1
2
3
4
5
clusnum_bp
clusnum_bp
1
2
3
4
5

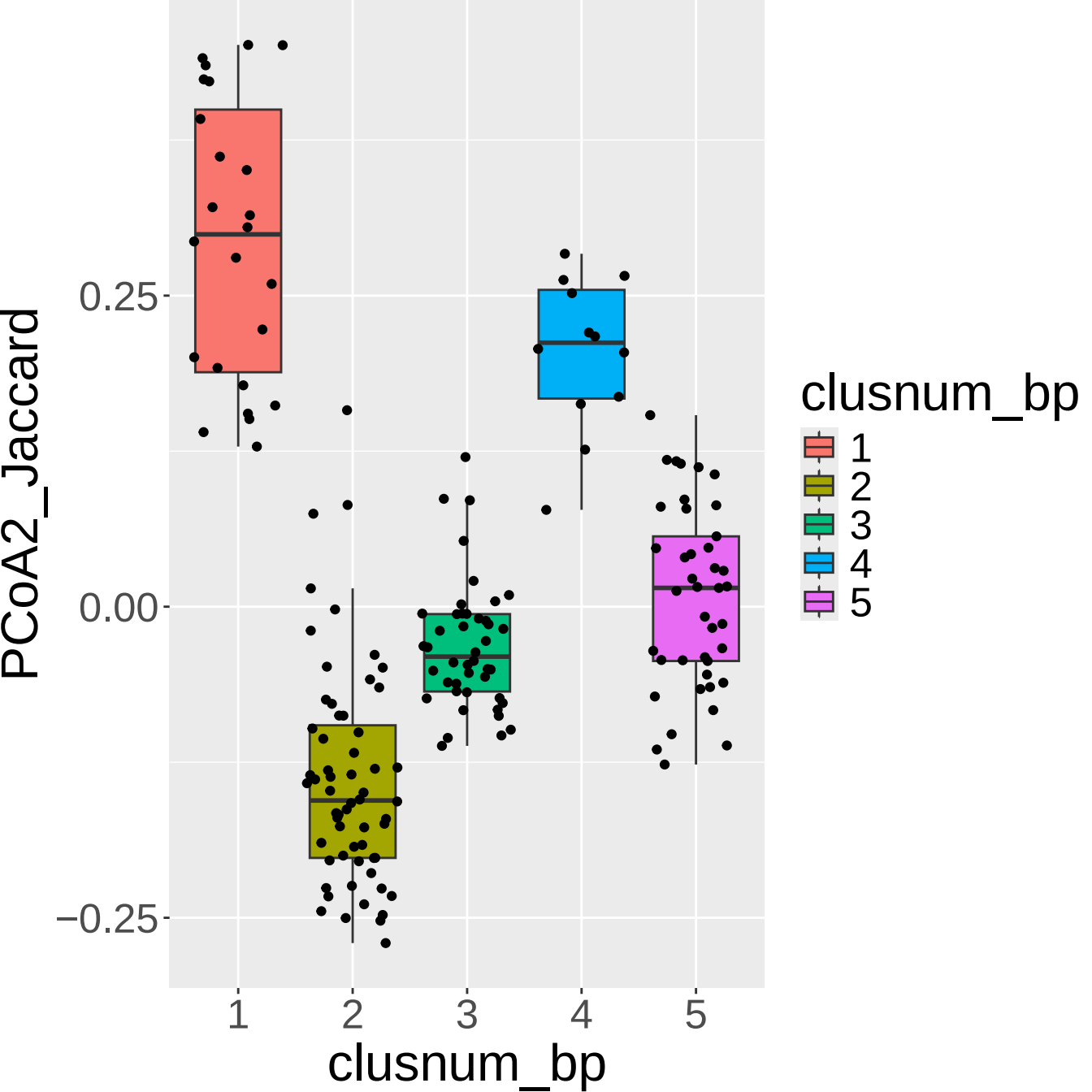

0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
PCoA1_Bray
−0.4
−0.2
0.0
0.2
clusnum_bp
1
2
3
4
5

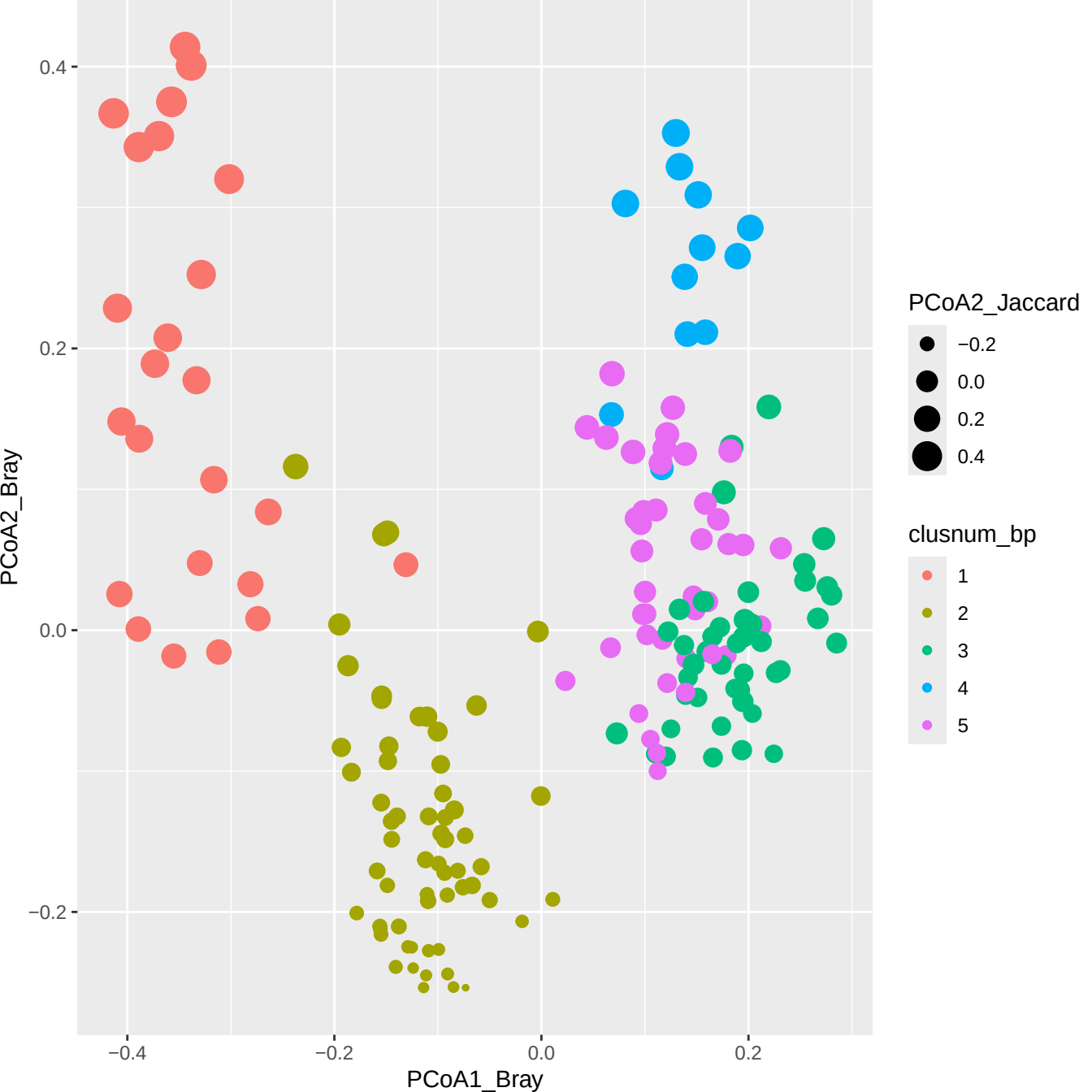

0.4
0.2
0.0
−0.2
PCoA2_Bray
1
2
3
4
5
clusnum_bp
clusnum_bp
1
2
3
4
5

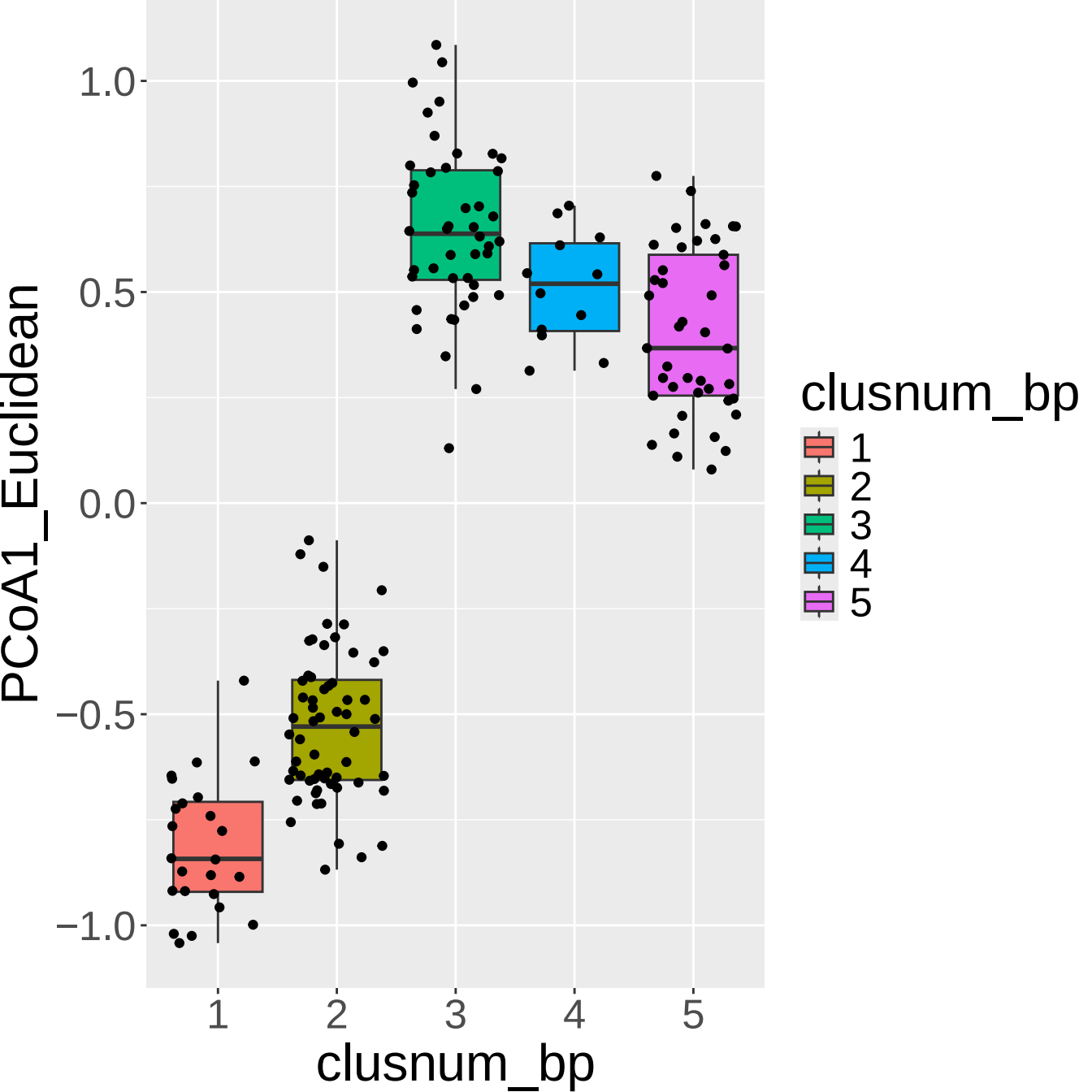

0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
clusnum_bp
1
2
3
4
5
PCoA2_Bray
−0.2
0.0
0.2
0.4

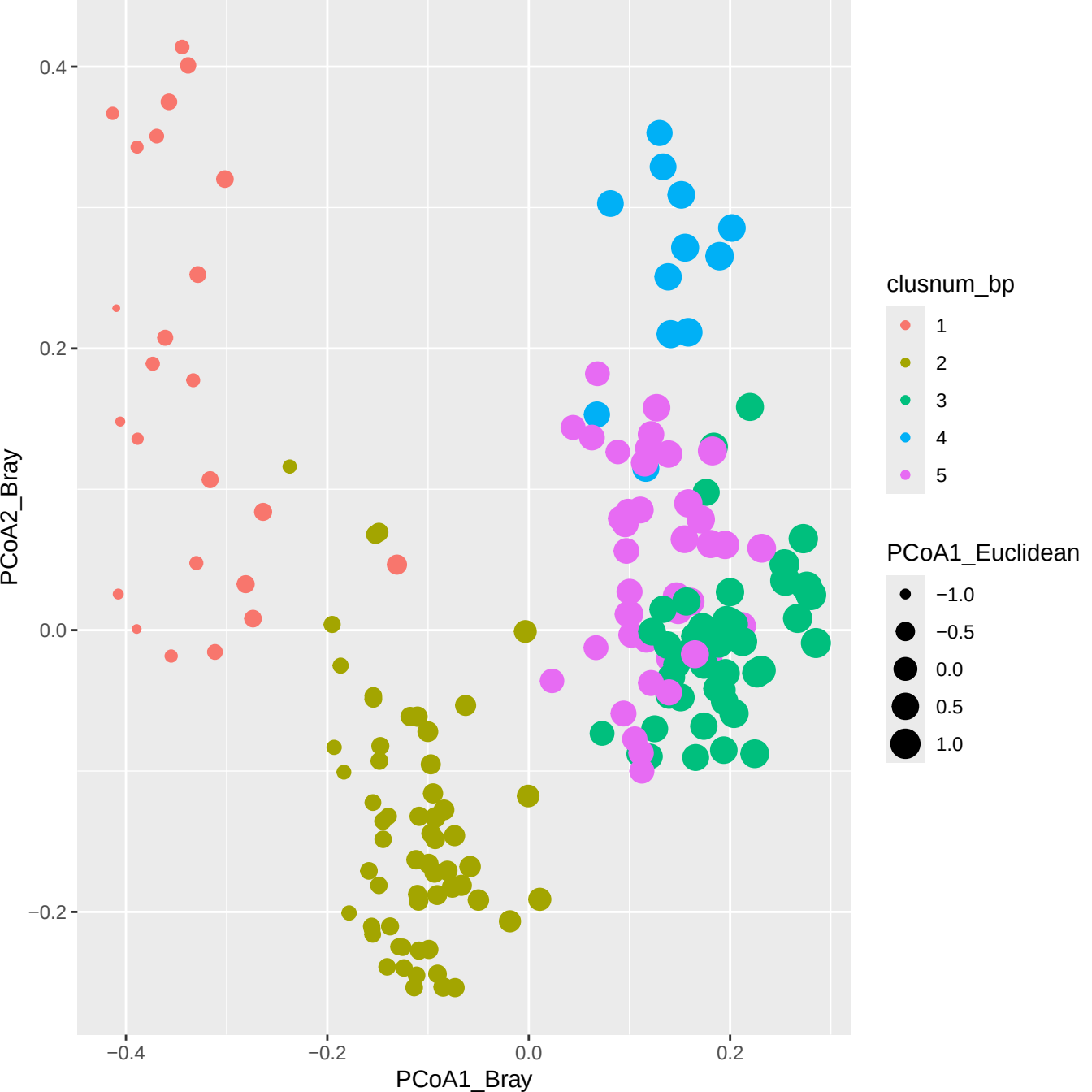

PCoA1_Jaccard
0.2
0.0
−0.2
1
2
3
4
5
clusnum_bp
clusnum_bp
1
2
3
4
5

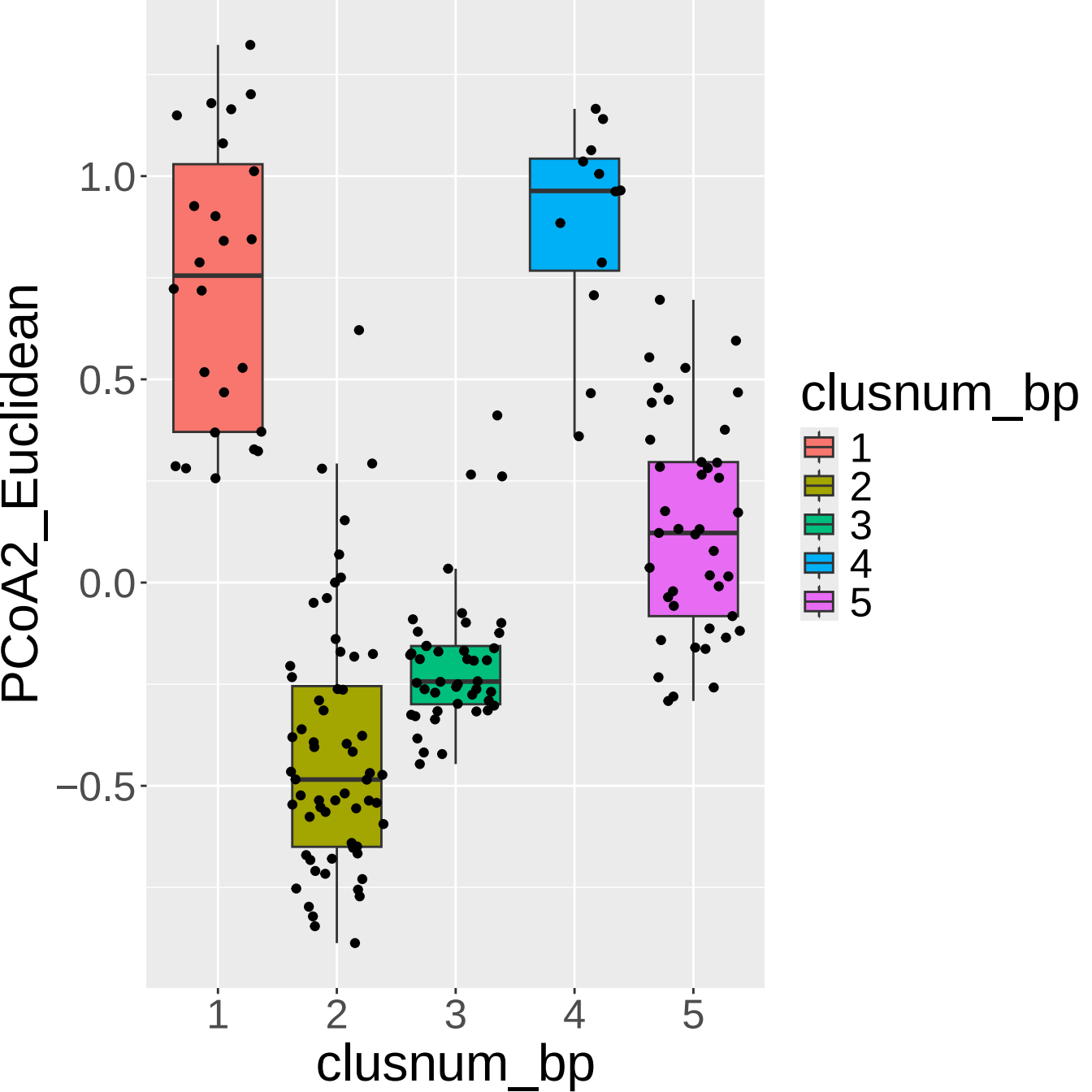

0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
clusnum_bp
1
2
3
4
5
PCoA1_Jaccard
−0.3
−0.2
−0.1
0.0
0.1
0.2

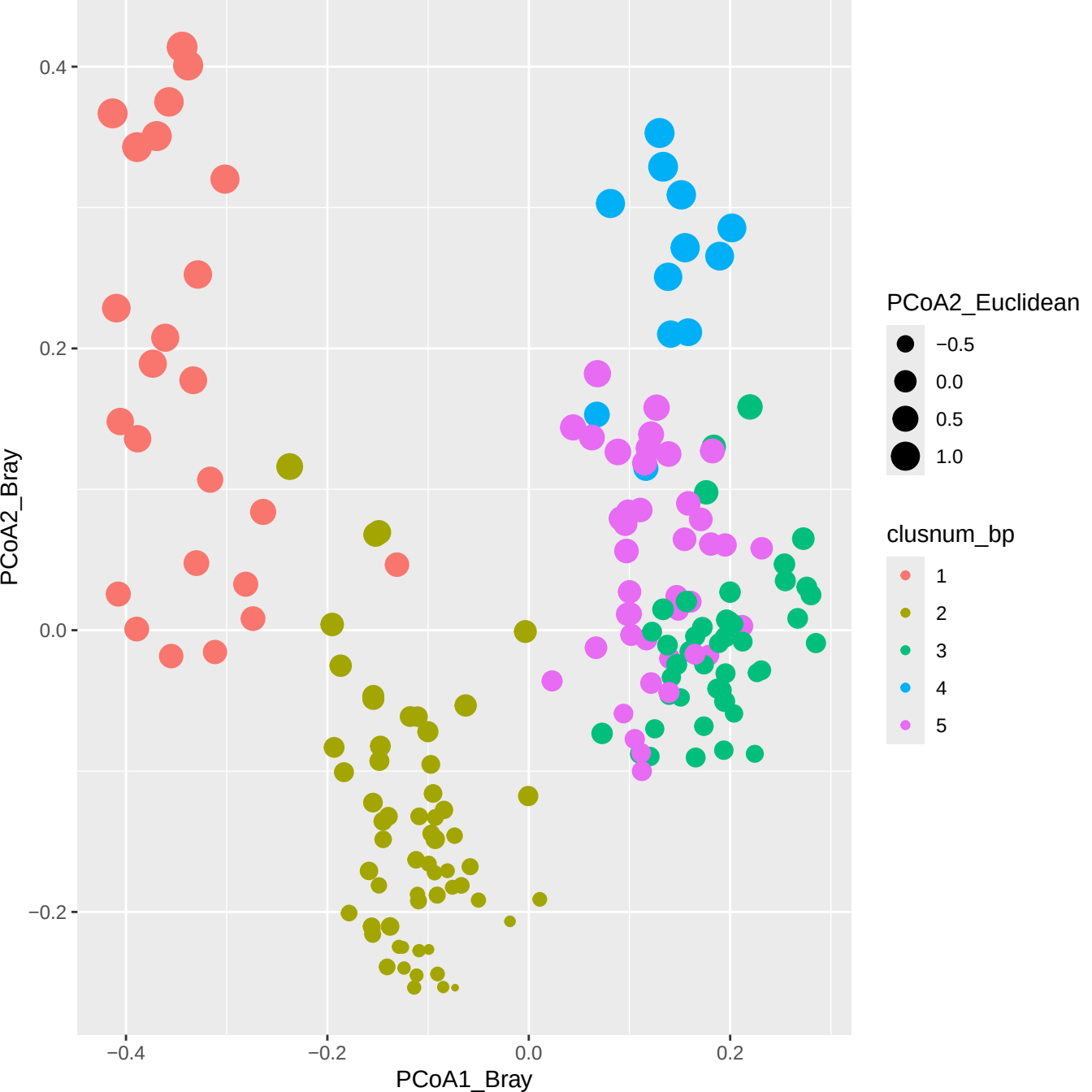

PCoA2_Jaccard
0.25
0.00
−0.25
1
2
3
4
5
clusnum_bp
clusnum_bp
1
2
3
4
5

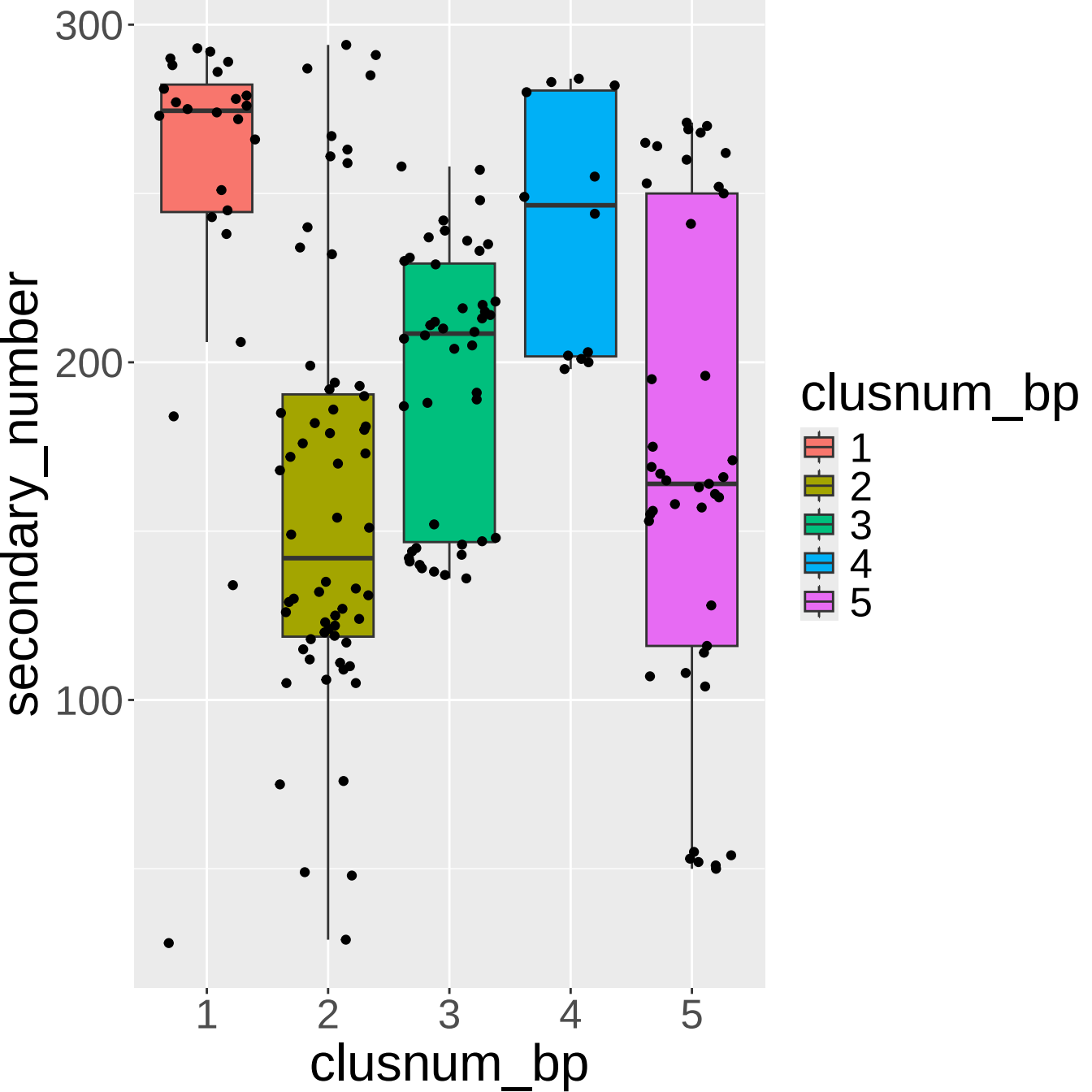

0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
PCoA2_Jaccard
−0.2
0.0
0.2
0.4
clusnum_bp
1
2
3
4
5

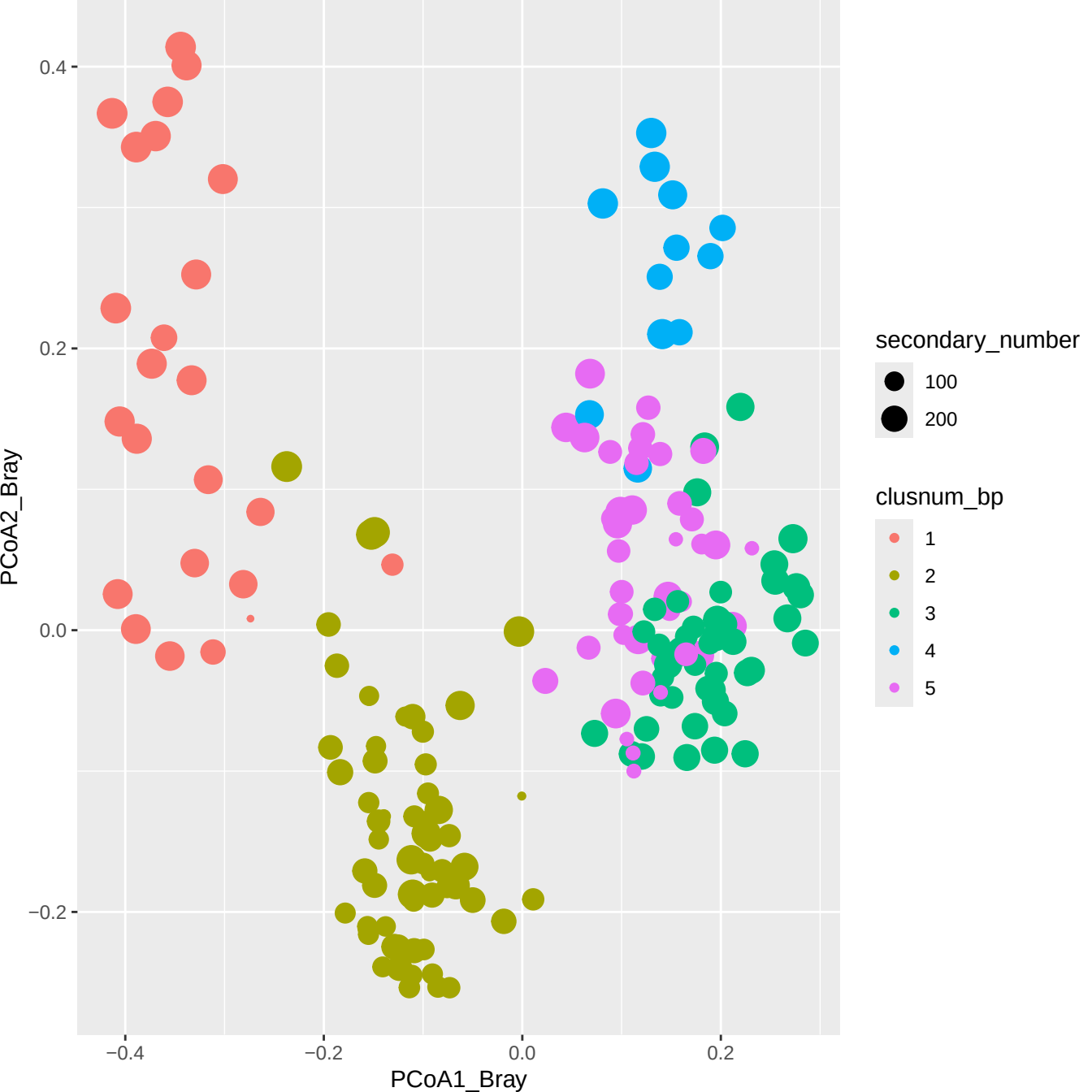

PCoA1_Euclidean
1.0
0.5
0.0
−0.5
−1.0
1
2
3
4
5
clusnum_bp
clusnum_bp
1
2
3
4
5

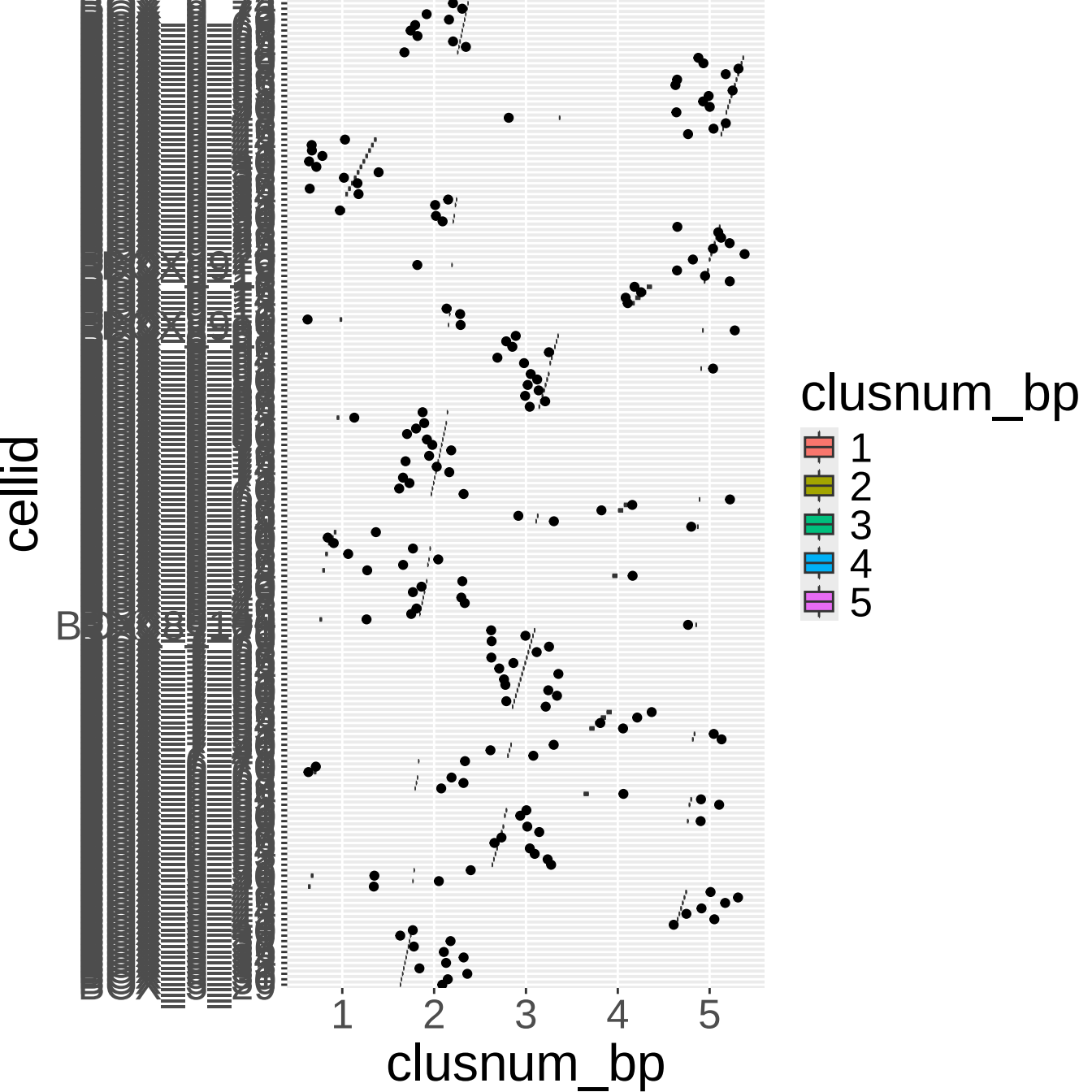

0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
clusnum_bp
1
2
3
4
5
PCoA1_Euclidean
−1.0
−0.5
0.0
0.5
1.0


PCoA2_Euclidean
1.0
0.5
0.0
−0.5
1
2
3
4
5
clusnum_bp
clusnum_bp
1
2
3
4
5

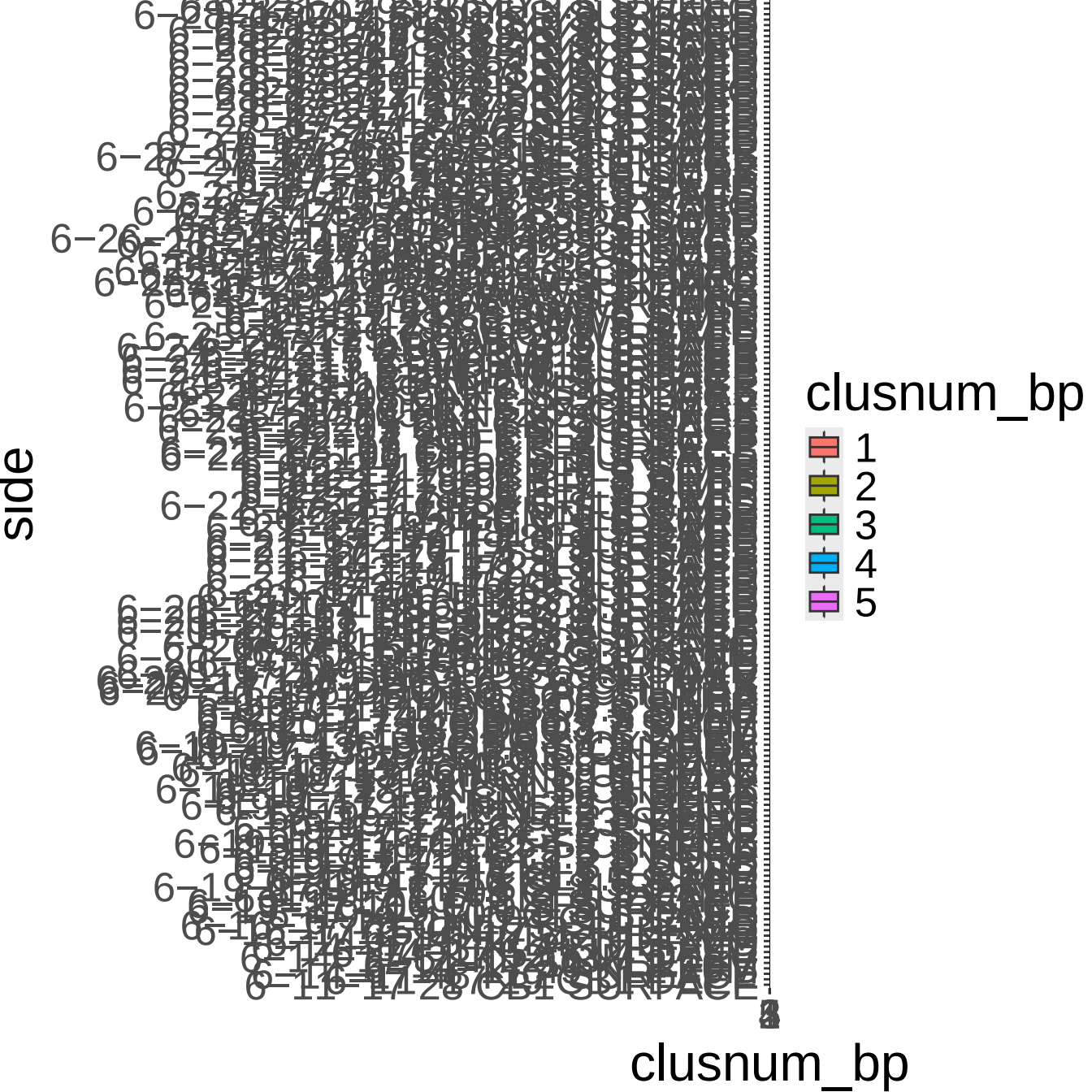

0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
PCoA2_Euclidean
−0.5
0.0
0.5
1.0
clusnum_bp
1
2
3
4
5


300
200
100
secondary_number
1
2
3
4
5
clusnum_bp
clusnum_bp
1
2
3
4
5

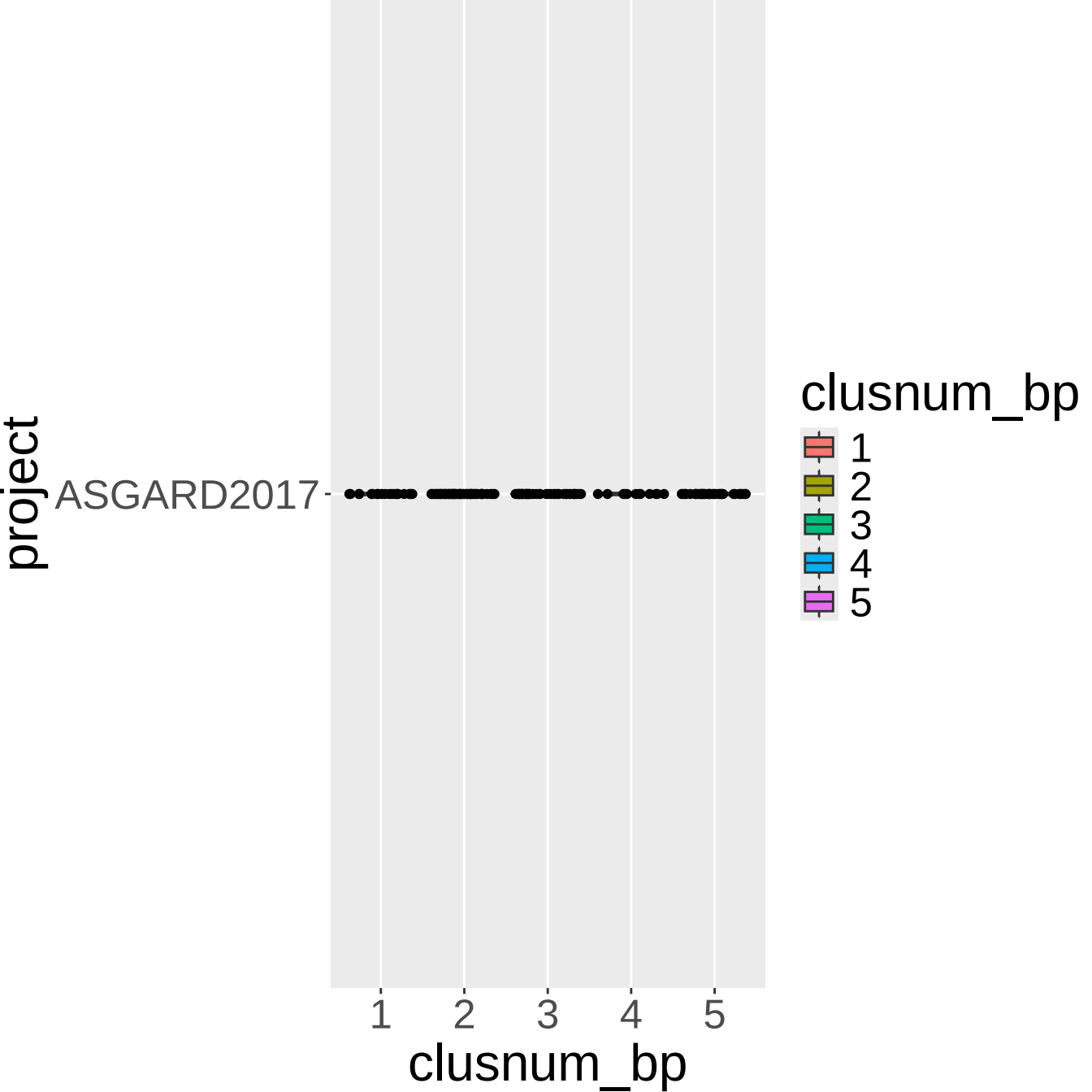

0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
secondary_number
100
200
clusnum_bp
1
2
3
4
5

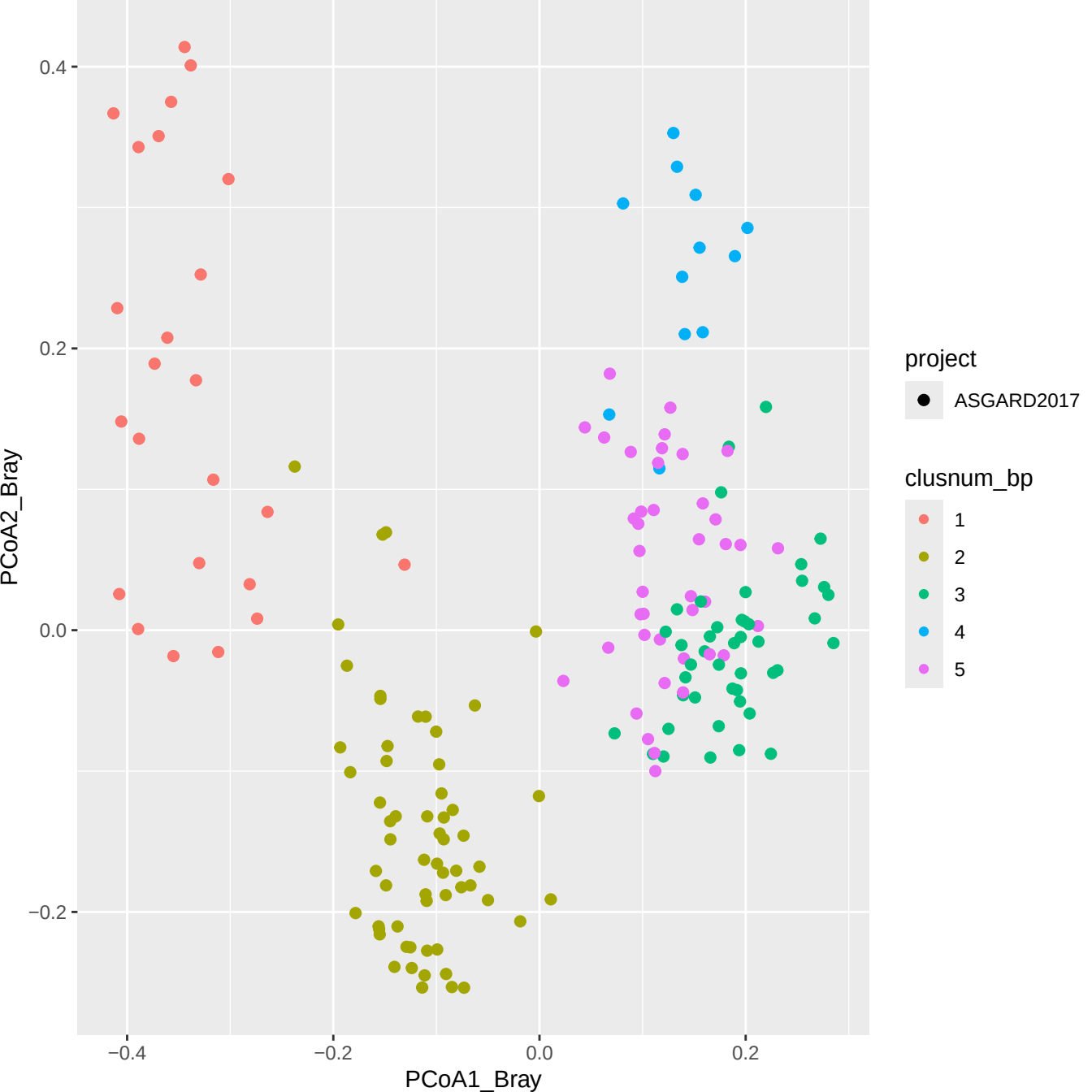

cellid
clusnum_bp
1
2
3
4
5
clusnum_bp
1
2
3
4
5

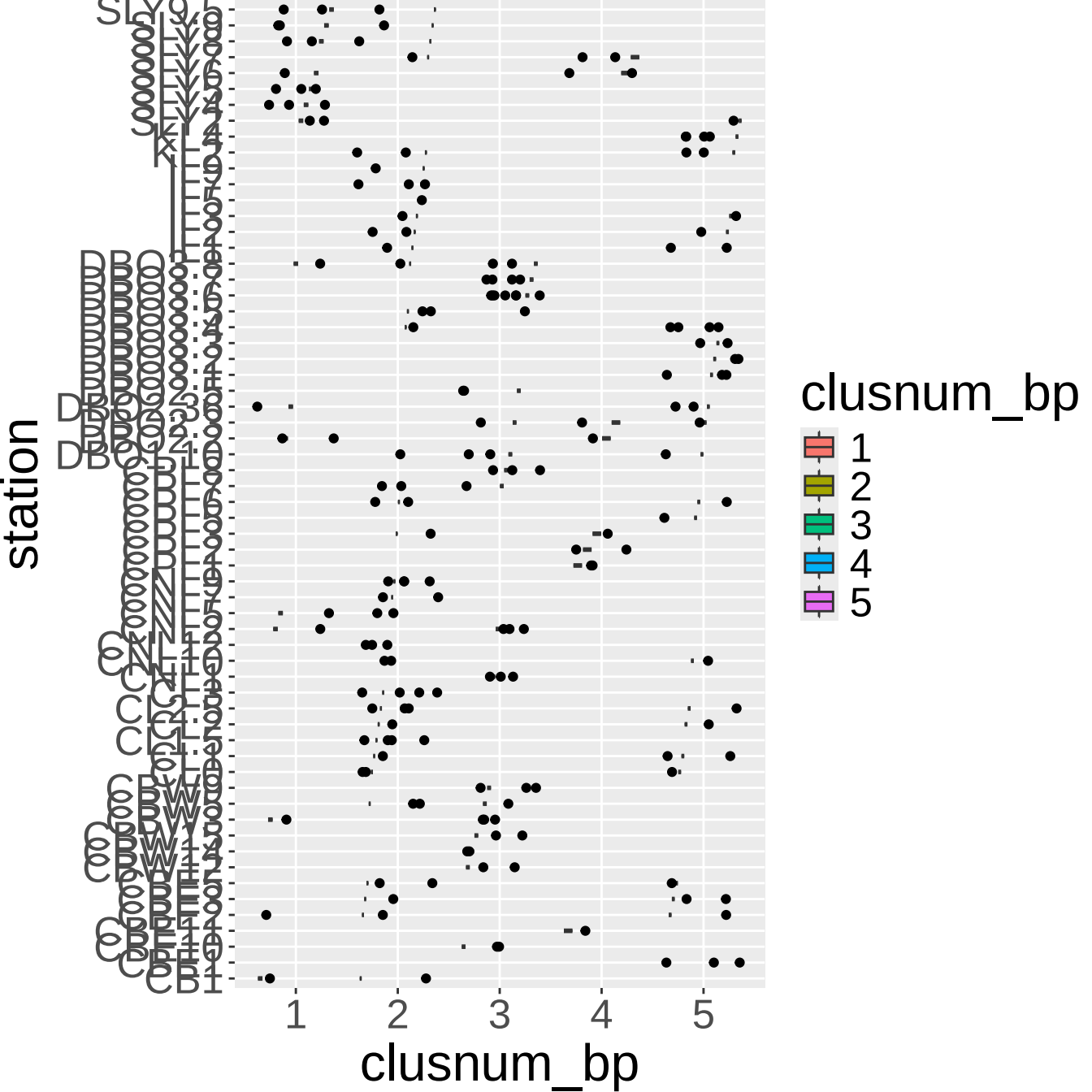
●	BOX_5_49	●	BOX_6_68	●	BOX_7_63	●	BOX_8_54	●	BOX_8_74	●	BOX_8_93	●	BOX_9_20	●
●	BOX_5_50	●	BOX_6_69	●	BOX_7_64	●	BOX_8_55	●	BOX_8_75	●	BOX_8_94	●	BOX_9_23	●
●	BOX_5_51	●	BOX_6_70	●	BOX_7_65	●	BOX_8_56	●	BOX_8_76	●	BOX_8_95	●	BOX_9_24	●
●	BOX_6_51	●	BOX_6_71	●	BOX_7_66	●	BOX_8_57	●	BOX_8_77	●	BOX_8_96	●	BOX_9_25	●
●	BOX_6_52	●	BOX_7_48	●	BOX_7_67	●	BOX_8_58	●	BOX_8_78	●	BOX_8_97	●	BOX_9_26	●
●	BOX_6_53	●	BOX_7_49	●	BOX_7_68	●	BOX_8_59	●	BOX_8_79	●	BOX_8_98	●	BOX_9_27	●
●	BOX_6_54	●	BOX_7_50	●	BOX_7_69	●	BOX_8_60	●	BOX_8_80	●	BOX_8_99	●	BOX_9_28	●
●	BOX_6_55	●	BOX_7_51	●	BOX_7_70	●	BOX_8_61	●	BOX_8_81	●	BOX_9_1	●	BOX_9_29	●
●	BOX_6_56	●	BOX_7_52	●	BOX_7_71	●	BOX_8_62	●	BOX_8_82	●	BOX_9_10	●	BOX_9_30	●
●	BOX_6_57	●	BOX_7_53	●	BOX_8_100	●	BOX_8_64	●	BOX_8_83	●	BOX_9_11	●	BOX_9_31	●
●	BOX_6_58	●	BOX_7_54	●	BOX_8_44	●	BOX_8_65	●	BOX_8_84	●	BOX_9_12	●	BOX_9_32	●
●	BOX_6_59	●	BOX_7_55	●	BOX_8_45	●	BOX_8_66	●	BOX_8_85	●	BOX_9_13	●	BOX_9_33	●
●	BOX_6_60	●	BOX_7_56	●	BOX_8_46	●	BOX_8_67	●	BOX_8_86	●	BOX_9_14	●	BOX_9_34	●
●	BOX_6_61	●	BOX_7_57	●	BOX_8_47	●	BOX_8_68	●	BOX_8_87	●	BOX_9_15	●	BOX_9_35	●
●	BOX_6_62	●	BOX_7_58	●	BOX_8_48	●	BOX_8_69	●	BOX_8_88	●	BOX_9_16	●	BOX_9_36	●
●	BOX_6_63	●	BOX_7_59	●	BOX_8_49	●	BOX_8_70	●	BOX_8_89	●	BOX_9_17	●	BOX_9_37	●
●	BOX_6_64	●	BOX_7_60	●	BOX_8_50	●	BOX_8_71	●	BOX_8_90	●	BOX_9_18	●	BOX_9_38	●
●	BOX_6_66	●	BOX_7_61	●	BOX_8_52	●	BOX_8_72	●	BOX_8_91	●	BOX_9_19	●	BOX_9_39	●
●	BOX_6_67	●	BOX_7_62	●	BOX_8_53	●	BOX_8_73	●	BOX_8_92	●	BOX_9_2	●	BOX_9_40	●

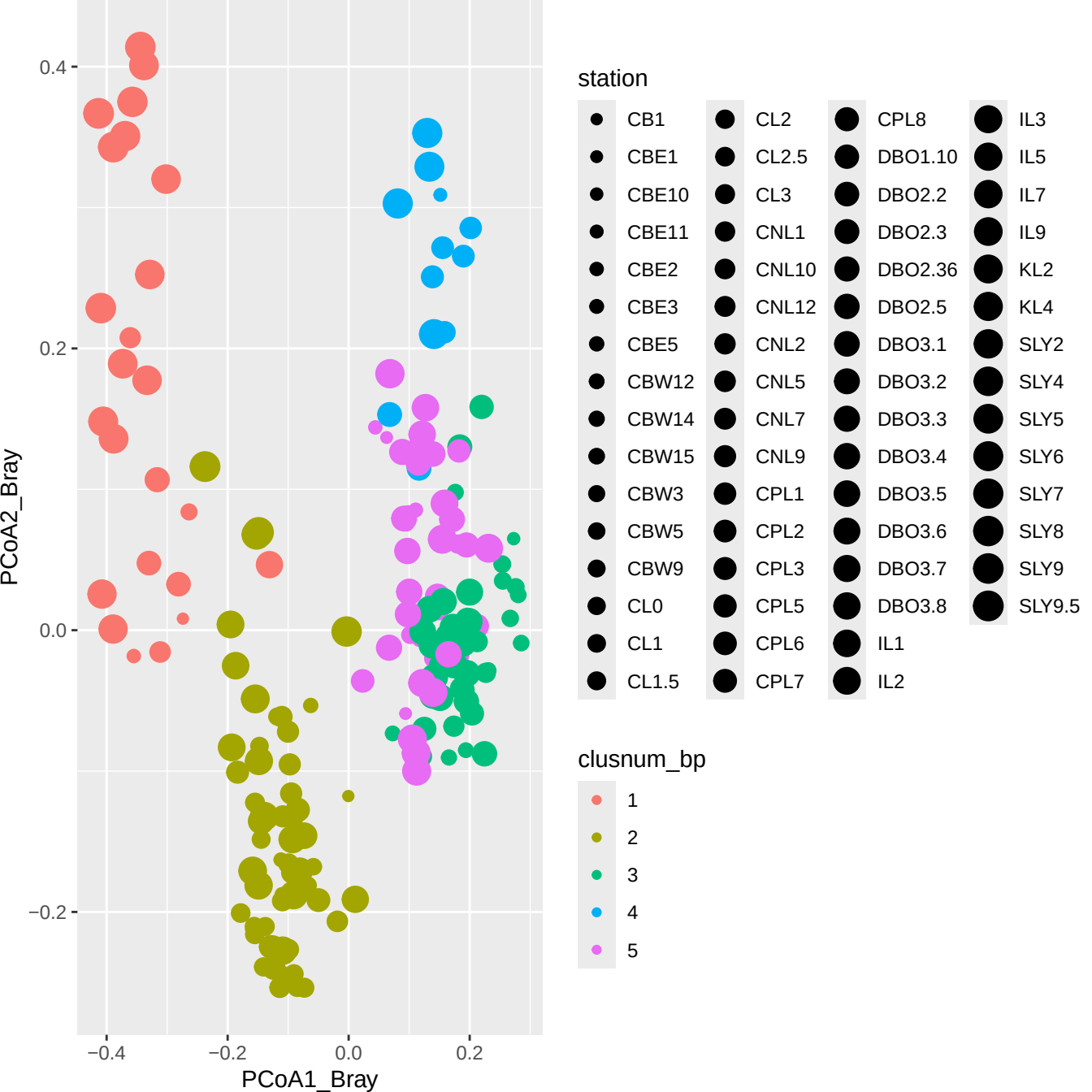

clusnum_bp
1
2
3
4
5
side
clusnum_bp

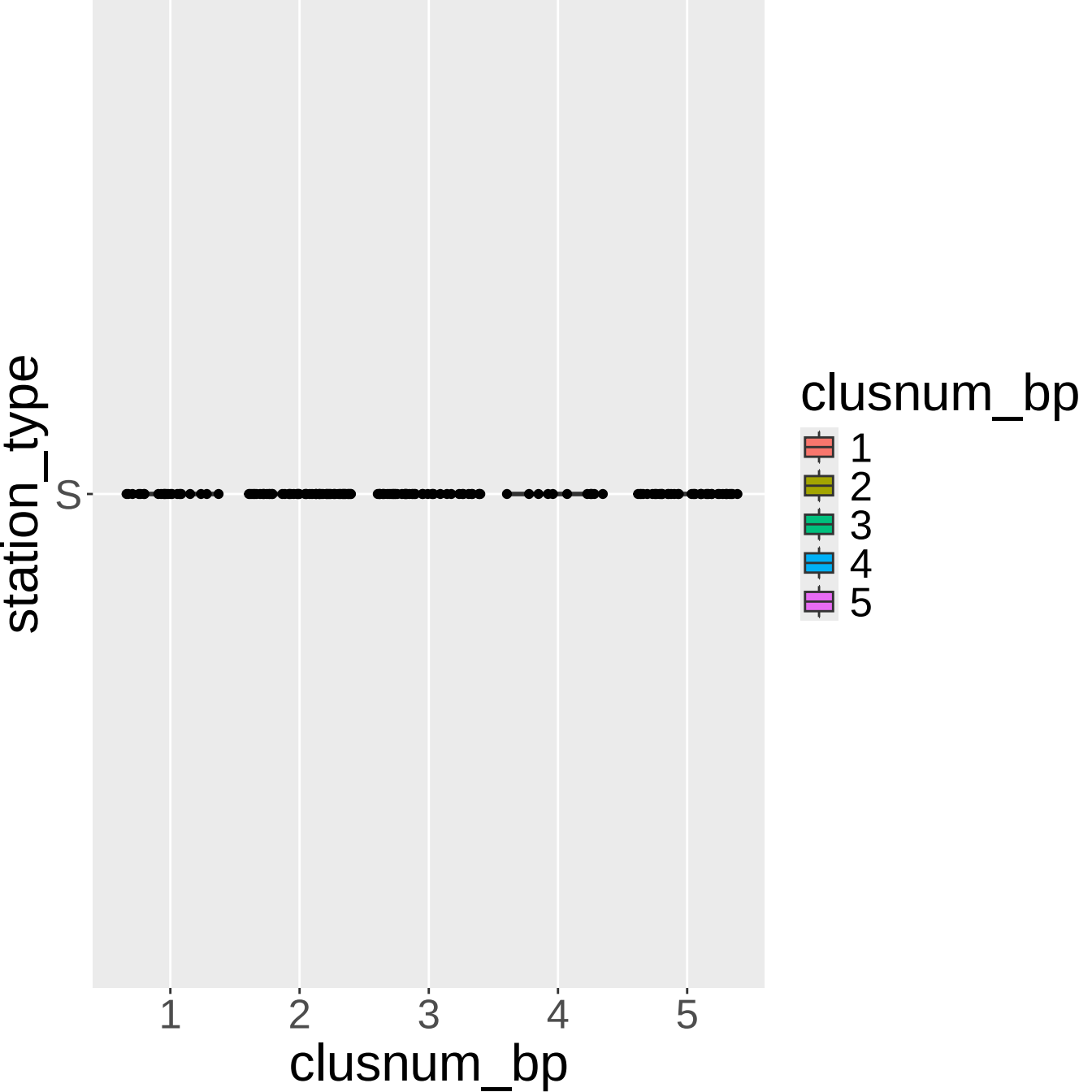
9 DBO3.5 S SURFACE	●	6−21−17 171 IL2 S MID	●	6−22−17 194 CPL6 S MID	●	6−24−17 214	
1 DBO3.5 S MID	●	6−21−17 172 IL2 S DEEP	●	6−22−17 195 CPL6 S DEEP	●	6−24−17 215	
2 DBO3.5 S DEEP	●	6−21−17 173 IL3 S SURFACE	●	6−22−17 196 CPL5 S SURFACE	●	6−24−17 216	
3 DBO3.4 S SURFACE	●	6−21−17 175 IL3 S DEEP	●	6−22−17 198 CPL3 S SURFACE	●	6−24−17 217	
4 DBO3.4 S ML	●	6−21−17 176 IL5 S SURFACE	●	6−22−17 199 CPL3 S DEEP	●	6−24−17 218	
5 DBO3.4 S PYCNO	●	6−21−17 179 IL7 S SURFACE	●	6−22−17 200 CPL2 S SURF	●	6−25−17 229	
6 DBO3.4 S MID	●	6−21−17 180 IL7 S MID	●	6−22−17 201 CPL2 S DEEP	●	6−25−17 230	
7 DBO3.4 S DEEP	●	6−21−17 181 IL7 S DEEP	●	6−23−17 202 CNL1 S SURFACE	●	6−25−17 231	
8 DBO3.3 S SURFACE	●	6−21−17 182 IL9 S SURFACE	●	6−23−17 203 CNL1 S DEEP	●	6−25−17 232	
0 DBO3.3 S DEEP	●	6−22−17 184 CNL5 S SURF	●	6−23−17 204 CNL2 S SURFACE	●	6−25−17 233	
1 DBO3.2 S SURFACE	●	6−22−17 185 CNL5 S MID	●	6−23−17 205 CNL2 S CHLMAX	●	6−25−17 234	
3 DBO3.2 S DEEP	●	6−22−17 186 CNL5 S DEEP	●	6−23−17 206 CNL2 S 2x CHLMAX	●	6−25−17 235	
4 DBO3.1 S SURFACE	●	6−22−17 187 CPL8 S SURFACE	●	6−23−17 207 CNL2 S DEEP	●	6−25−17 236	
5 DBO3.1 S MID	●	6−22−17 188 CPL8 S MID	●	6−23−17 208 CNL1 S SURFACE	●	6−25−17 237	
6 DBO3.1 S DEEP	●	6−22−17 189 CPL8 S DEEP	●	6−23−17 209 CNL1 S CHLMAX	●	6−25−17 238	
7 IL1 S SURFACE	●	6−22−17 190 CPL7 S SURF	●	6−23−17 210 CNL1 S DEEP	●	6−25−17 239	
8 IL1 S MID	●	6−22−17 191 CPL7 S MID	●	6−24−17 211 CBW15 S SURFACE	●	6−25−17 240	
9 IL1 S DEEP	●	6−22−17 192 CPL7 S DEEP	●	6−24−17 212 CBW15 S DEEP	●	6−25−17 241	
0 IL2 S SURFACE	●	6−22−17 193 CPL6 S SURF	●	6−24−17 213 CBW14 S SURFACE	●	6−25−17 242	

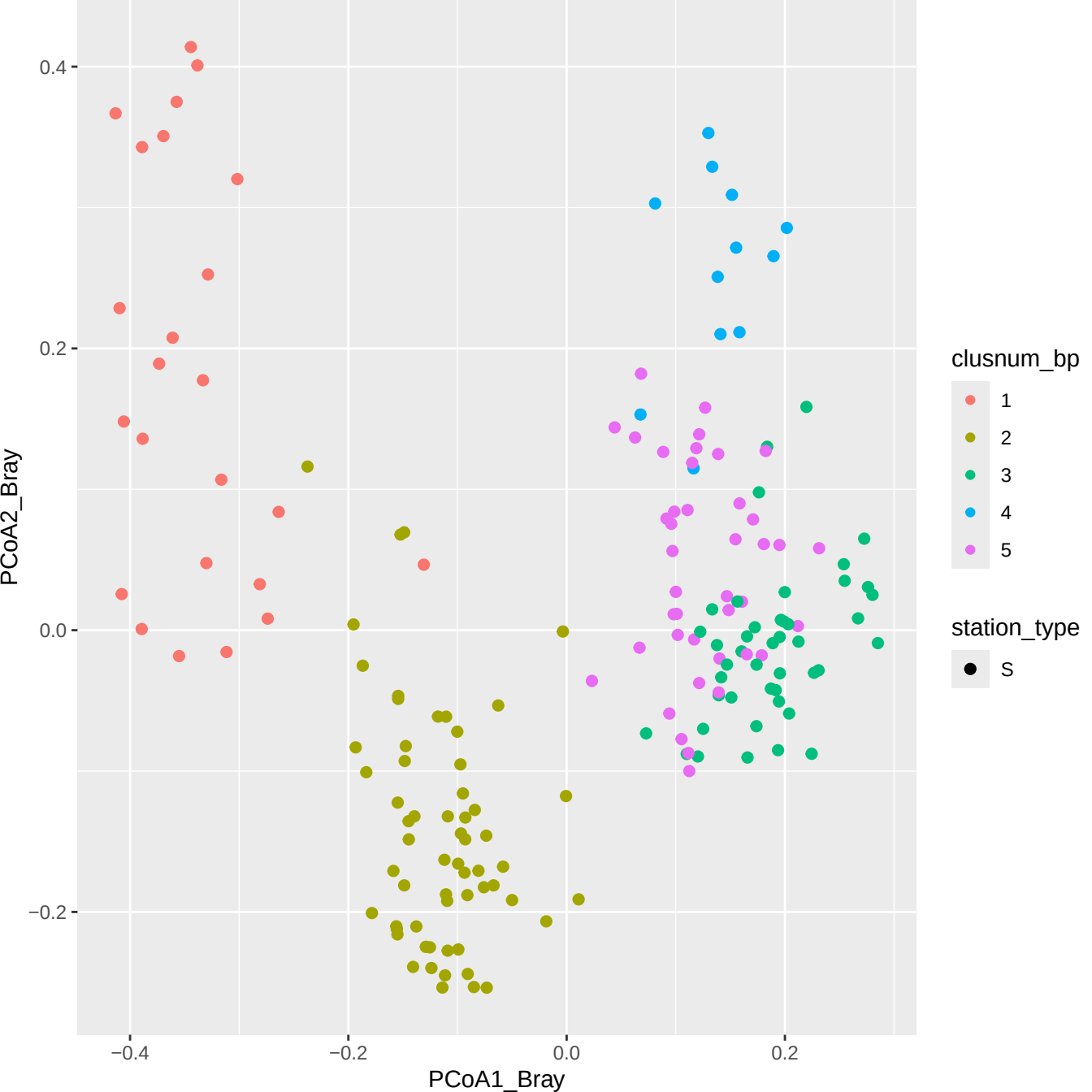

project
ASGARD2017
1
2
3
4
5
clusnum_bp
clusnum_bp
1
2
3
4
5

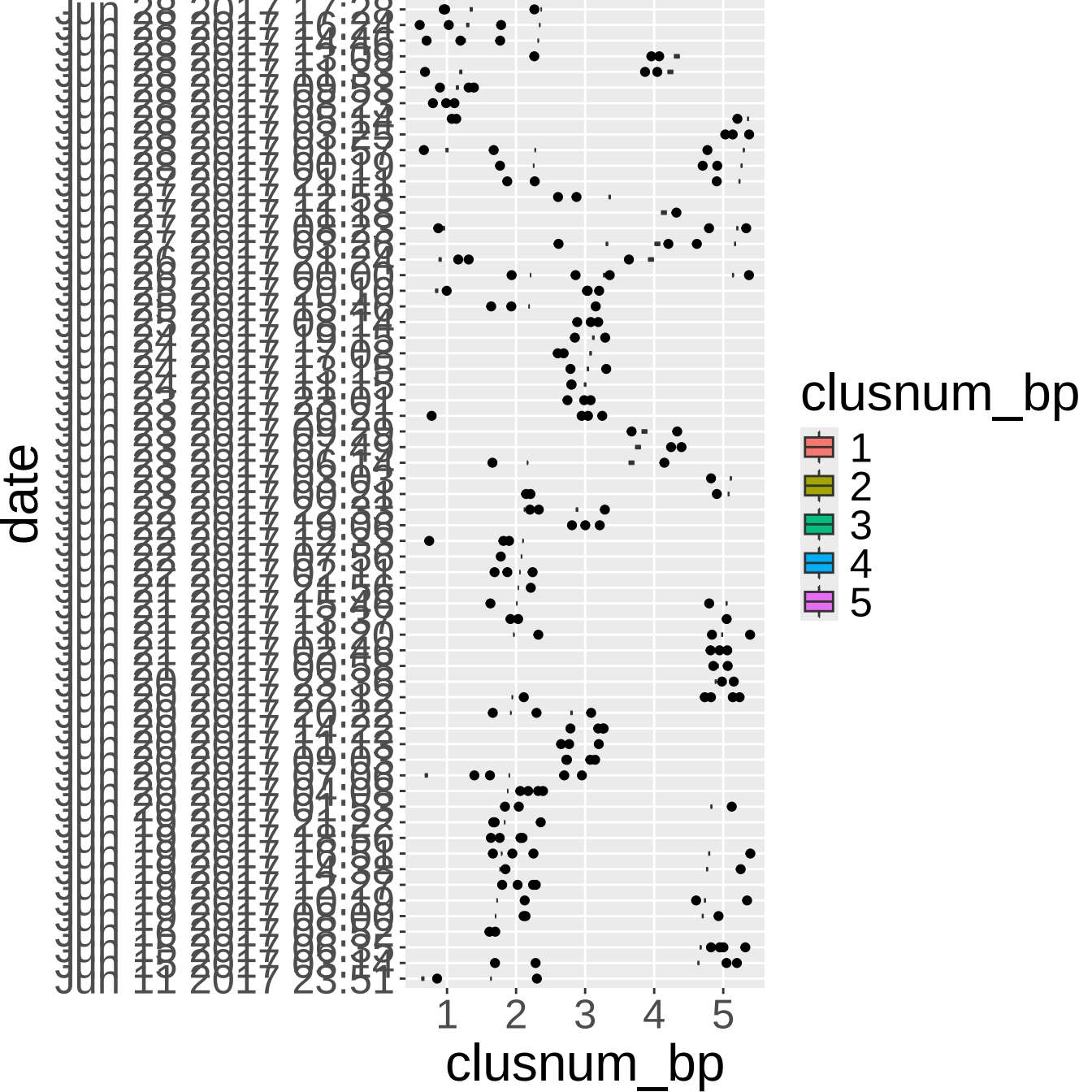

0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
project
ASGARD2017
clusnum_bp
1
2
3
4
5

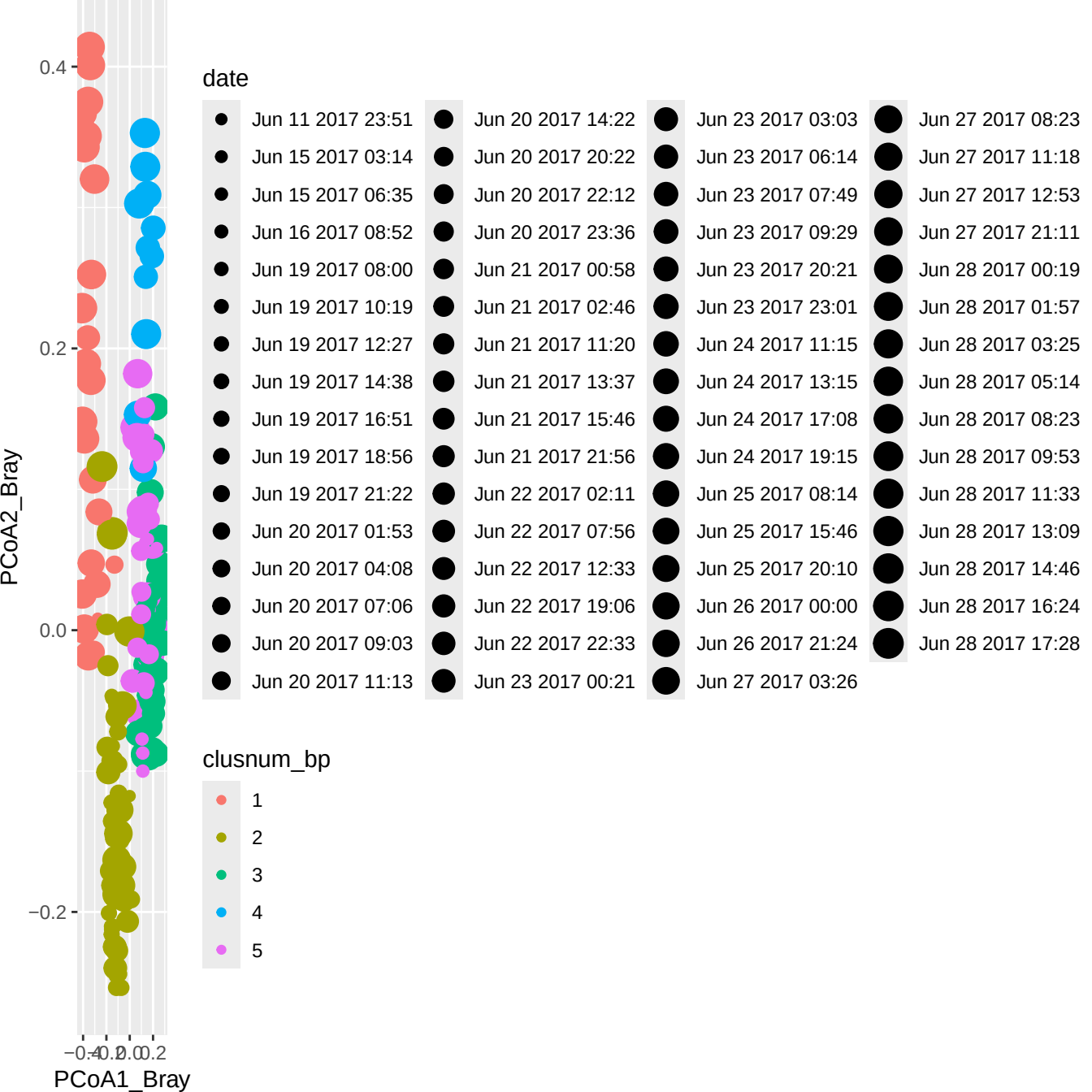

station
clusnum_bp
1
2
3
4
5
clusnum_bp
1
2
3
4
5

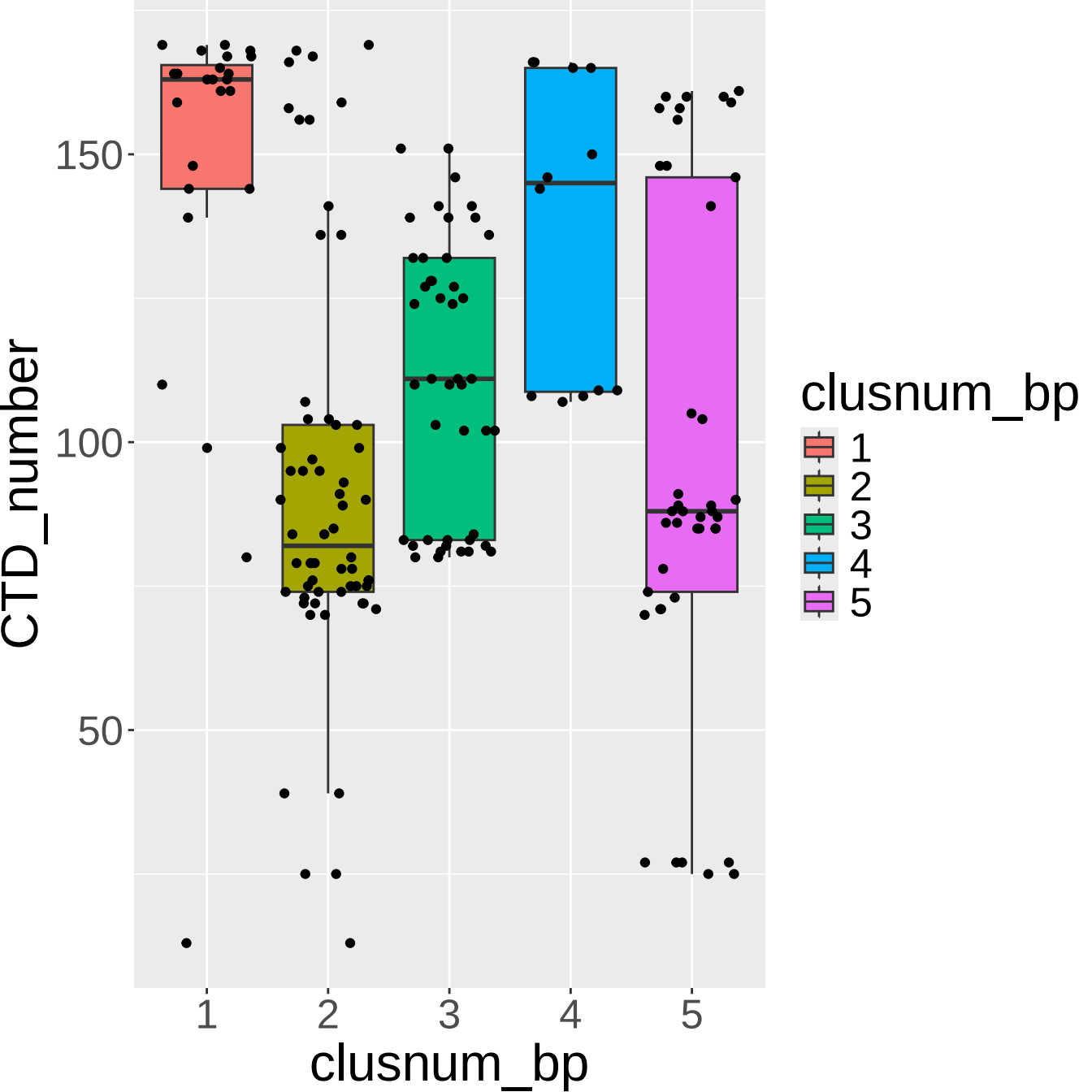

0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
station
CB1
CBE1
CBE10
CBE11
CBE2
CBE3
CBE5
CBW12
CBW14
CBW15
CBW3
CBW5
CBW9
CL0
CL1
CL1.5
CL2
CL2.5
CL3
CNL1
CNL10
CNL12
CNL2
CNL5
CNL7
CNL9
CPL1
CPL2
CPL3
CPL5
CPL6
CPL7
CPL8
DBO1.10
DBO2.2
DBO2.3
DBO2.36
DBO2.5
DBO3.1
DBO3.2
DBO3.3
DBO3.4
DBO3.5
DBO3.6
DBO3.7
DBO3.8
IL1
IL2
IL3
IL5
IL7
IL9
KL2
KL4
SLY2
SLY4
SLY5
SLY6
SLY7
SLY8
SLY9
SLY9.5
clusnum_bp
1
2
3
4
5

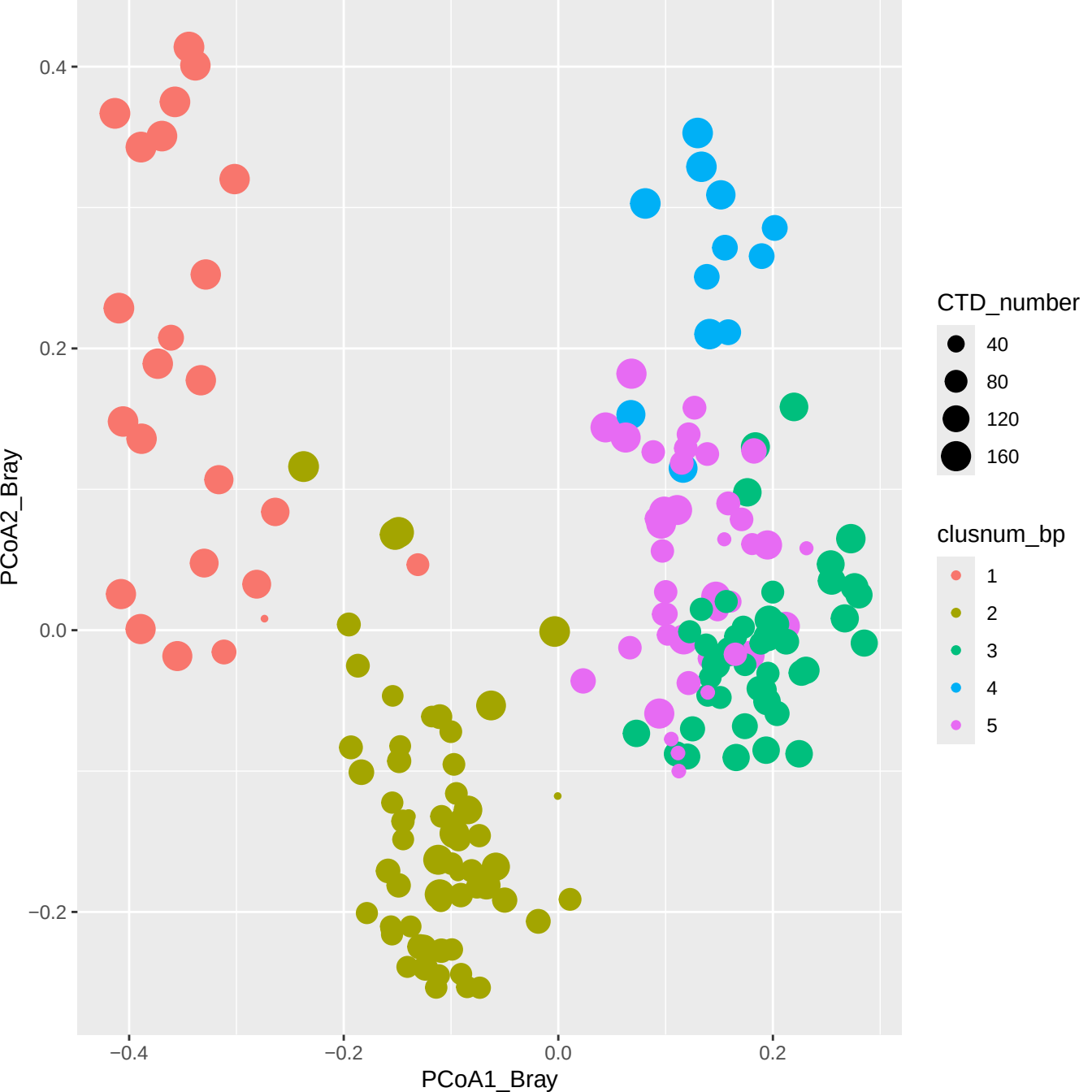

station_type
S
1
2
3
4
5
clusnum_bp
clusnum_bp
1
2
3
4
5

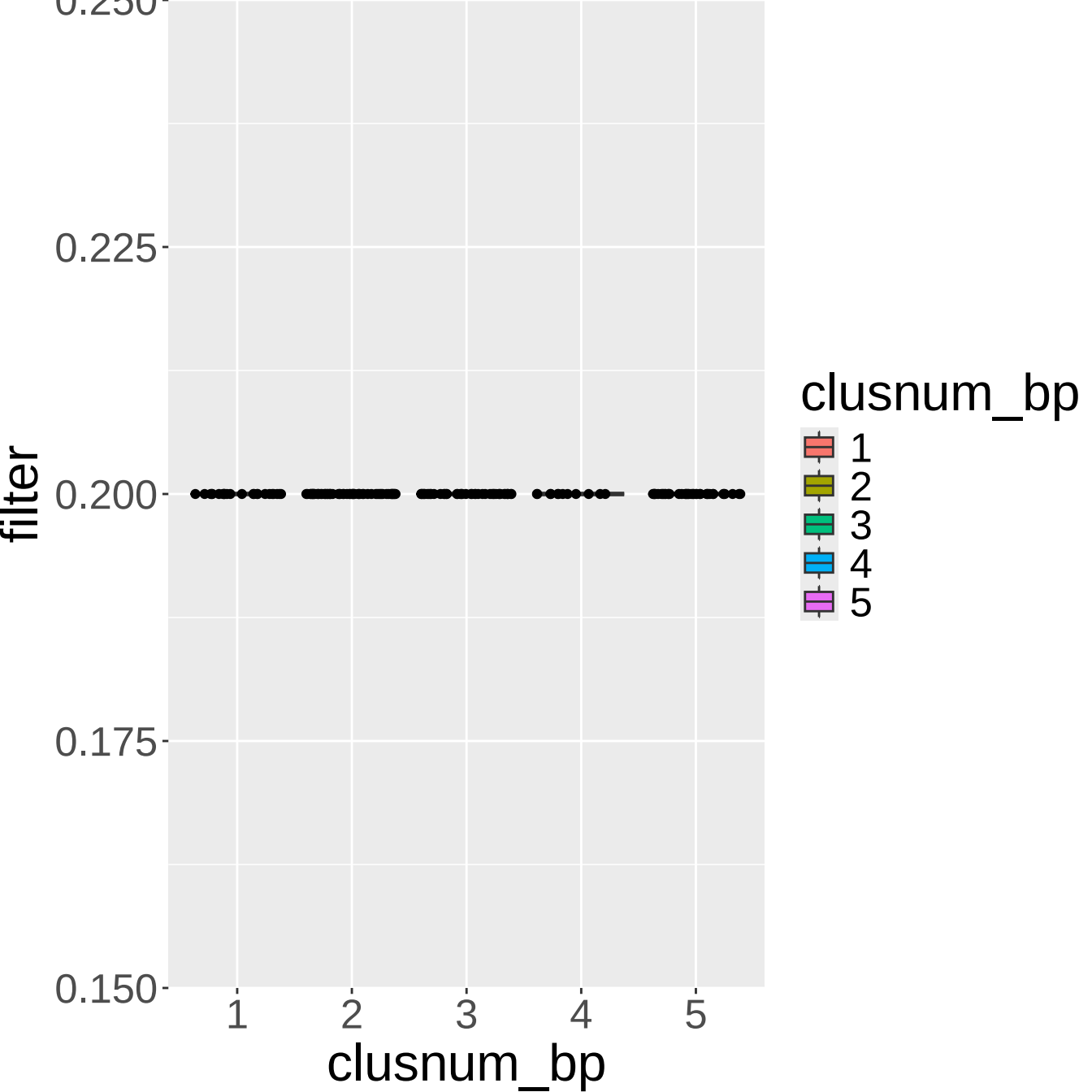

0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
clusnum_bp
1
2
3
4
5
station_type
S

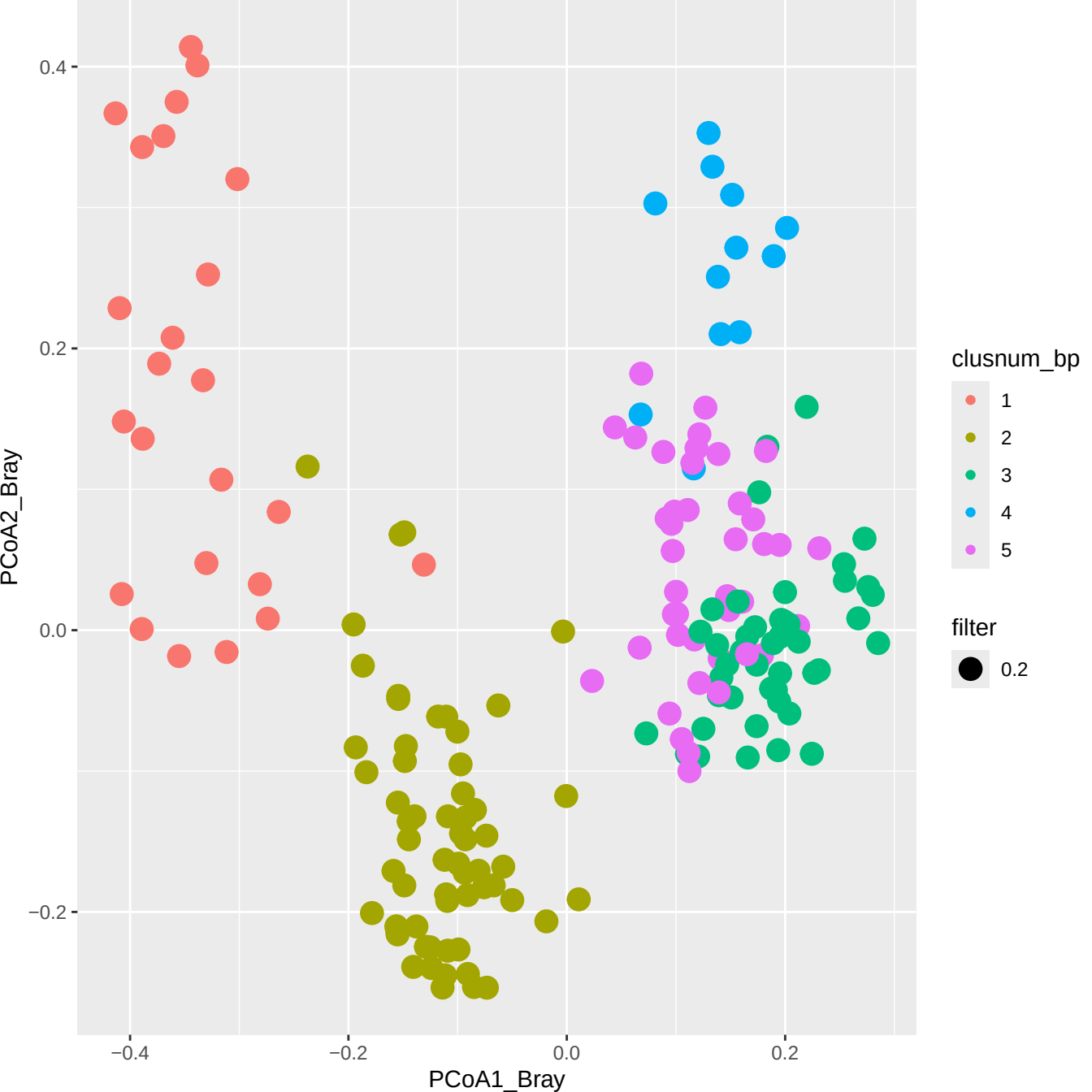

date
clusnum_bp
1
2
3
4
5
clusnum_bp
1
2
3
4
5

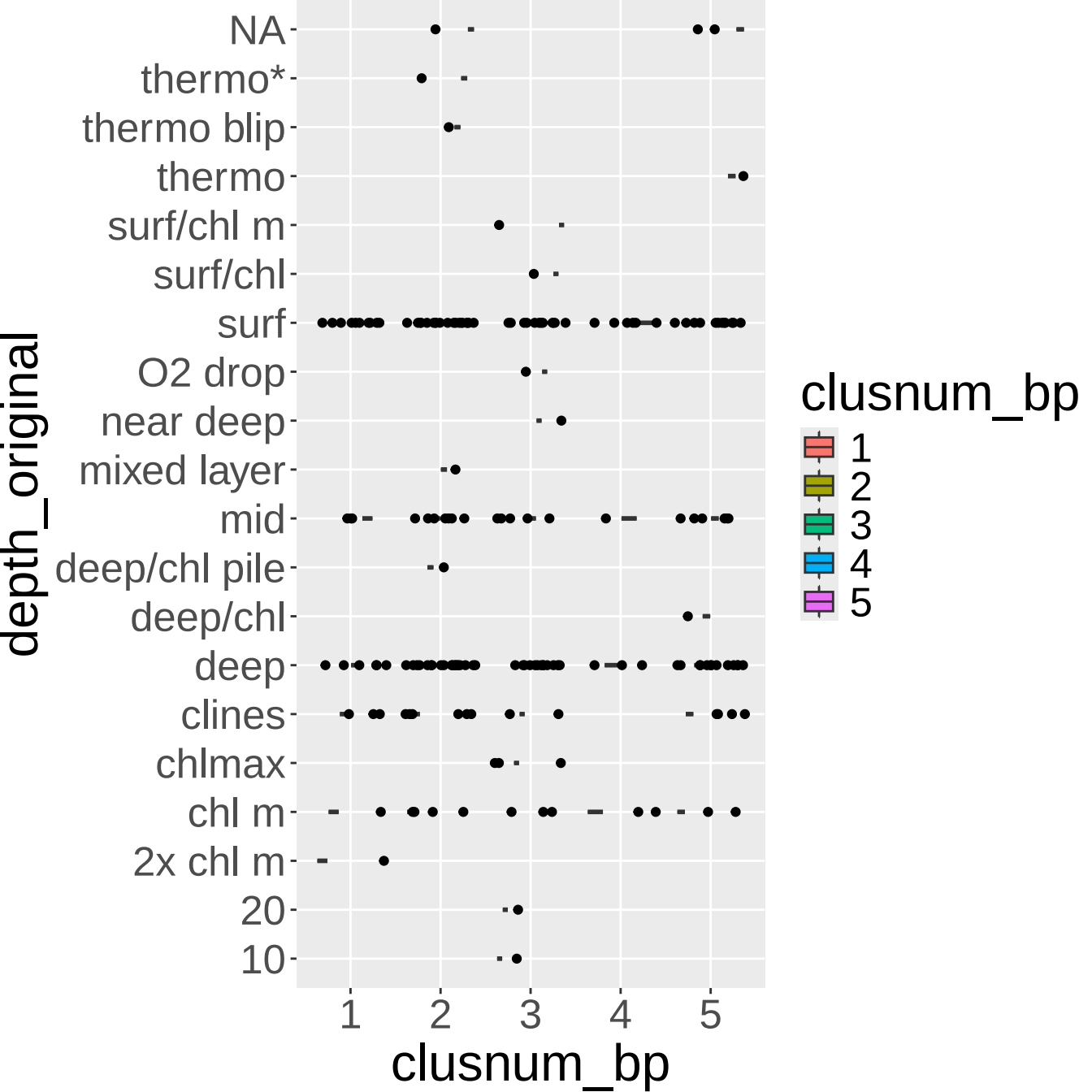

0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
date
Jun 11 2017 23:51
Jun 15 2017 03:14
Jun 15 2017 06:35
Jun 16 2017 08:52
Jun 19 2017 08:00
Jun 19 2017 10:19
Jun 19 2017 12:27
Jun 19 2017 14:38
Jun 19 2017 16:51
Jun 19 2017 18:56
Jun 19 2017 21:22
Jun 20 2017 01:53
Jun 20 2017 04:08
Jun 20 2017 07:06
Jun 20 2017 09:03
Jun 20 2017 11:13
Jun 20 2017 14:22
Jun 20 2017 20:22
Jun 20 2017 22:12
Jun 20 2017 23:36
Jun 21 2017 00:58
Jun 21 2017 02:46
Jun 21 2017 11:20
Jun 21 2017 13:37
Jun 21 2017 15:46
Jun 21 2017 21:56
Jun 22 2017 02:11
Jun 22 2017 07:56
Jun 22 2017 12:33
Jun 22 2017 19:06
Jun 22 2017 22:33
Jun 23 2017 00:21
Jun 23 2017 03:03
Jun 23 2017 06:14
Jun 23 2017 07:49
Jun 23 2017 09:29
Jun 23 2017 20:21
Jun 23 2017 23:01
Jun 24 2017 11:15
Jun 24 2017 13:15
Jun 24 2017 17:08
Jun 24 2017 19:15
Jun 25 2017 08:14
Jun 25 2017 15:46
Jun 25 2017 20:10
Jun 26 2017 00:00
Jun 26 2017 21:24
Jun 27 2017 03:26
Jun 27 2017 08:23
Jun 27 2017 11:18
Jun 27 2017 12:53
Jun 27 2017 21:11
Jun 28 2017 00:19
Jun 28 2017 01:57
Jun 28 2017 03:25
Jun 28 2017 05:14
Jun 28 2017 08:23
Jun 28 2017 09:53
Jun 28 2017 11:33
Jun 28 2017 13:09
Jun 28 2017 14:46
Jun 28 2017 16:24
Jun 28 2017 17:28
clusnum_bp
1
2
3
4
5

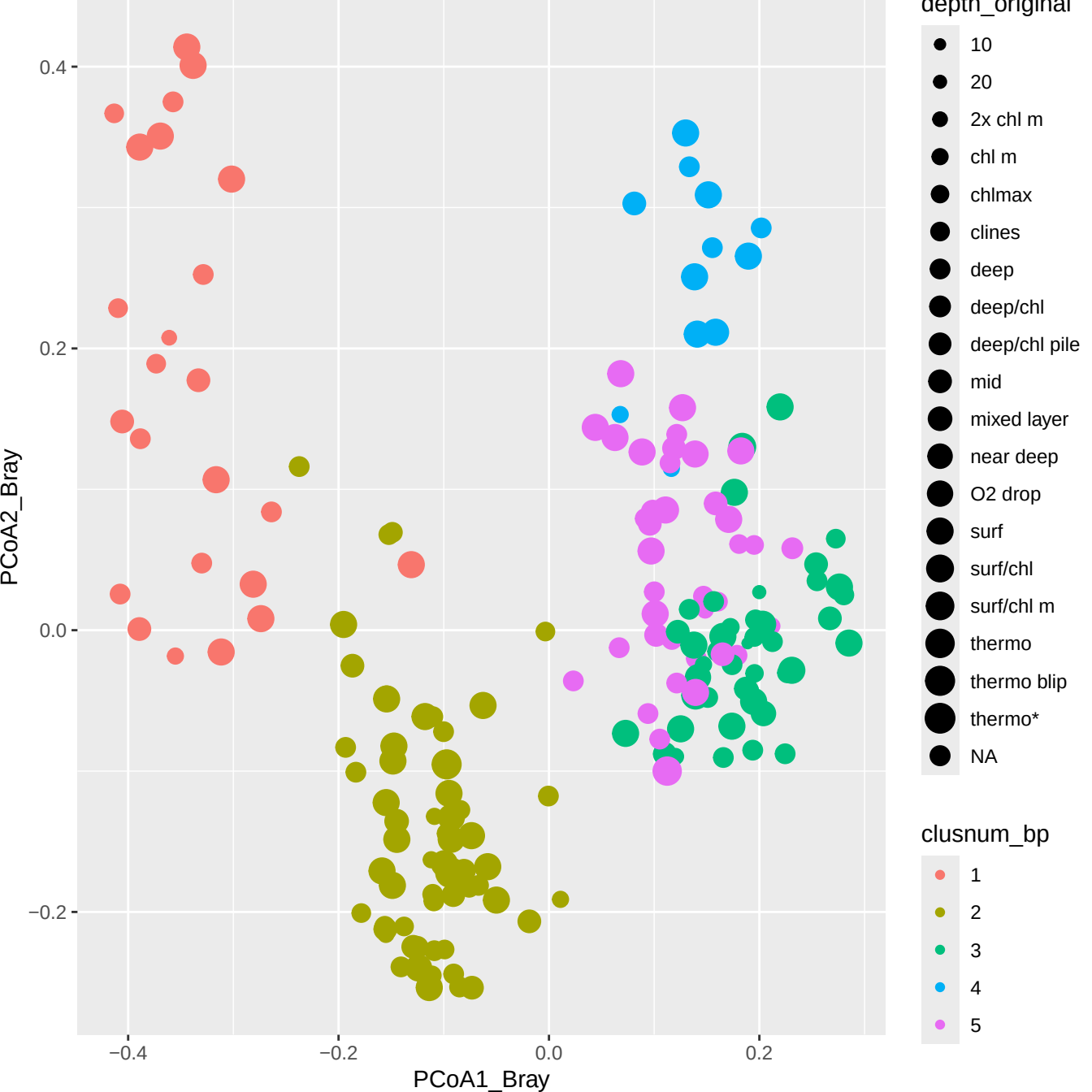

CTD_number
150
100
50
1
2
3
4
5
clusnum_bp
clusnum_bp
1
2
3
4
5

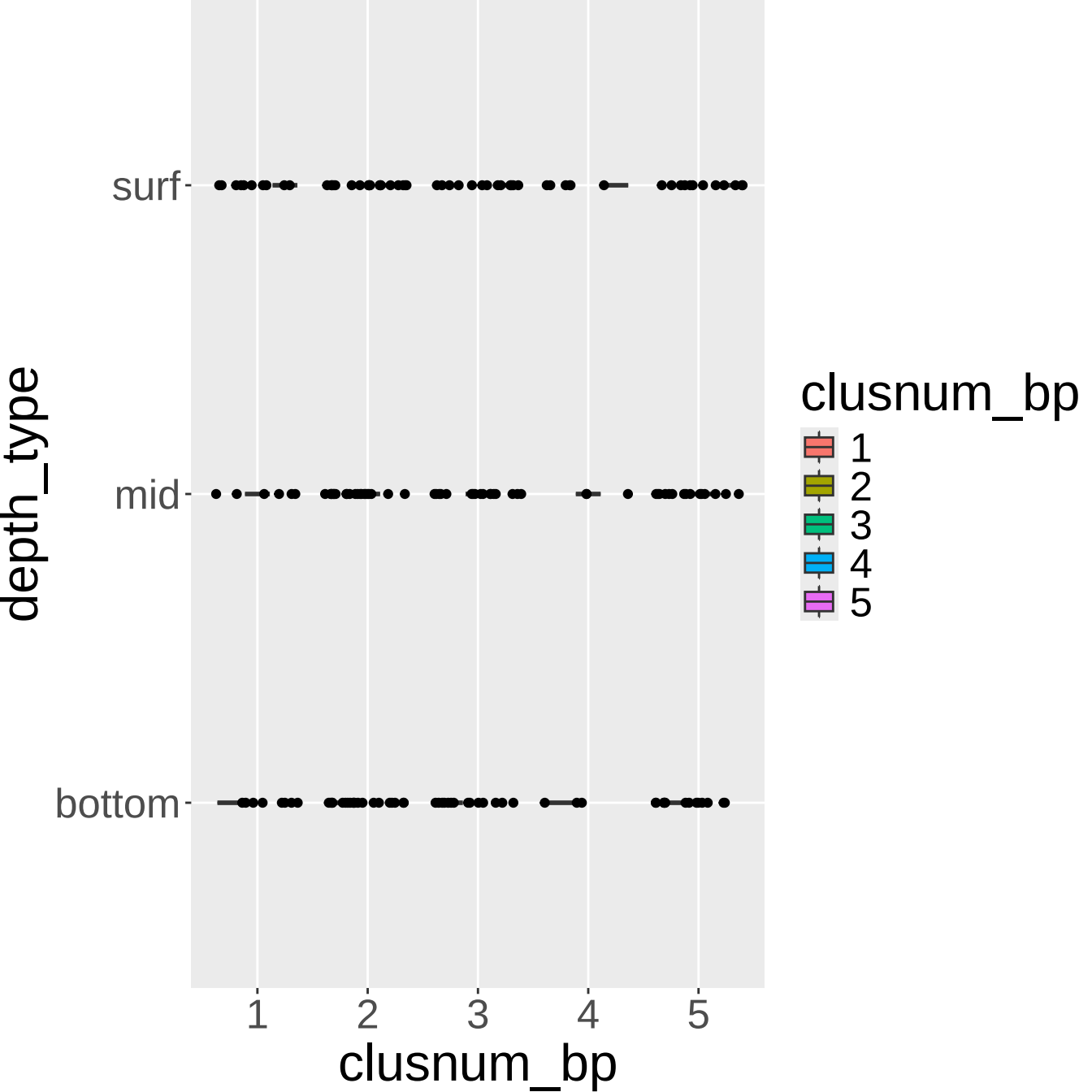

0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
CTD_number
40
80
120
160
clusnum_bp
1
2
3
4
5

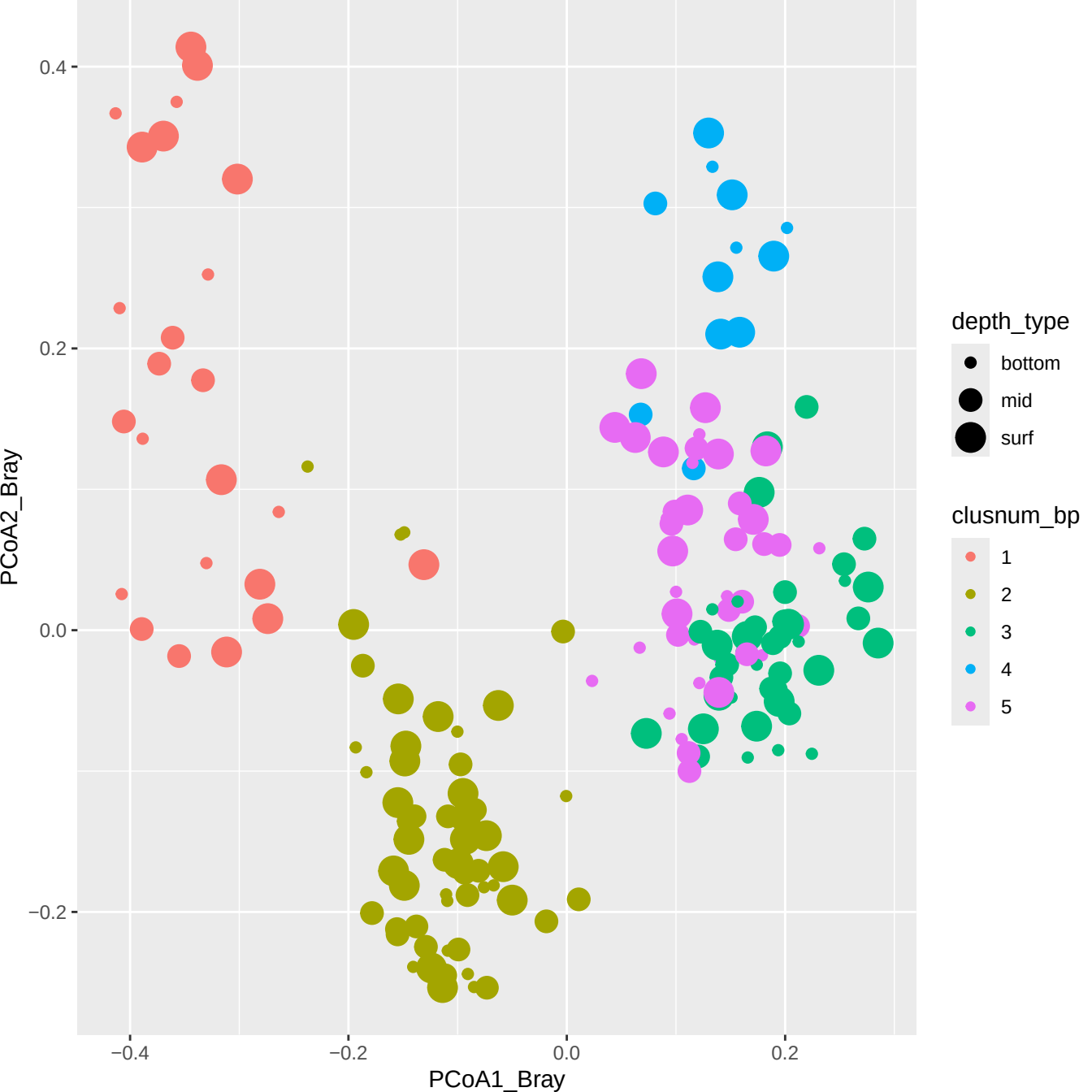

0.250
0.225
0.200
0.175
0.150
filter
1
2
3
4
5
clusnum_bp
clusnum_bp
1
2
3
4
5

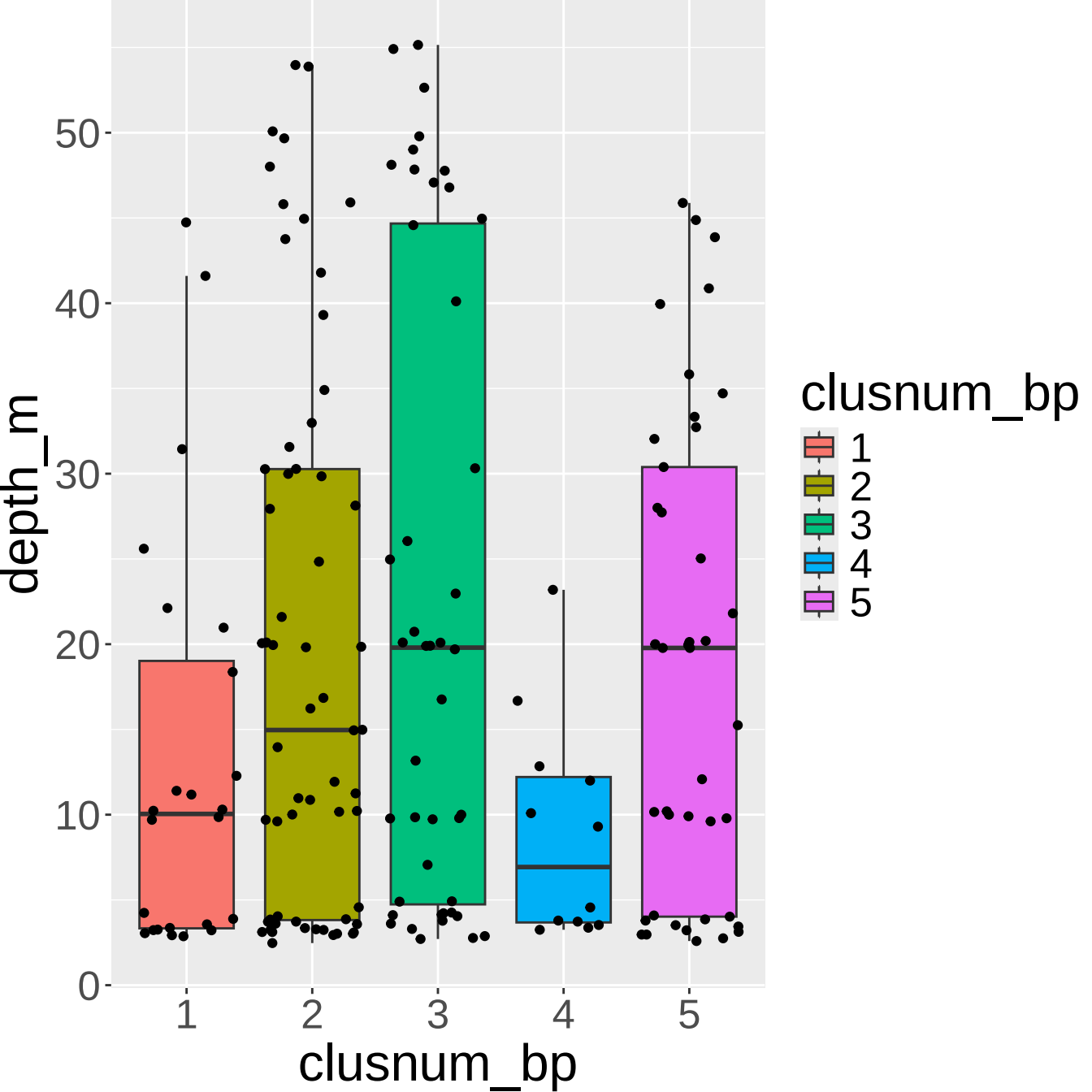

0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
clusnum_bp
1
2
3
4
5
filter
0.2

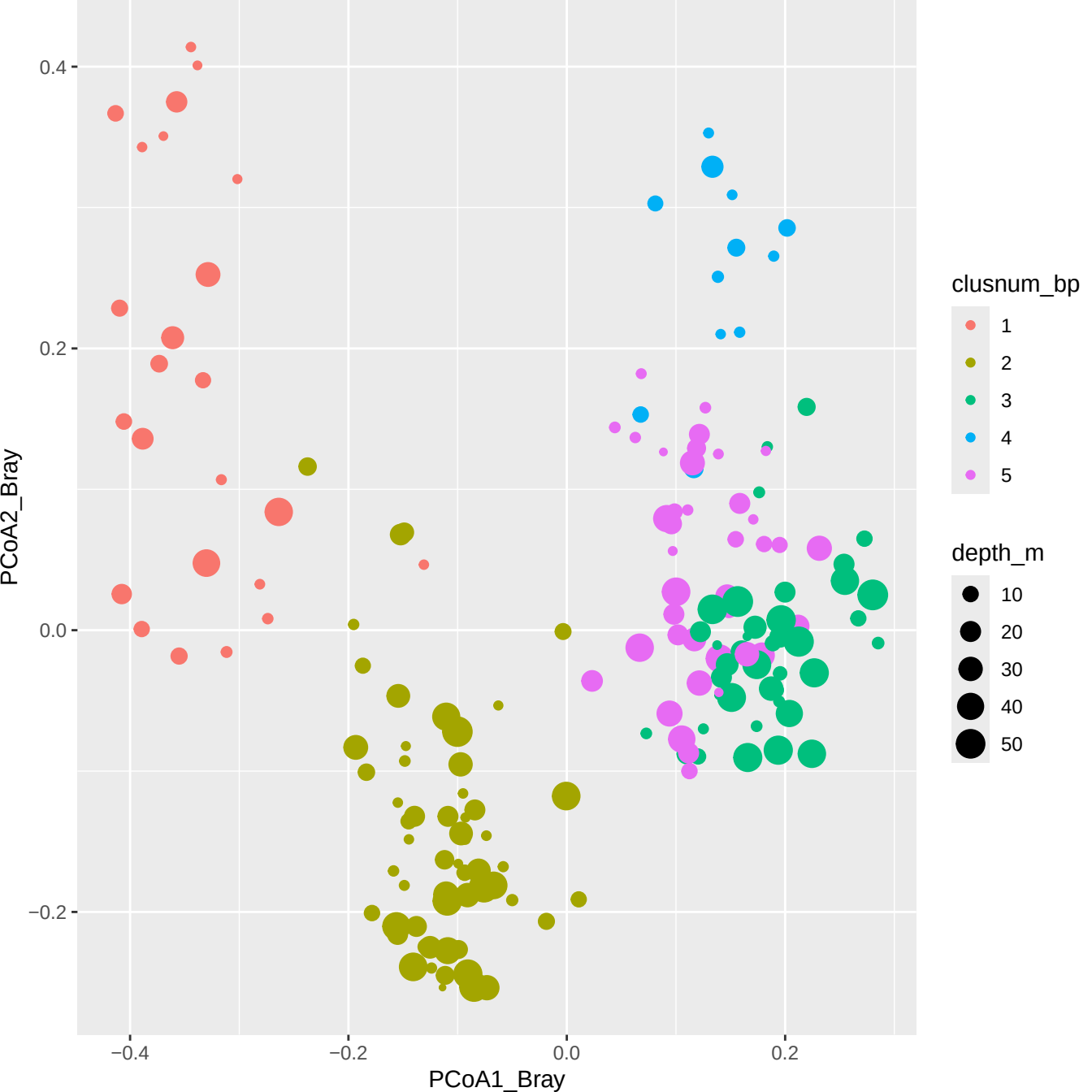

depth_original
NA
thermo*
thermo blip
thermo
surf/chl m
surf/chl
surf
O2 drop
near deep
mixed layer
mid
deep/chl pile
deep/chl
deep
clines
chlmax
chl m
2x chl m
20
10
1
2
3
4
5
clusnum_bp
clusnum_bp
1
2
3
4
5

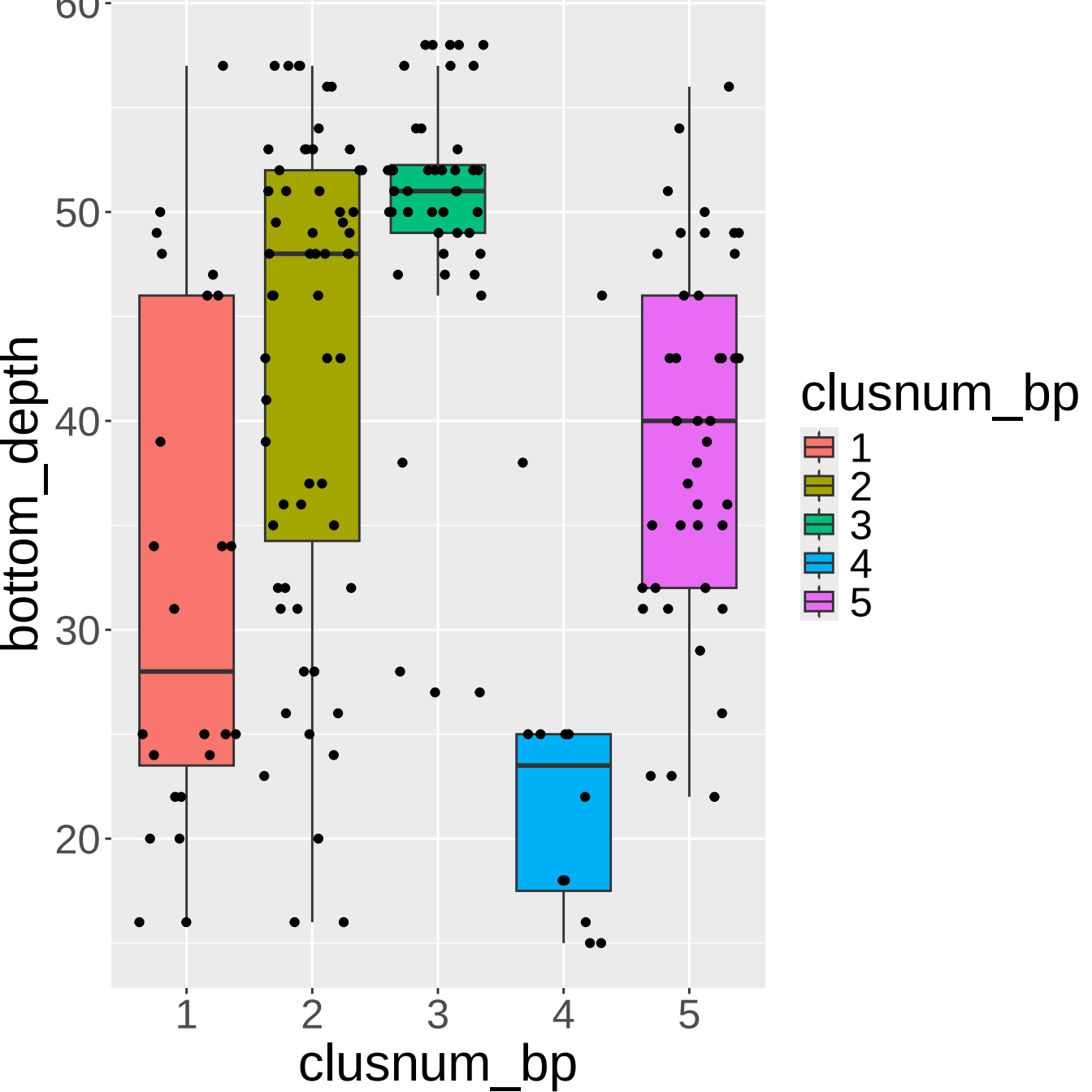

0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
depth_original
10
20
2x chl m
chl m
chlmax
clines
deep
deep/chl
deep/chl pile
mid
mixed layer
near deep
O2 drop
surf
surf/chl
surf/chl m
thermo
thermo blip
thermo*
NA
clusnum_bp
1
2
3
4
5

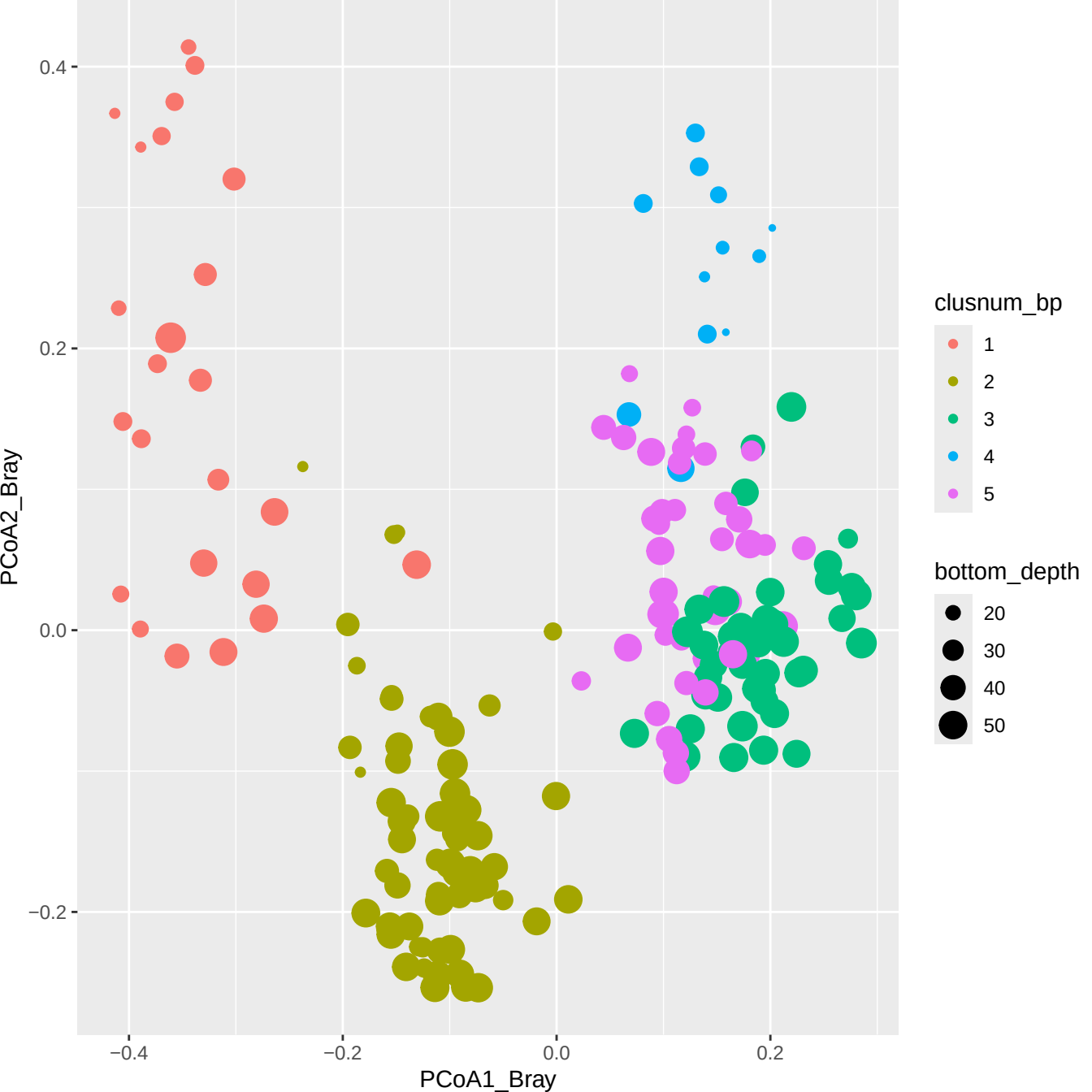

surf
mid
bottom
depth_type
1
2
3
4
5
clusnum_bp
clusnum_bp
1
2
3
4
5

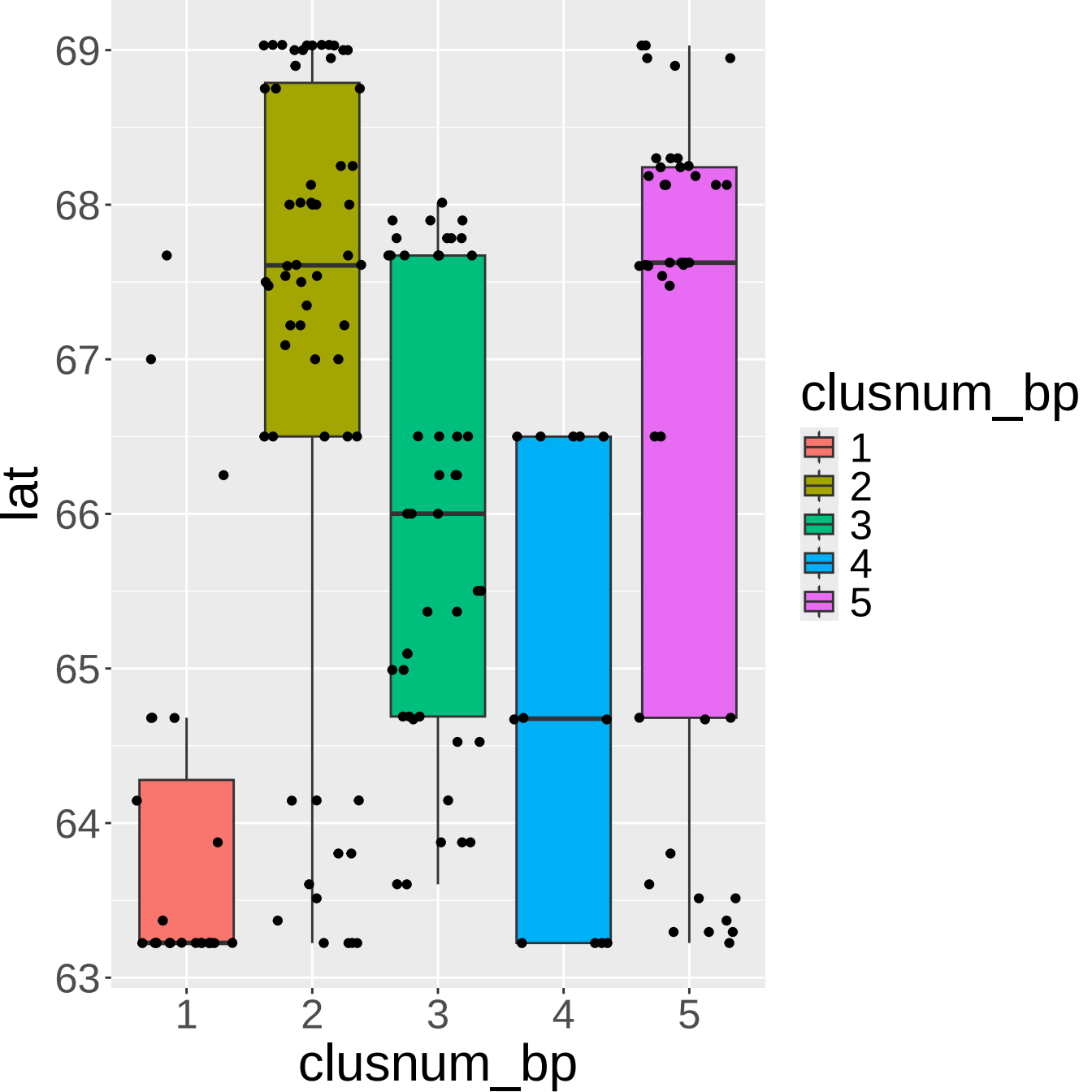

0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
depth_type
bottom
mid
surf
clusnum_bp
1
2
3
4
5

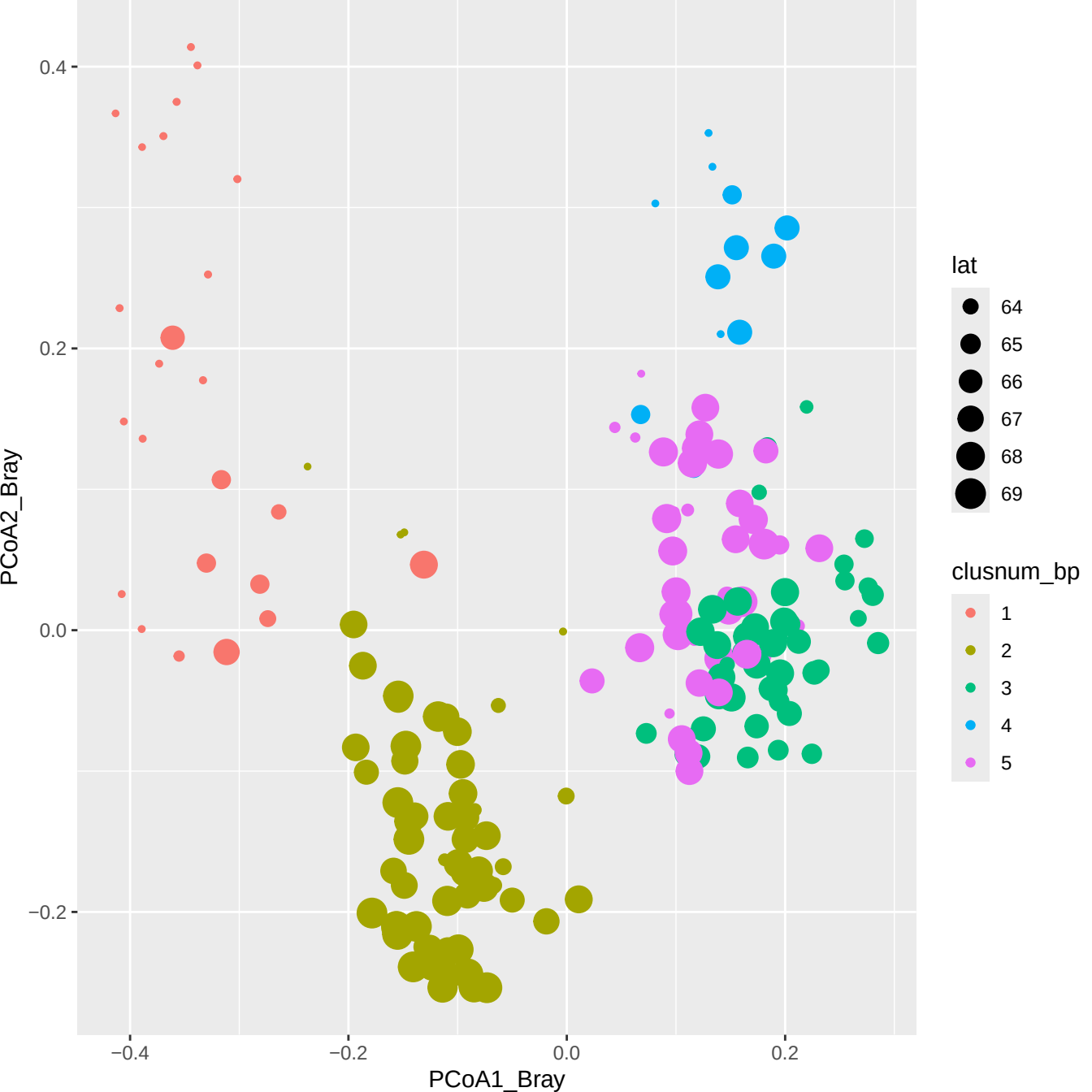

depth_m
0
10
20
30
40
50
1
2
3
4
5
clusnum_bp
clusnum_bp
1
2
3
4
5

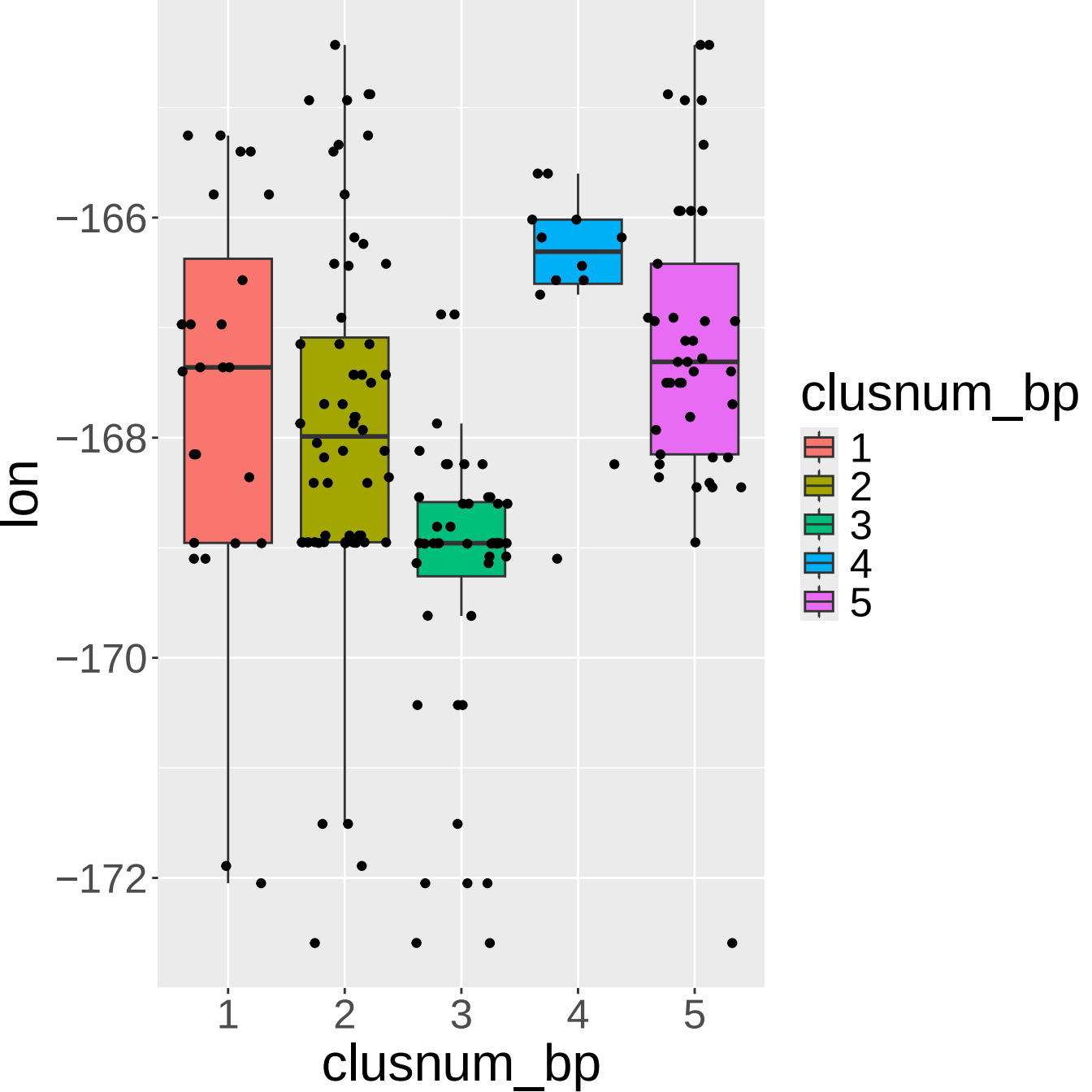

0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
clusnum_bp
1
2
3
4
5
depth_m
10
20
30
40
50

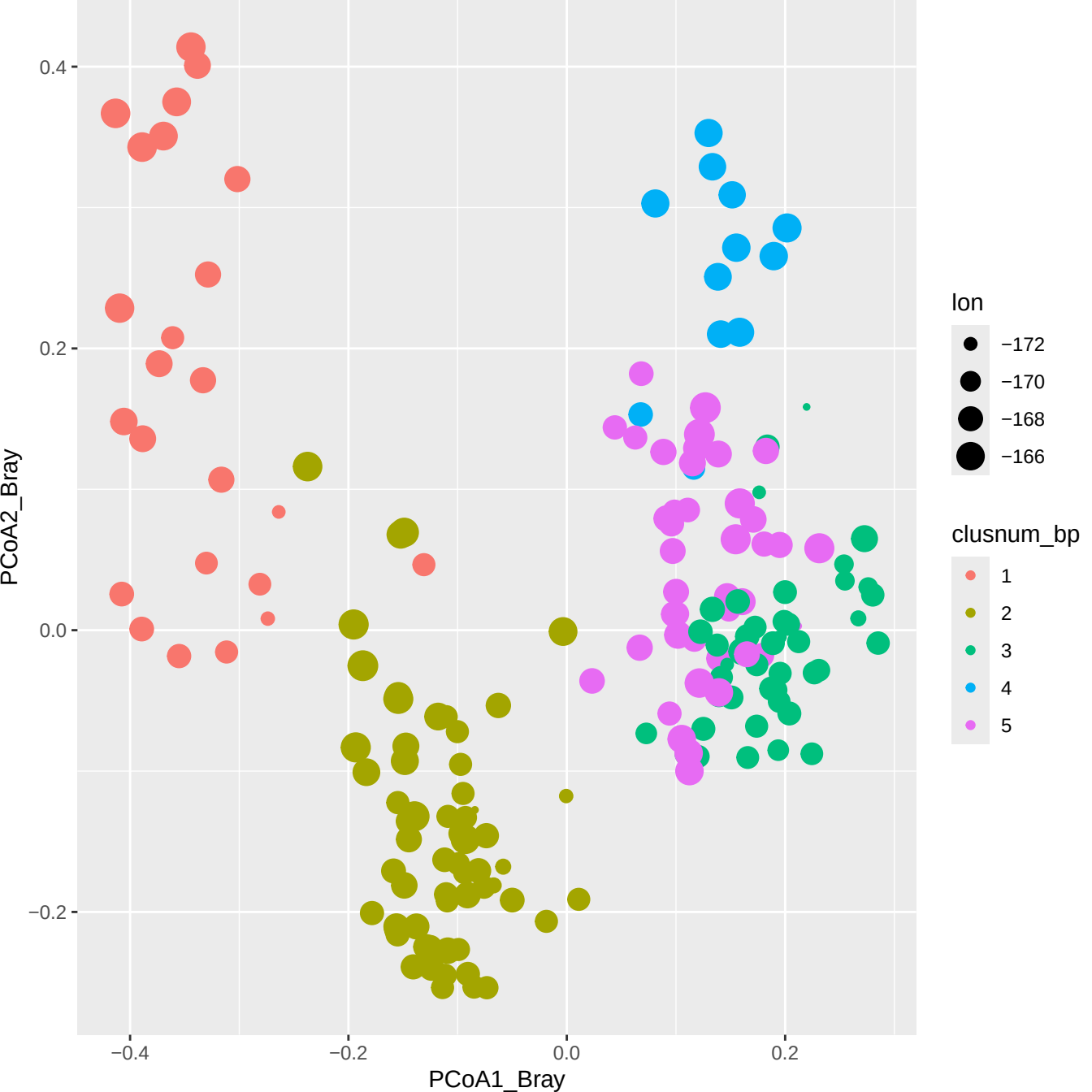

bottom_depth
60
50
40
30
20
1
2
3
4
5
clusnum_bp
clusnum_bp
1
2
3
4
5

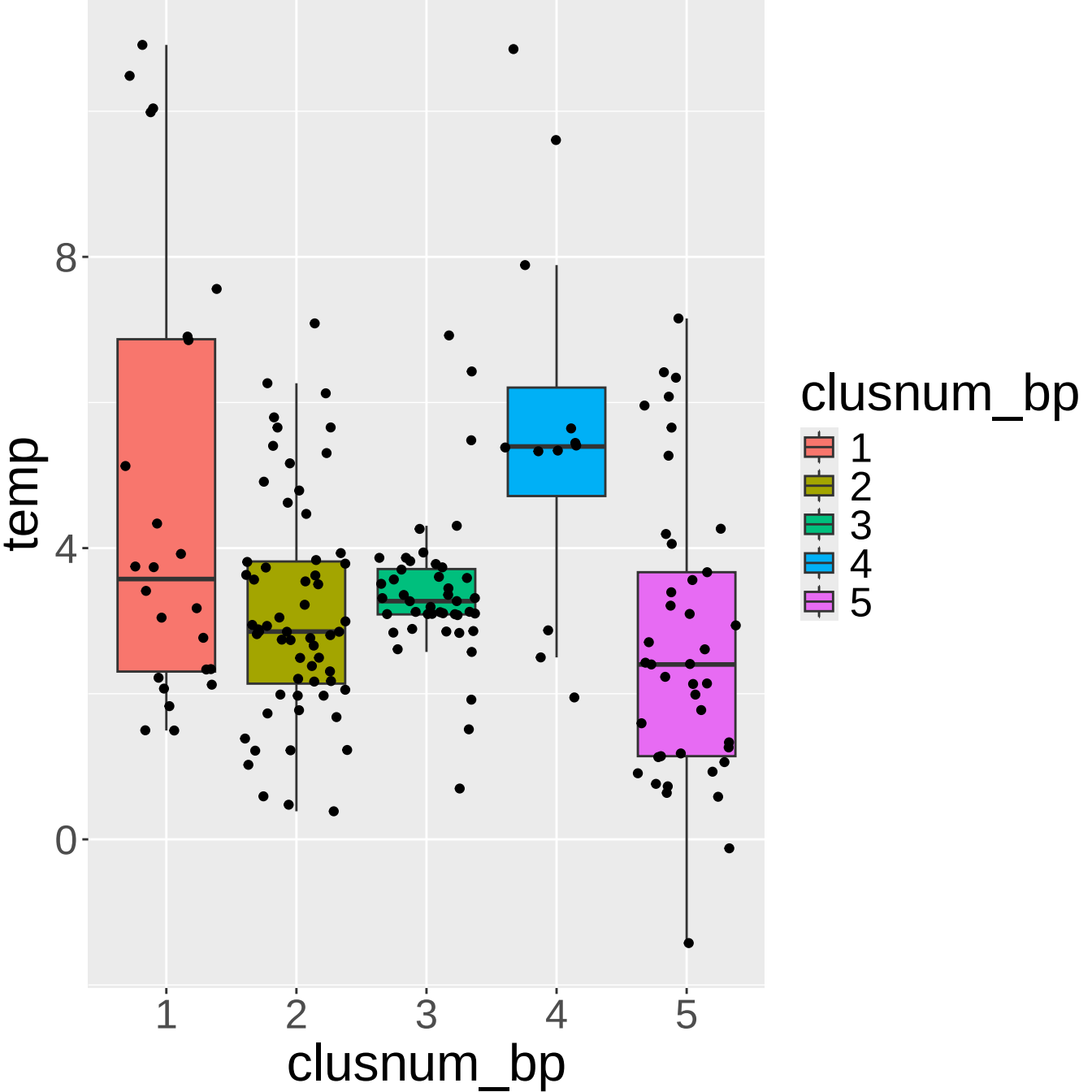

0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
clusnum_bp
1
2
3
4
5
bottom_depth
20
30
40
50

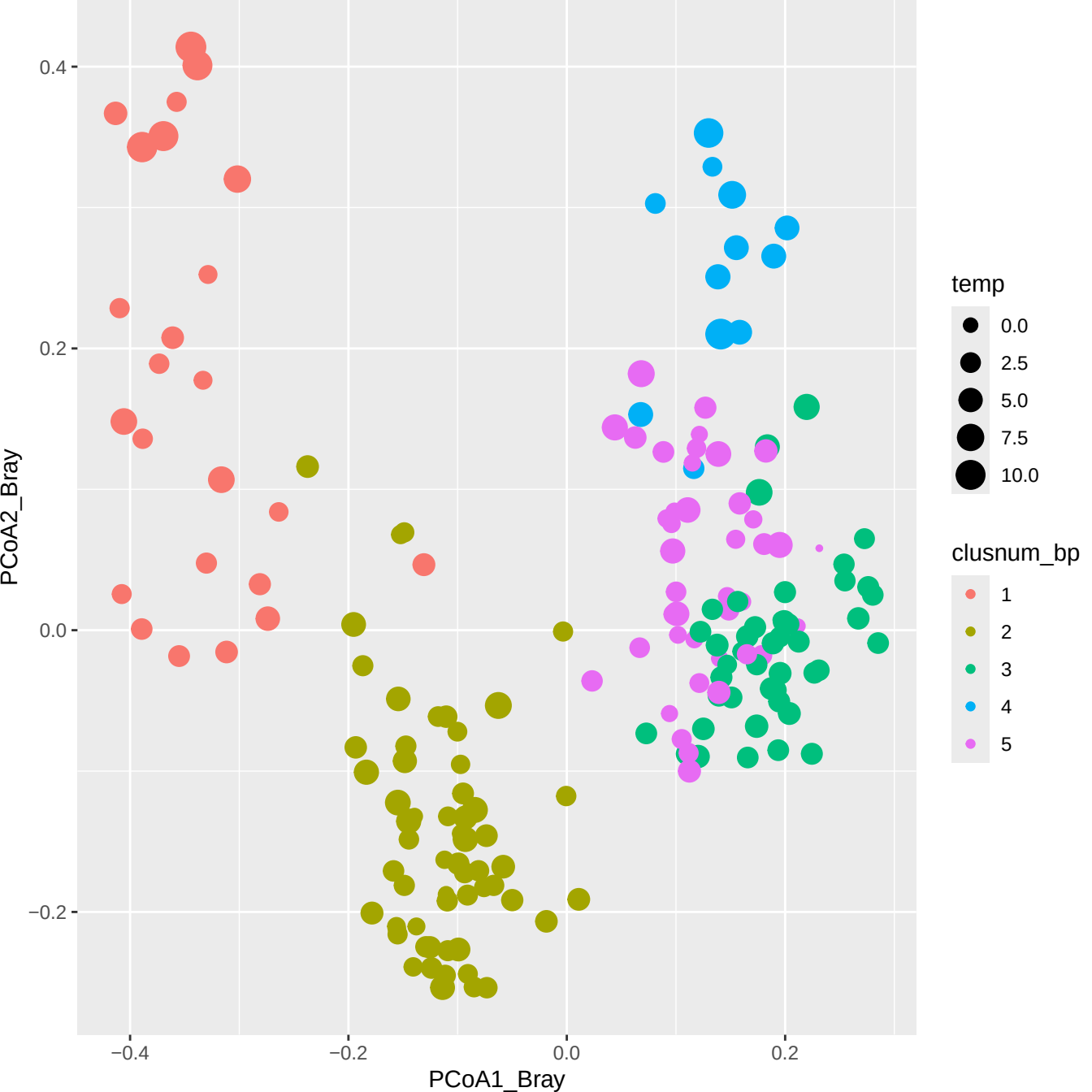

69
68
67
66
65
64
63
lat
1
2
3
4
5
clusnum_bp
clusnum_bp
1
2
3
4
5

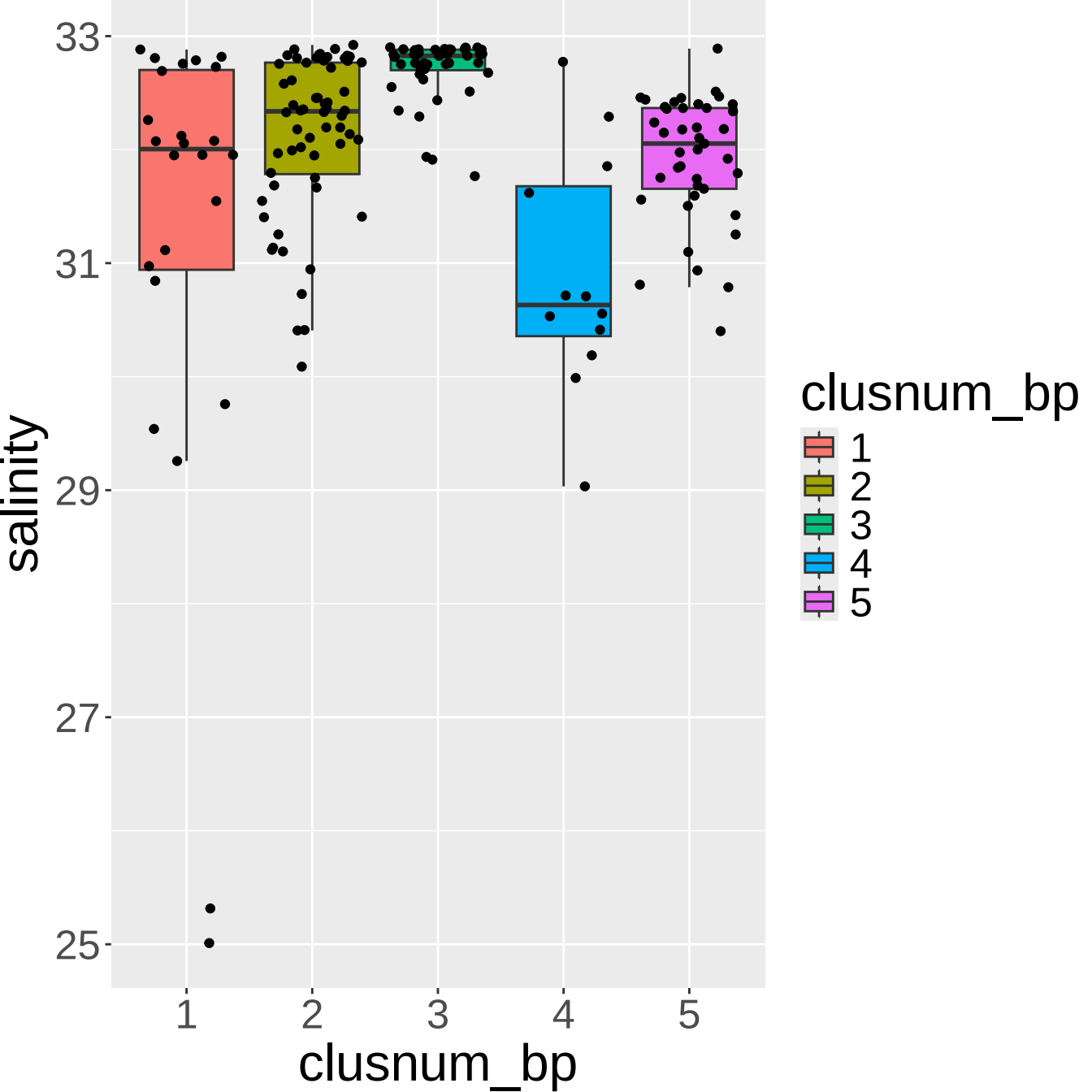

0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
lat
64
65
66
67
68
69
clusnum_bp
1
2
3
4
5

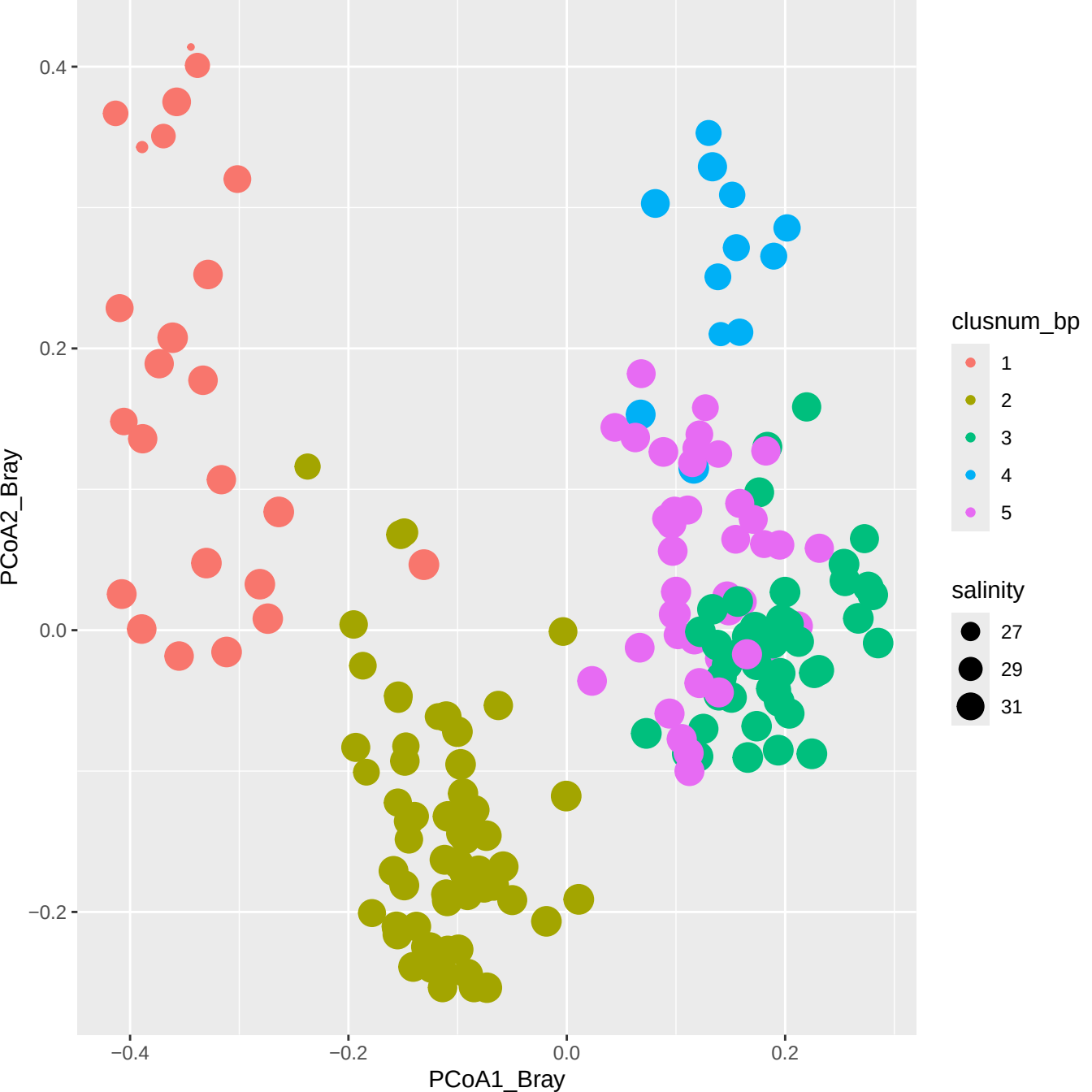

lon
−166
−168
−170
−172
1
2
3
4
5
clusnum_bp
clusnum_bp
1
2
3
4
5

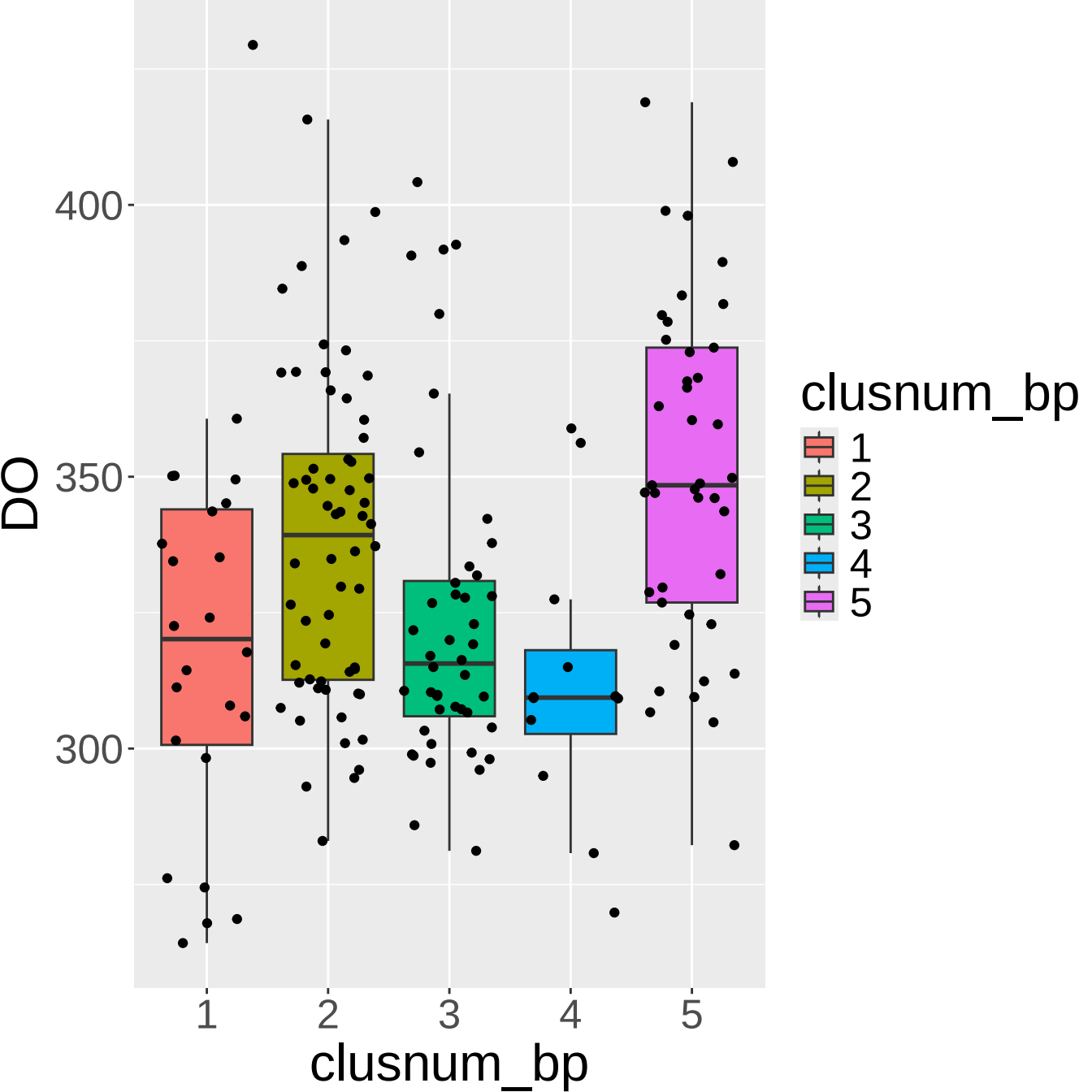

0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
lon
−172
−170
−168
−166
clusnum_bp
1
2
3
4
5

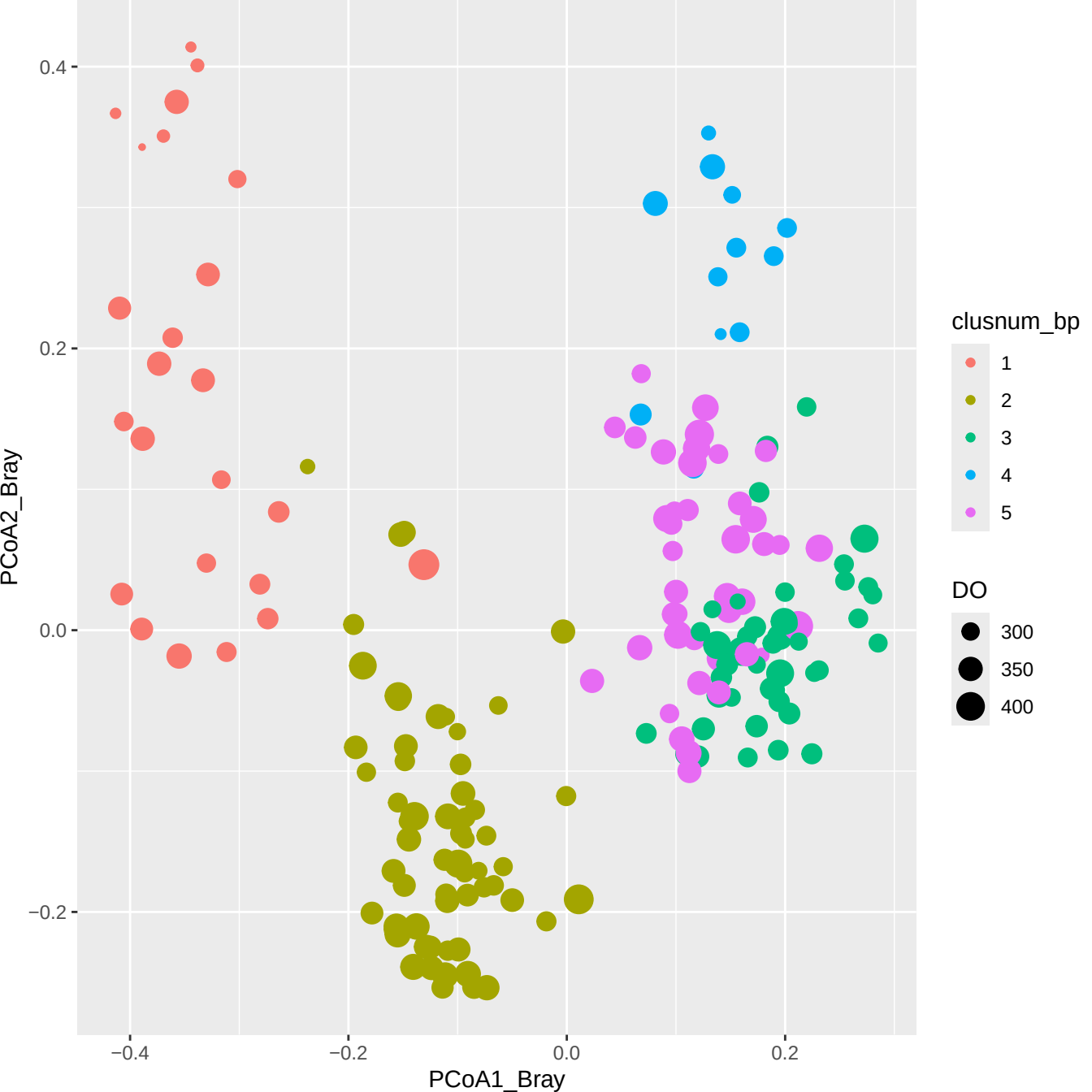

temp
8
4
0
1
2
3
4
5
clusnum_bp
clusnum_bp
1
2
3
4
5

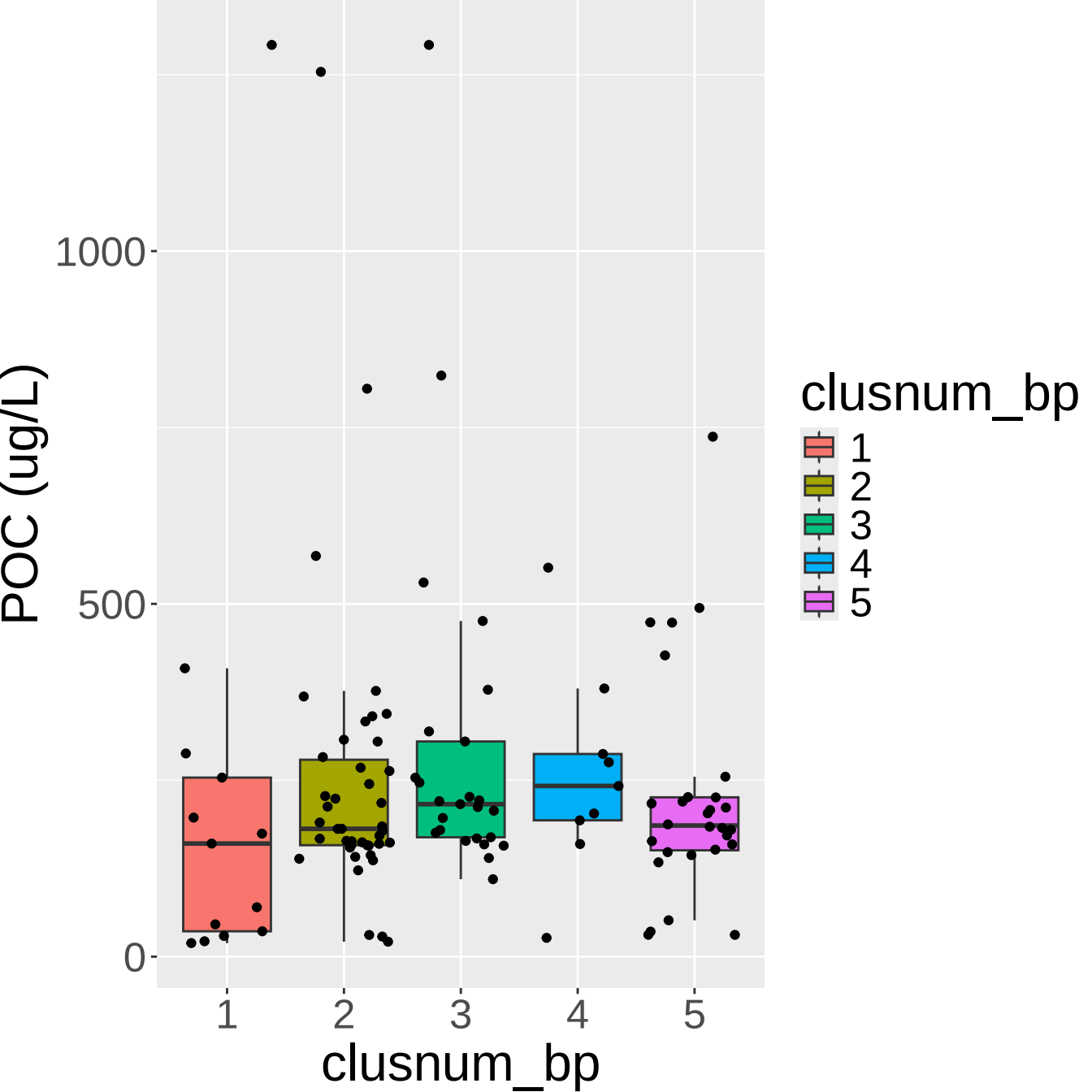

0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
temp
0.0
2.5
5.0
7.5
10.0
clusnum_bp
1
2
3
4
5

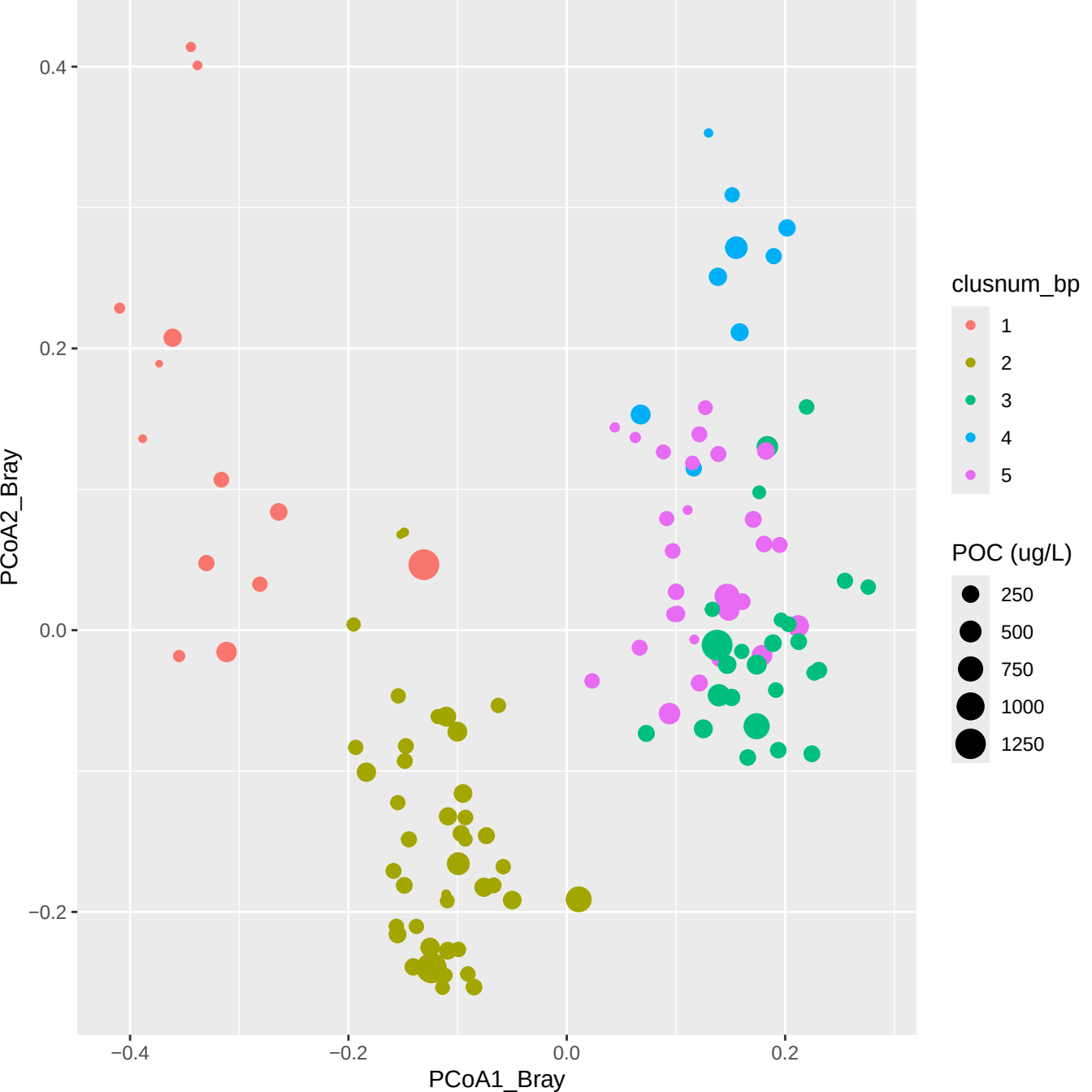

33
31
29
27
25
salinity
1
2
3
4
5
clusnum_bp
clusnum_bp
1
2
3
4
5

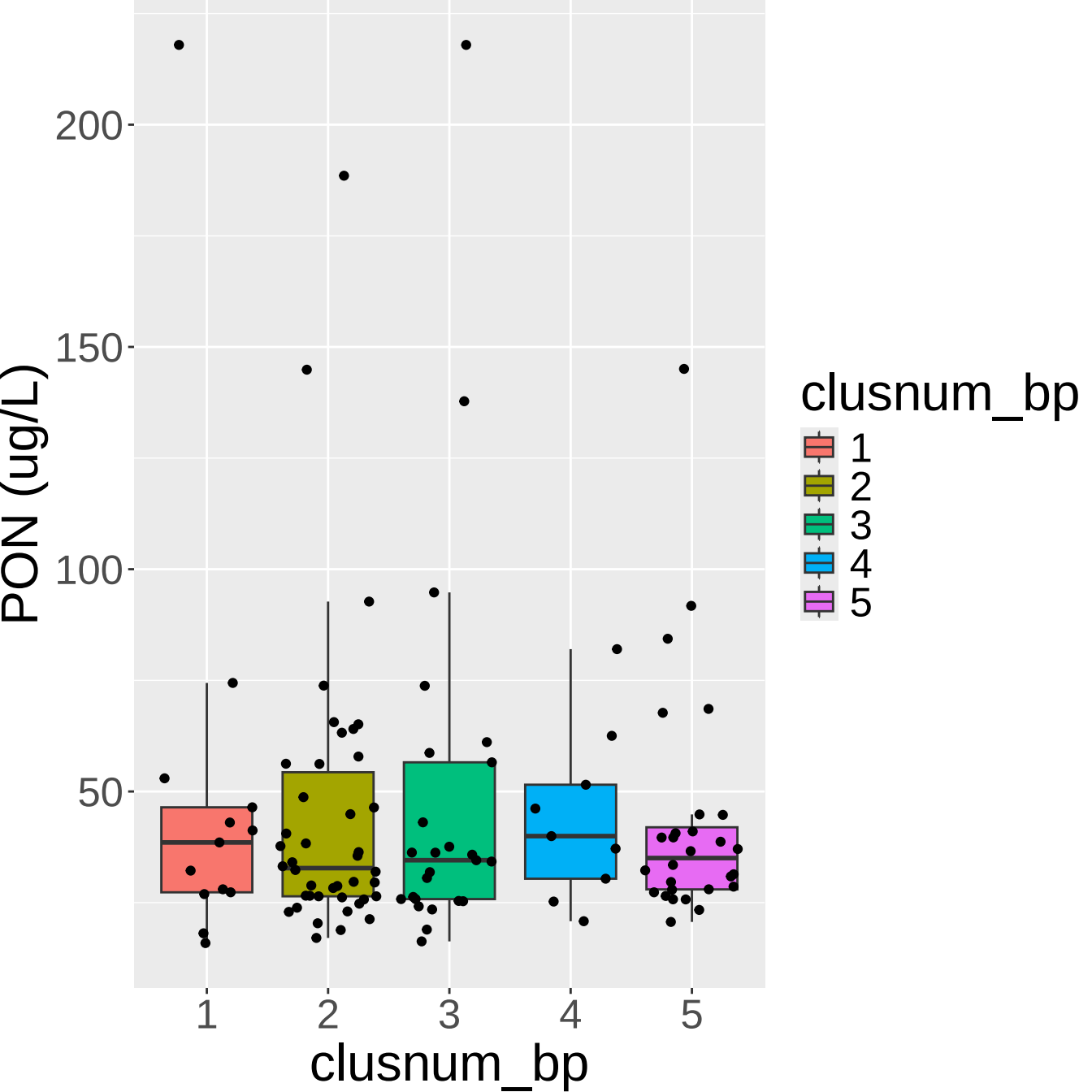

0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
clusnum_bp
1
2
3
4
5
salinity
27
29
31

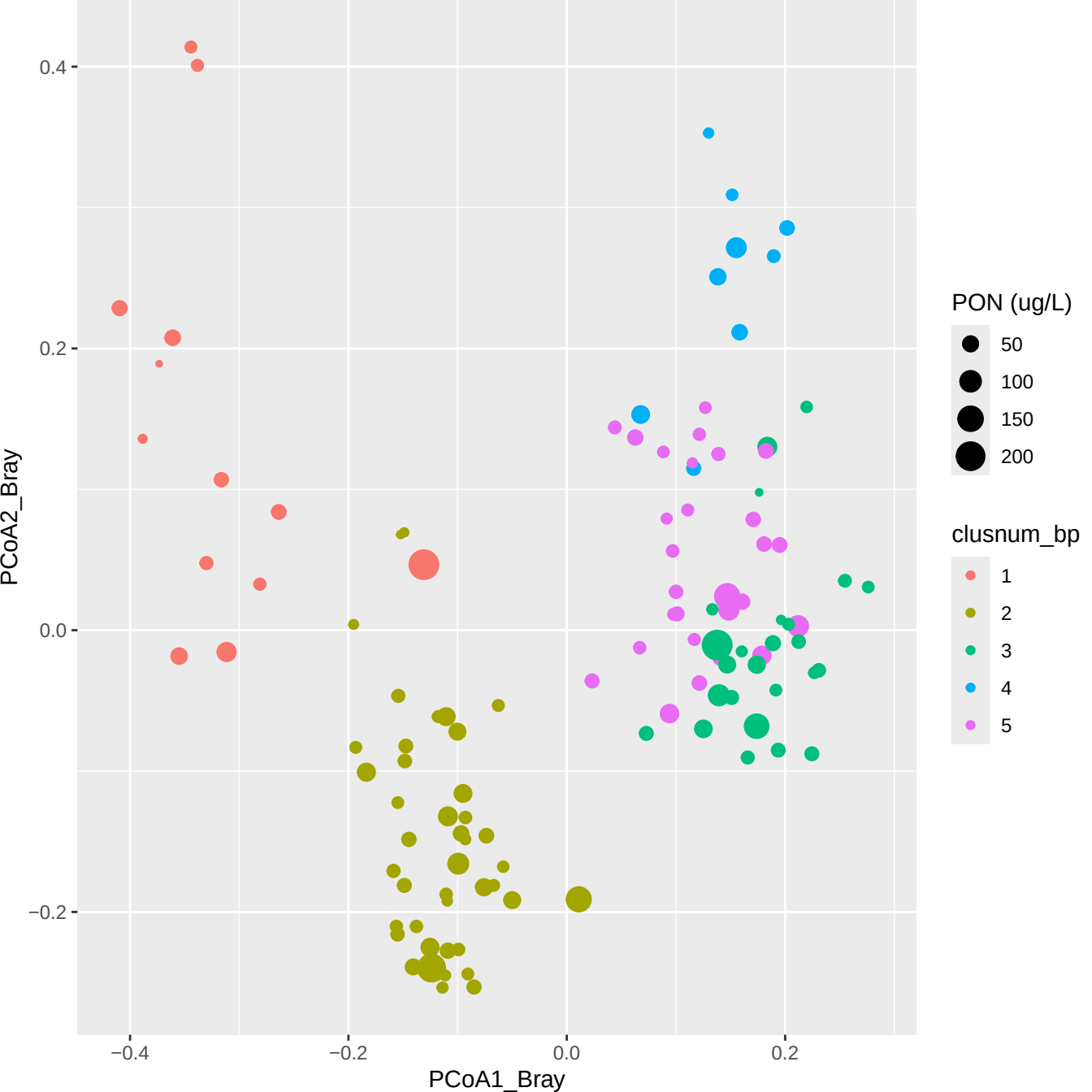

DO
400
350
300
1
2
3
4
5
clusnum_bp
clusnum_bp
1
2
3
4
5

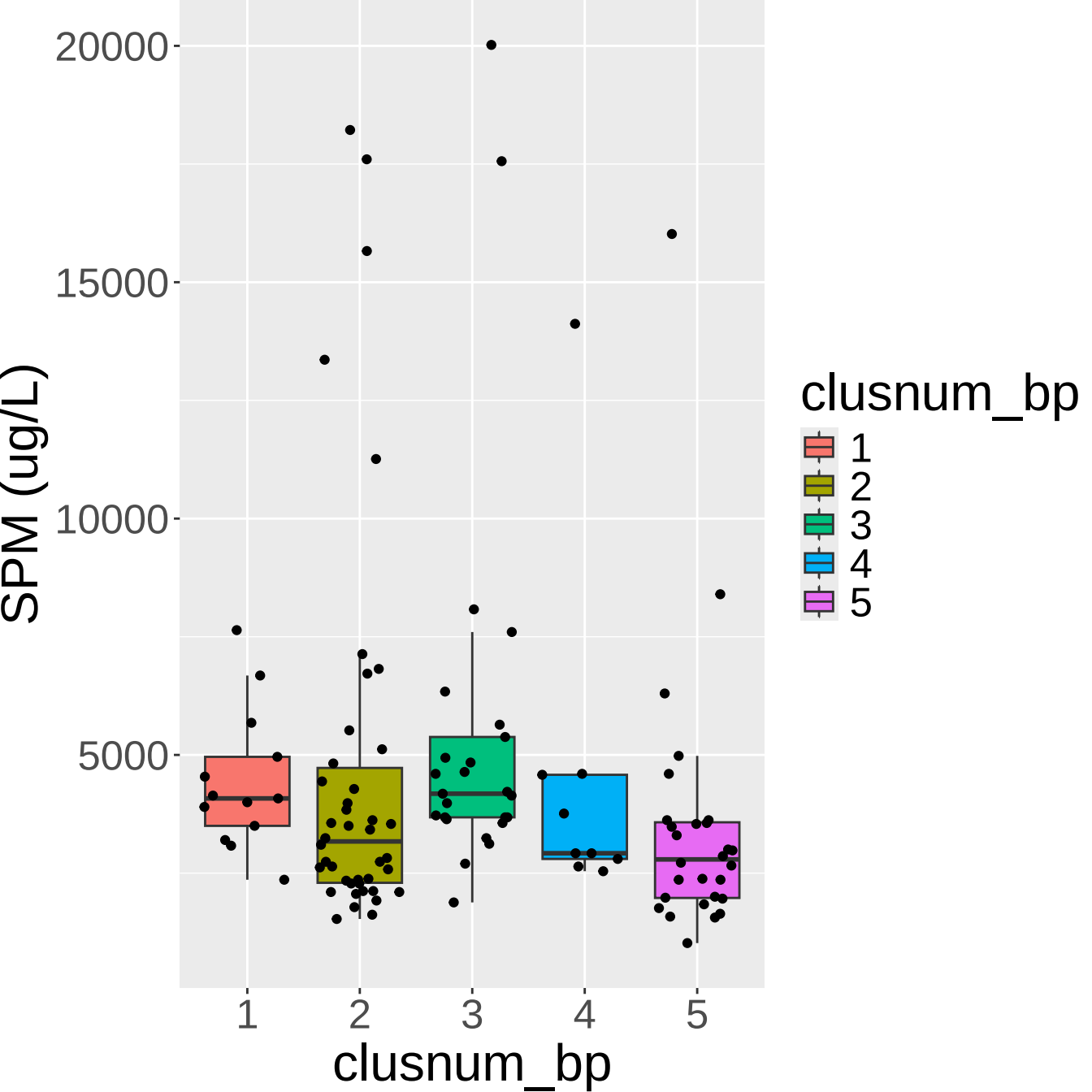

0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
clusnum_bp
1
2
3
4
5
DO
300
350
400

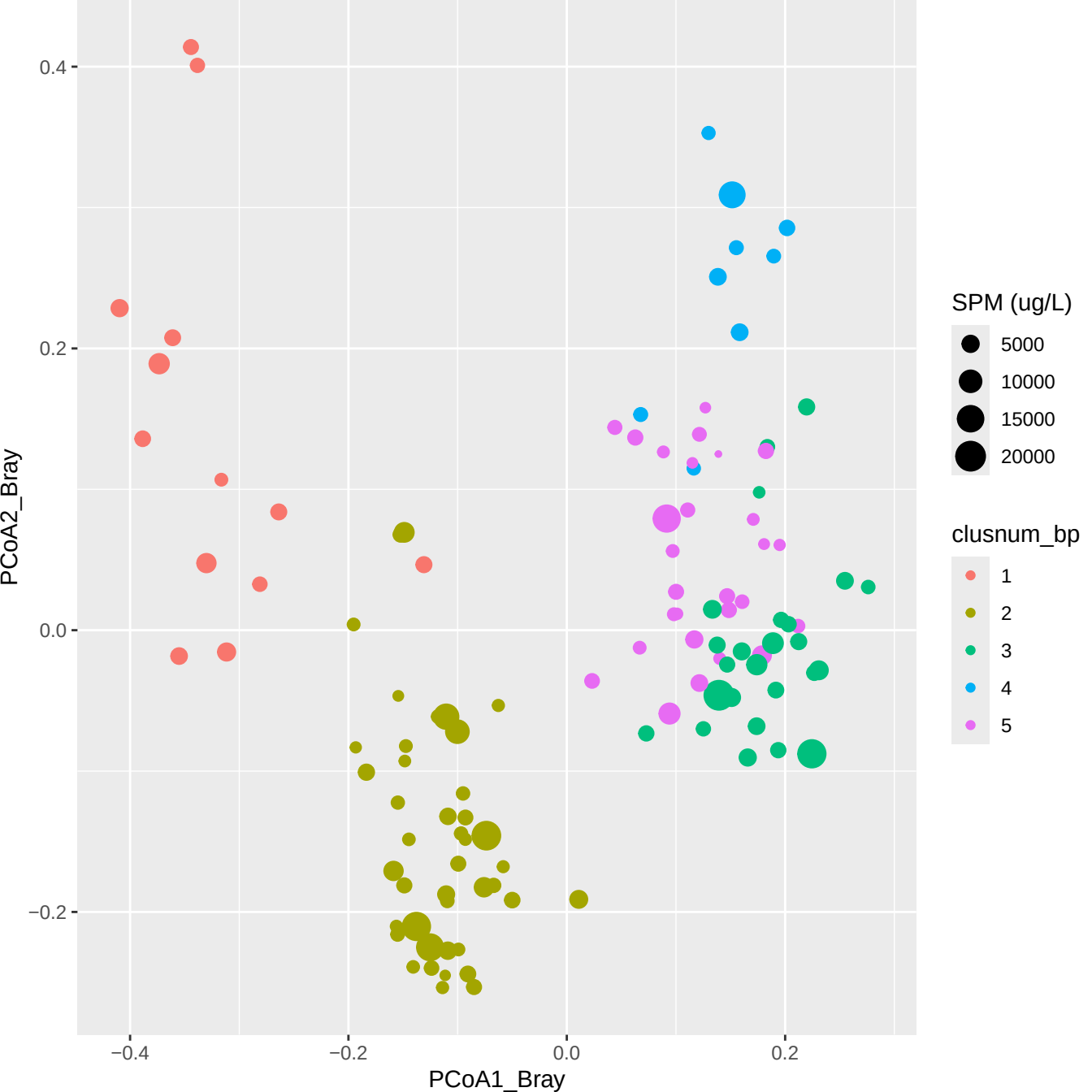

POC (ug/L)
1000
500
0
1
2
3
4
5
clusnum_bp
clusnum_bp
1
2
3
4
5

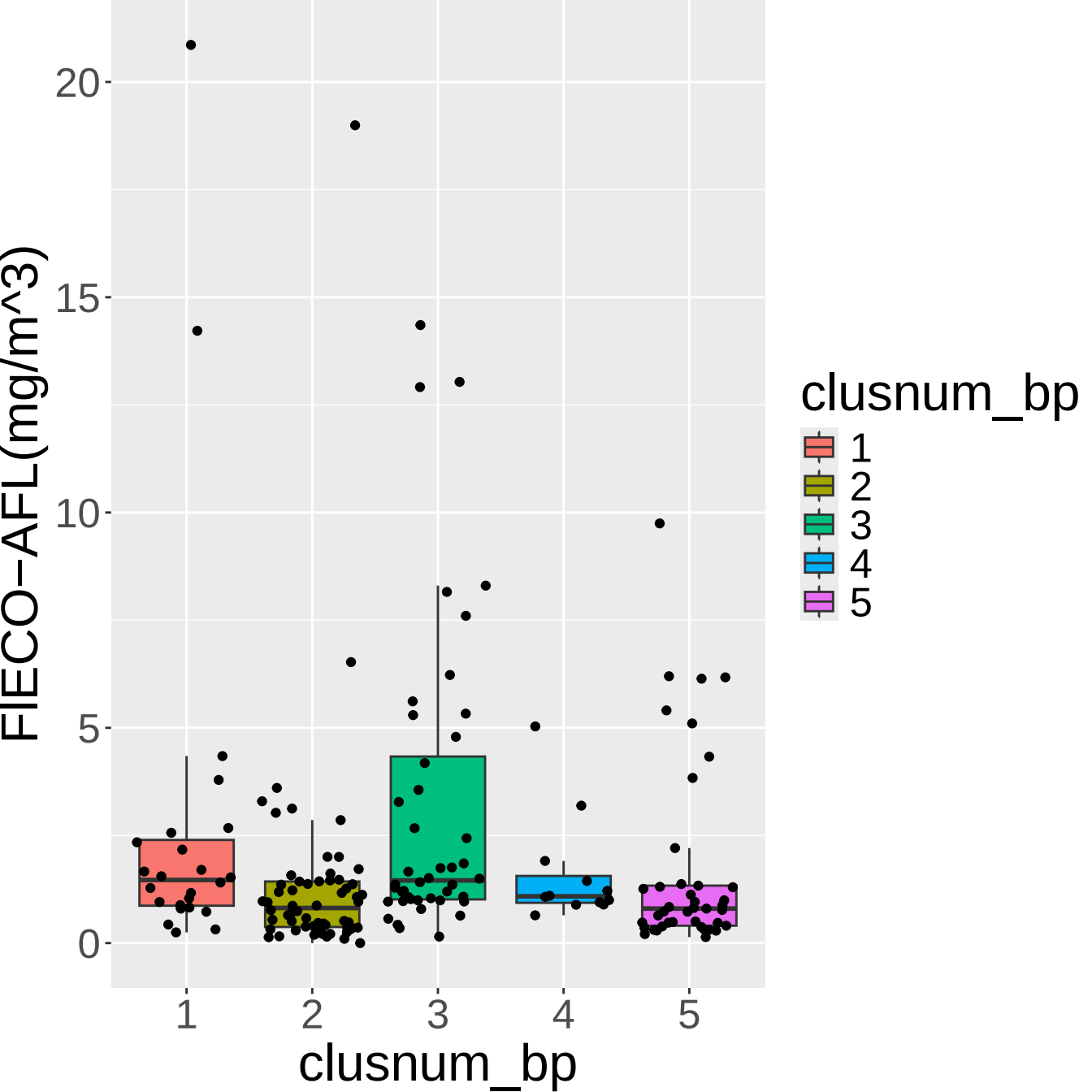

0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
clusnum_bp
1
2
3
4
5
POC (ug/L)
250
500
750
1000
1250

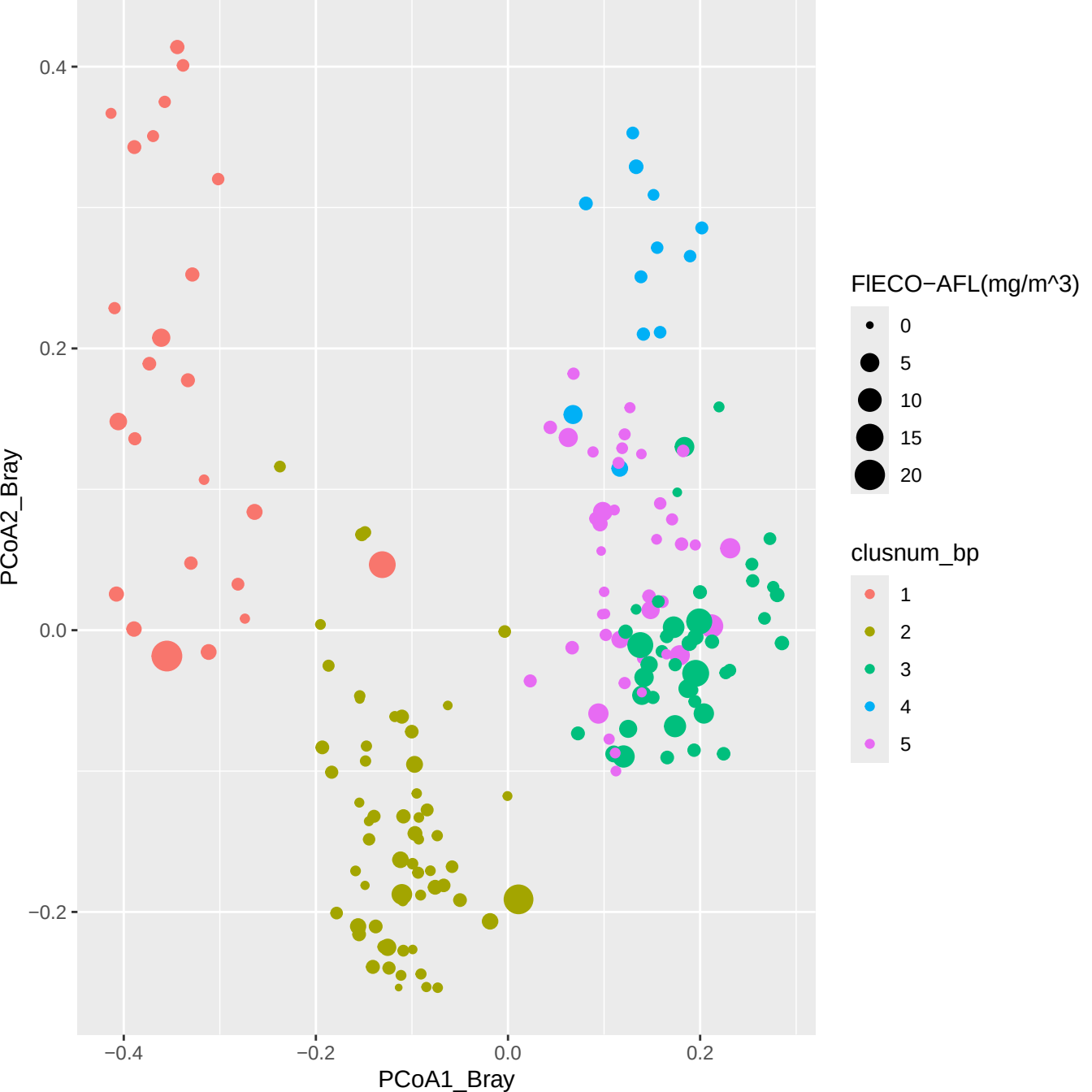

PON (ug/L)
200
150
100
50
1
2
3
4
5
clusnum_bp
clusnum_bp
1
2
3
4
5

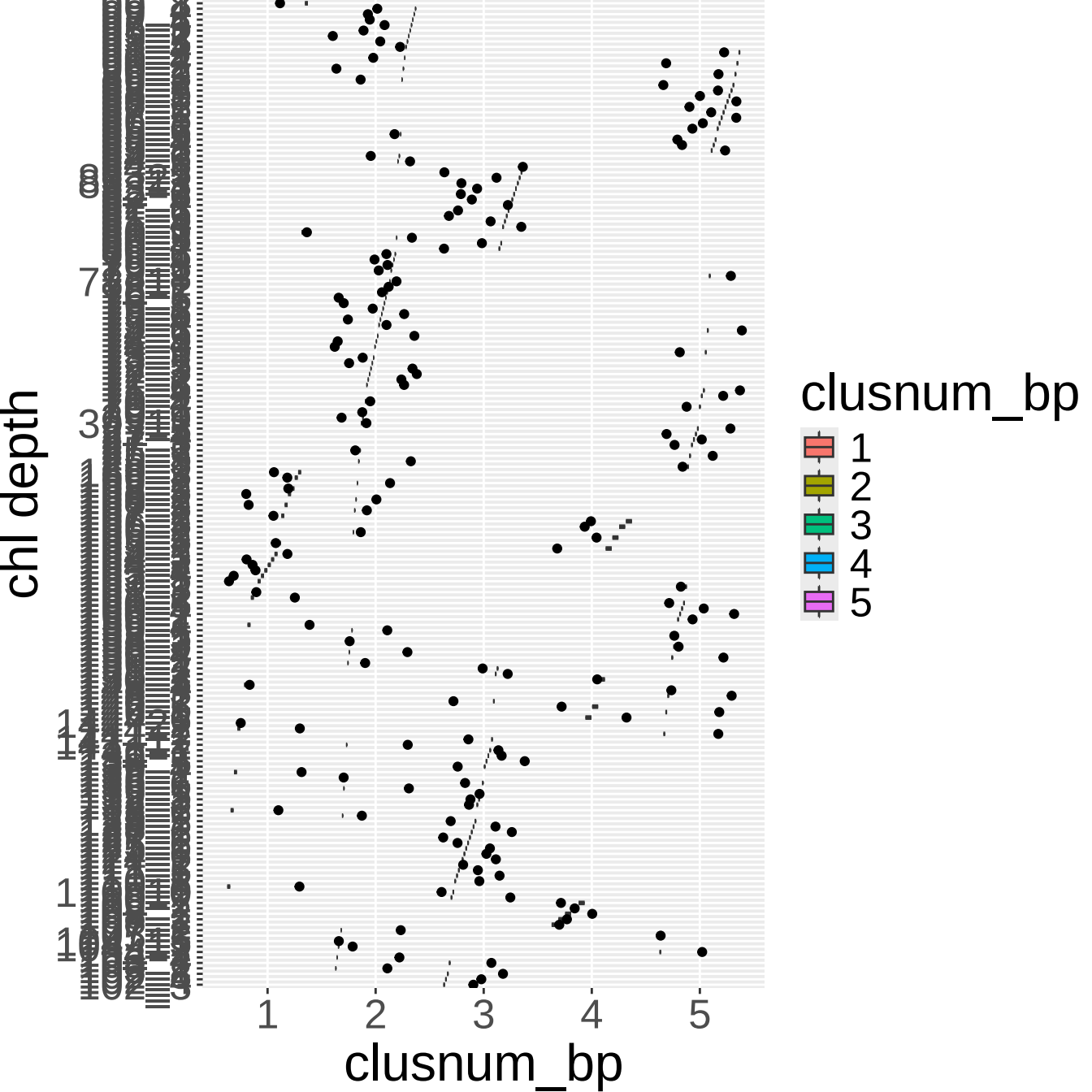

0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
PON (ug/L)
50
100
150
200
clusnum_bp
1
2
3
4
5


20000
15000
10000
5000
SPM (ug/L)
1
2
3
4
5
clusnum_bp
clusnum_bp
1
2
3
4
5


0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
SPM (ug/L)
5000
10000
15000
20000
clusnum_bp
1
2
3
4
5


FIECO−AFL(mg/m^3)
20
15
10
5
0
1
2
3
4
5
clusnum_bp
clusnum_bp
1
2
3
4
5


0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
FIECO−AFL(mg/m^3)
0
5
10
15
20
clusnum_bp
1
2
3
4
5


chl depth
clusnum_bp
1
2
3
4
5
clusnum_bp
1
2
3
4
5

chl depth

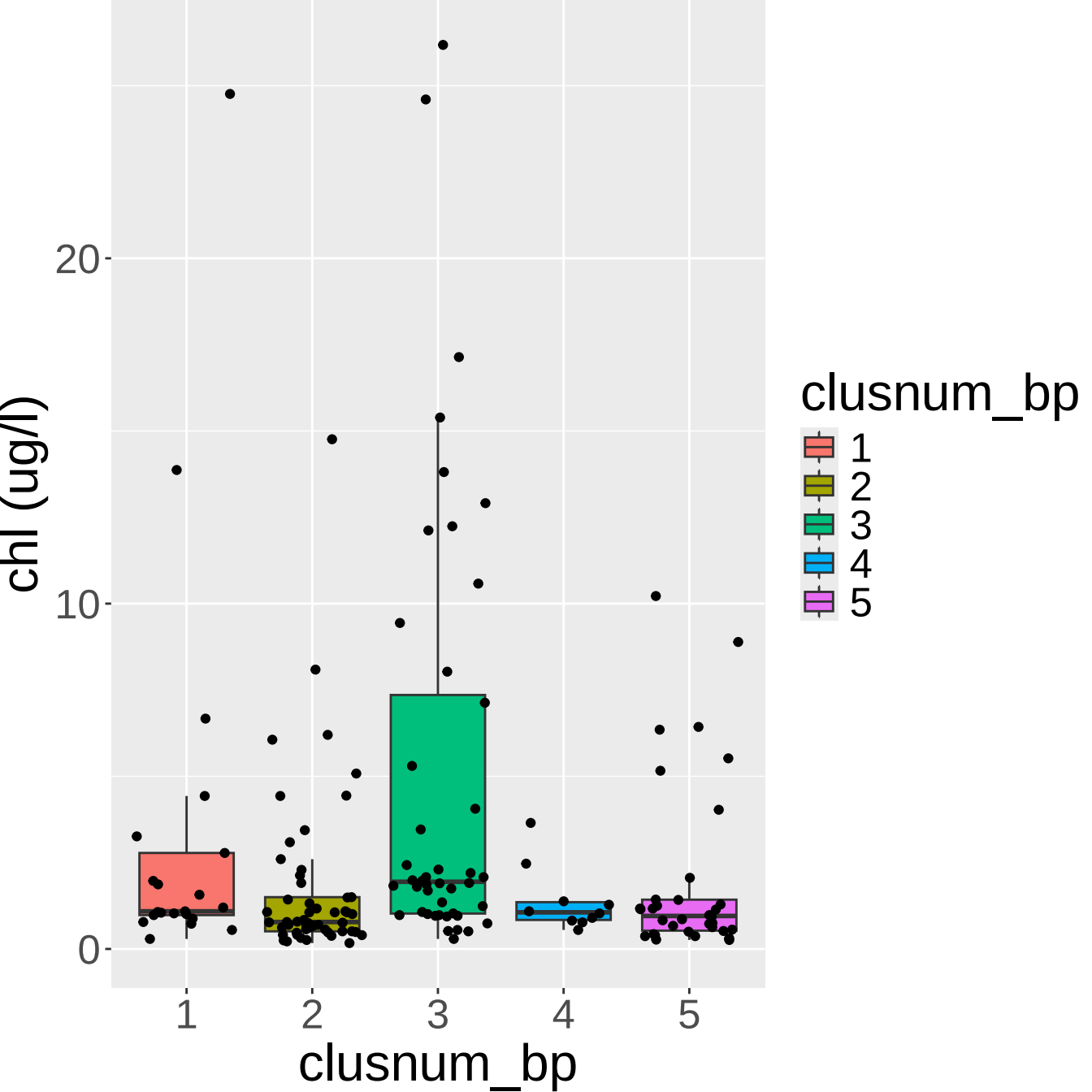
102_3	110_8	136_7	151_1	163_4	25_1	72_5	79_4	84_6	
102_4	111_1	139_1	151_2	164_1	25_3	73_1	79_6	85_1	
102_8	111_5	139_4	156_1	164_2	25_5	73_9	80_1	85_3	
103_1	111_7	139_5	156_2	164_3	25_7	74_1	80_4	85_4	
103_2	124_1	139_7	156_4	165_1	27_1	74_4	80_5	85_5	
103_4	124_6	141_1	158_1	165_2	27_3	74_5	80_9	85_6	
104_1	125_1	141_11	158_3	165_3	27_4	74_6	81_1	86_1	
104_11	125_6	141_12	158_5	166_1	27_5	75_1	81_4	86_5	
104_15	127_1	141_7	159_1	166_2	39_11	75_4	81_5	87_1	
105_3	127_6	144_1	159_4	166_3	39_9	75_6	81_6	87_5	
107_1	128_1	144_22	159_7	167_1	70_1	75_8	82_1	88_1	
107_2	128_5	144_6	160_1	167_2	70_2	76_1	82_4	88_3	
108_1	13_1	146_1	160_4	167_3	70_4	76_5	82_6	88_5	
108_2	13_6	146_5	160_5	168_1	71_1	76_7	83_1	89_1	
109_1	132_1	146_7	161_1	168_2	71_4	78_1	83_13	89_2	
109_2	132_3	148_1	161_2	168_3	71_6	78_11	83_22	89_3	
110_1	132_5	148_3	161_3	169_1	72_1	78_8	83_5	90_1	
110_10	136_1	148_4	163_1	169_2	72_2	79_1	84_1	90_2	
110_5	136_6	150_3	163_3	169_3	72_3	79_3	84_3	90_4	

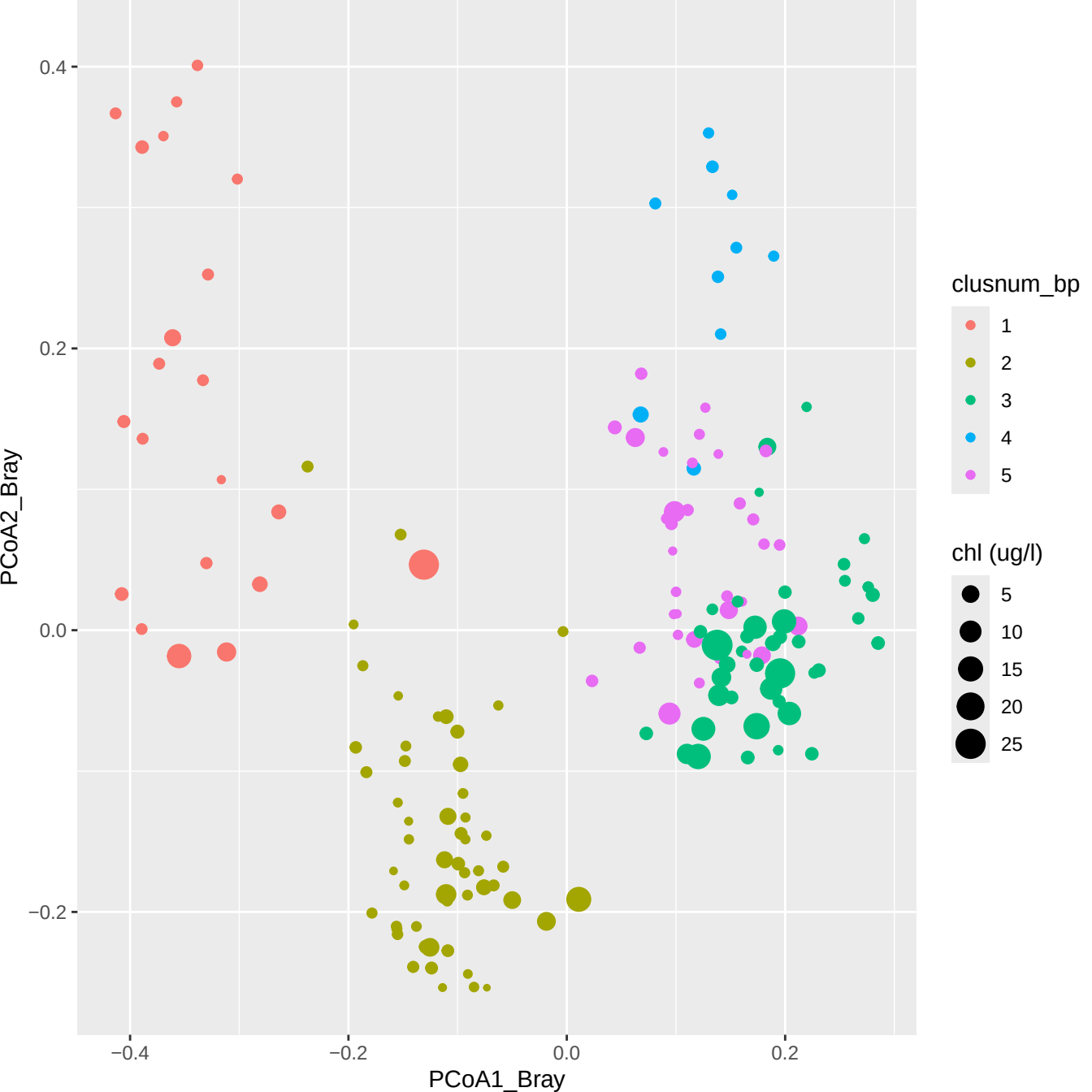
clusnum_bp

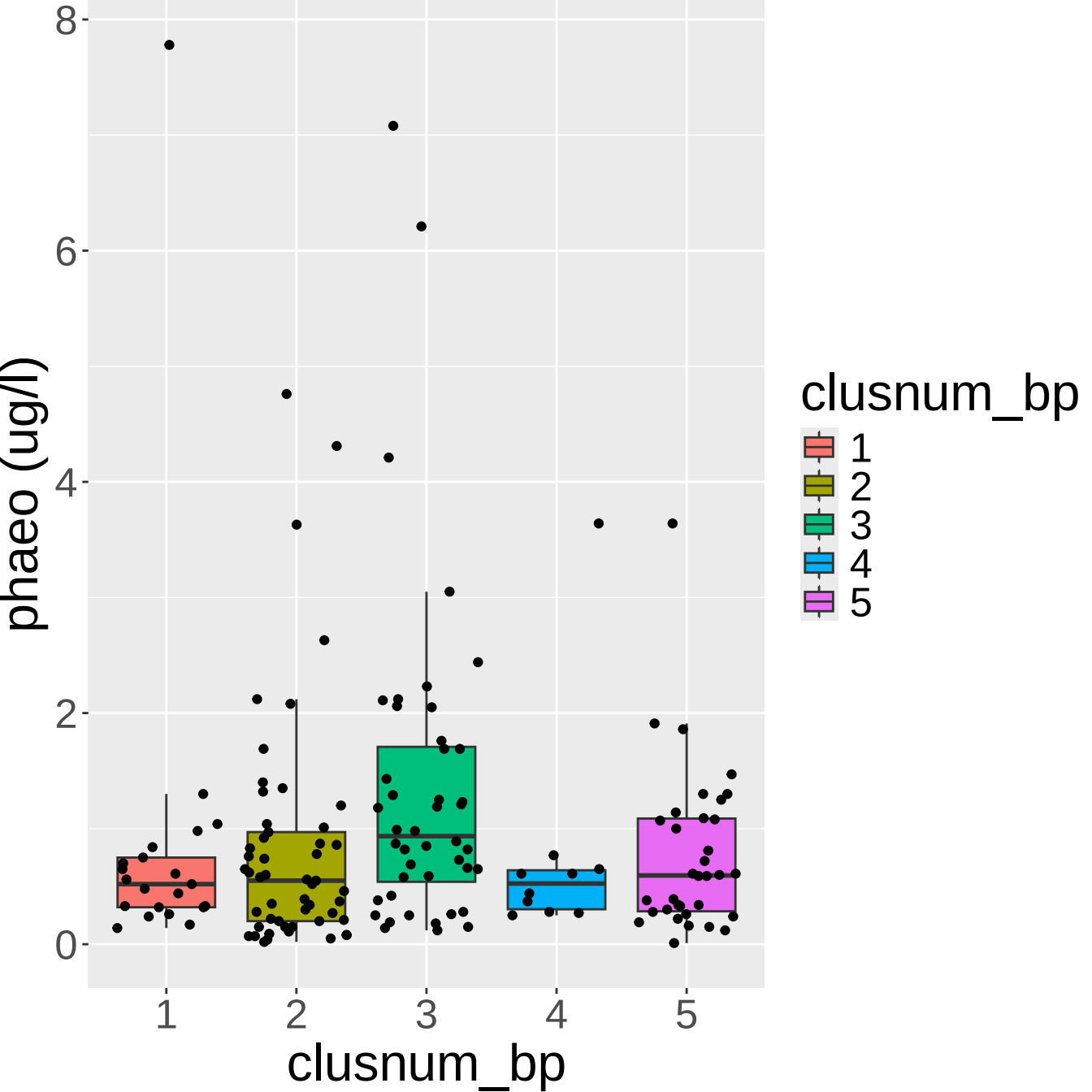
- 1
- 2
- 3
- 4
- 5

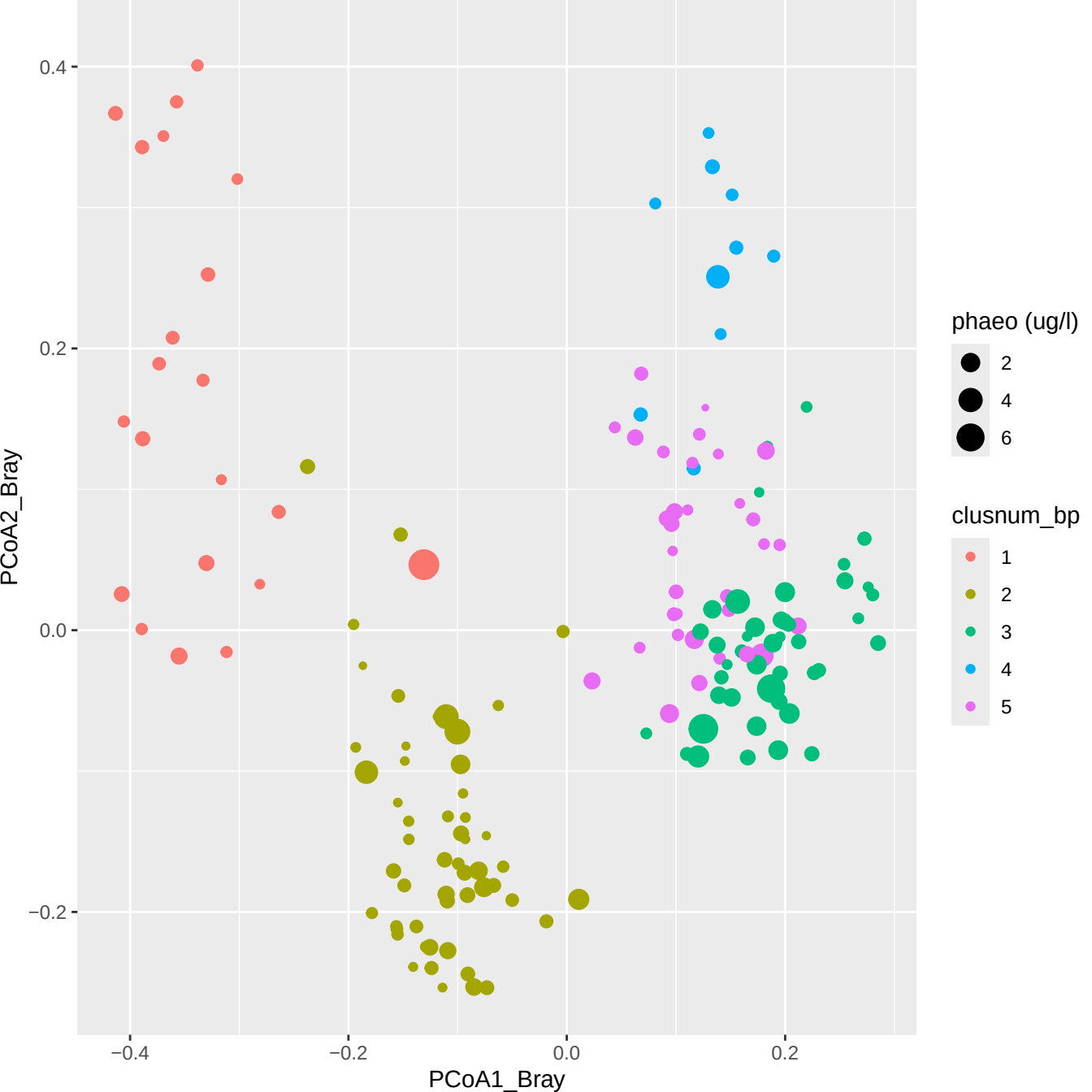
2

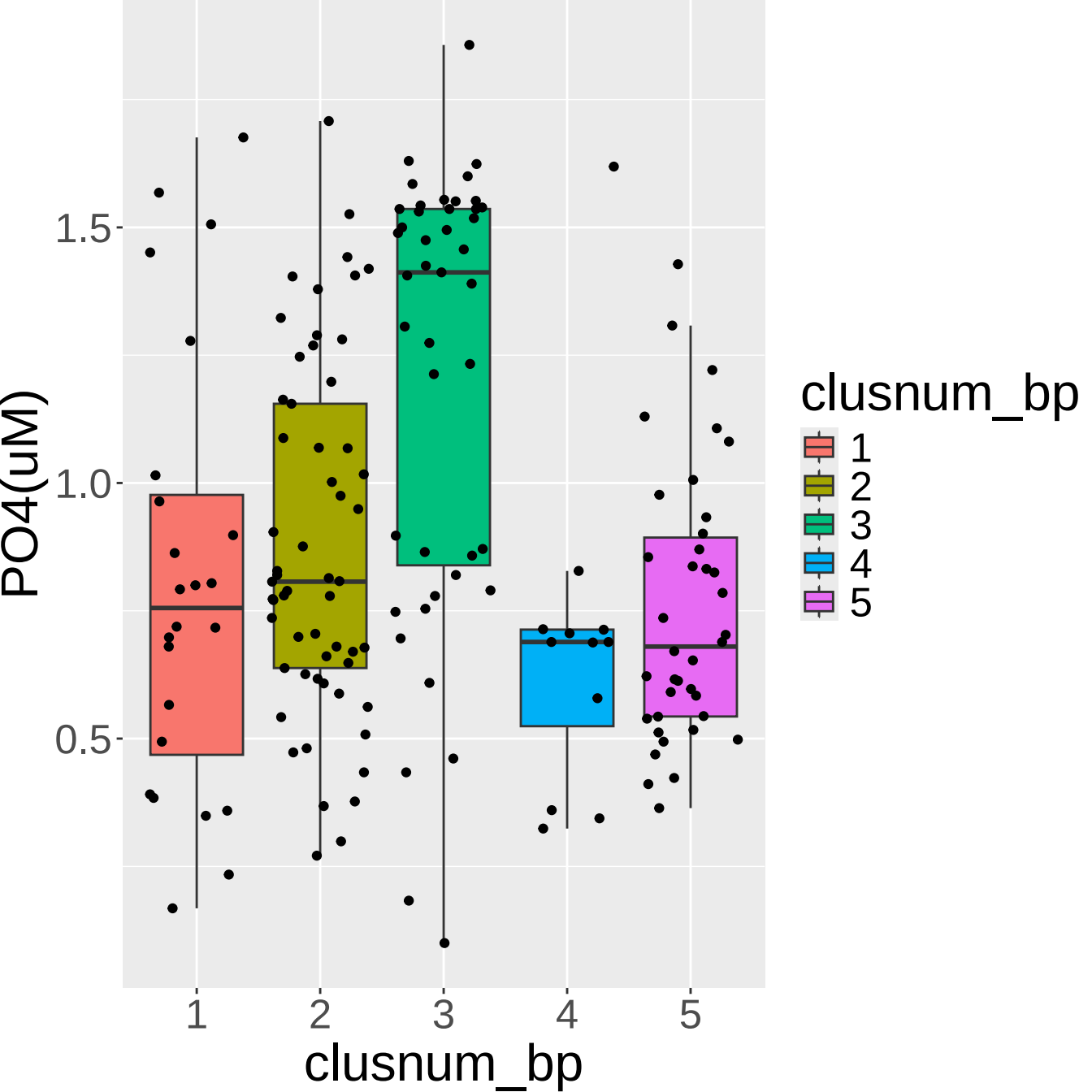
_Bray

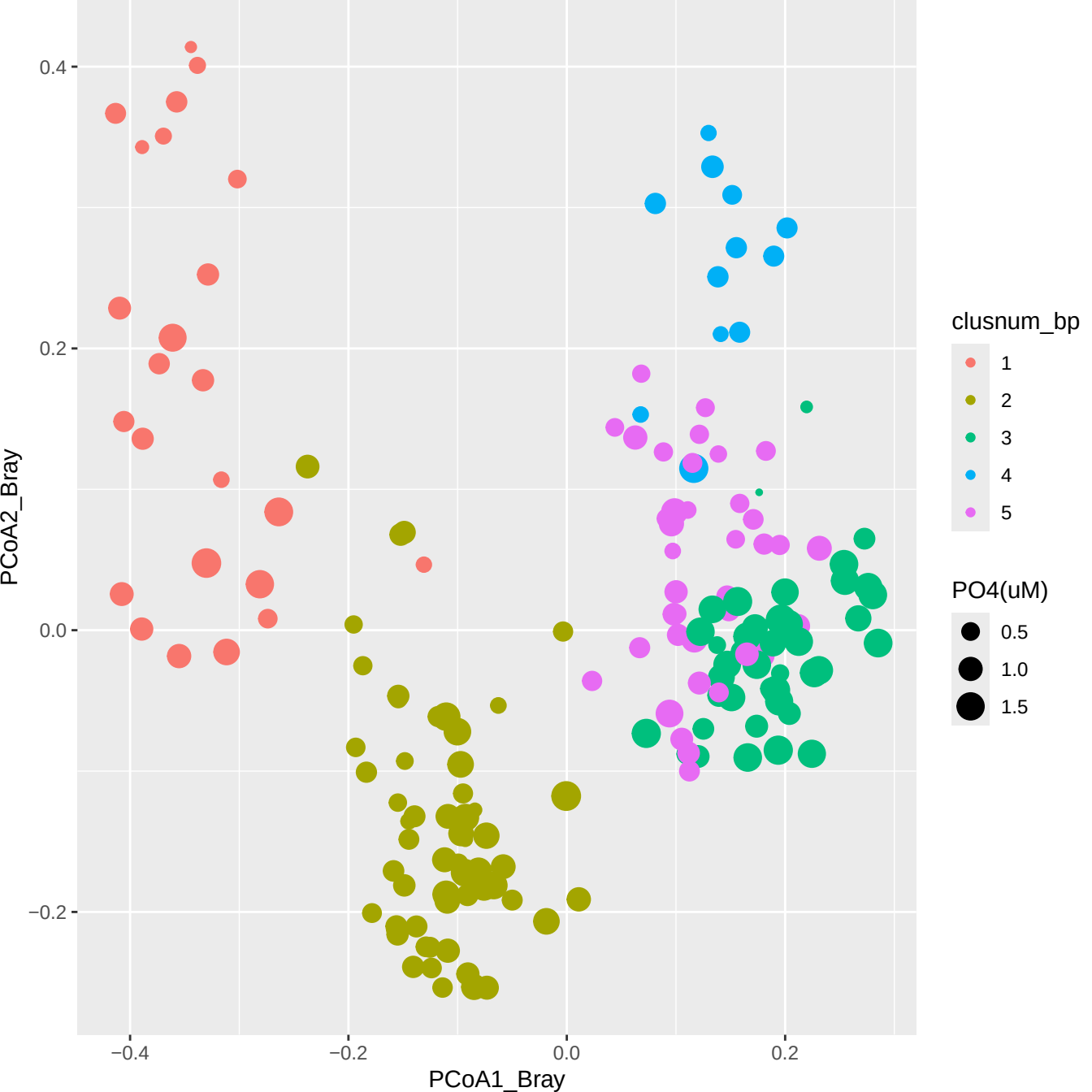

chl (ug/l)
20
10
0
1
2
3
4
5
clusnum_bp
clusnum_bp
1
2
3
4
5

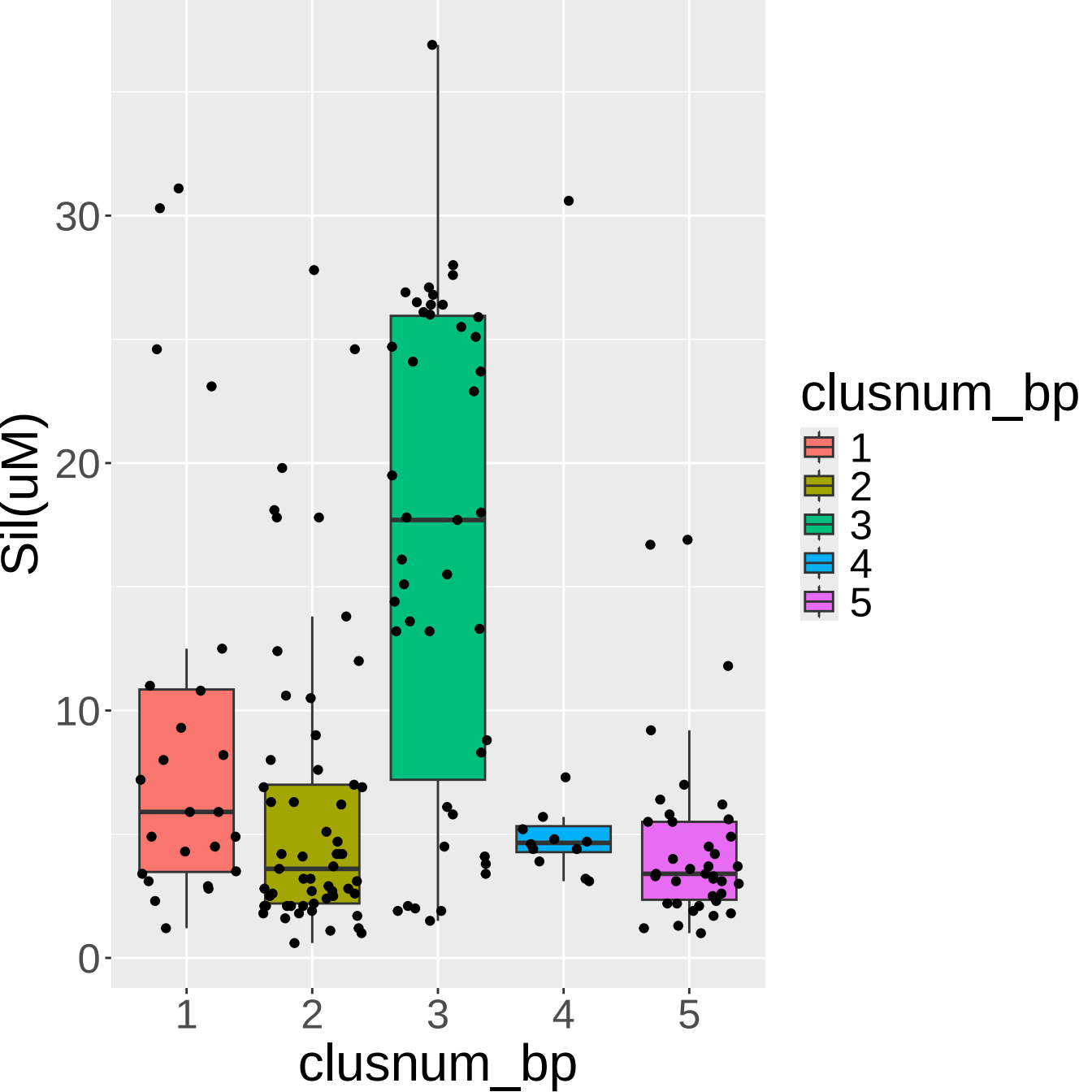

0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
clusnum_bp
1
2
3
4
5
chl (ug/l)
5
10
15
20
25

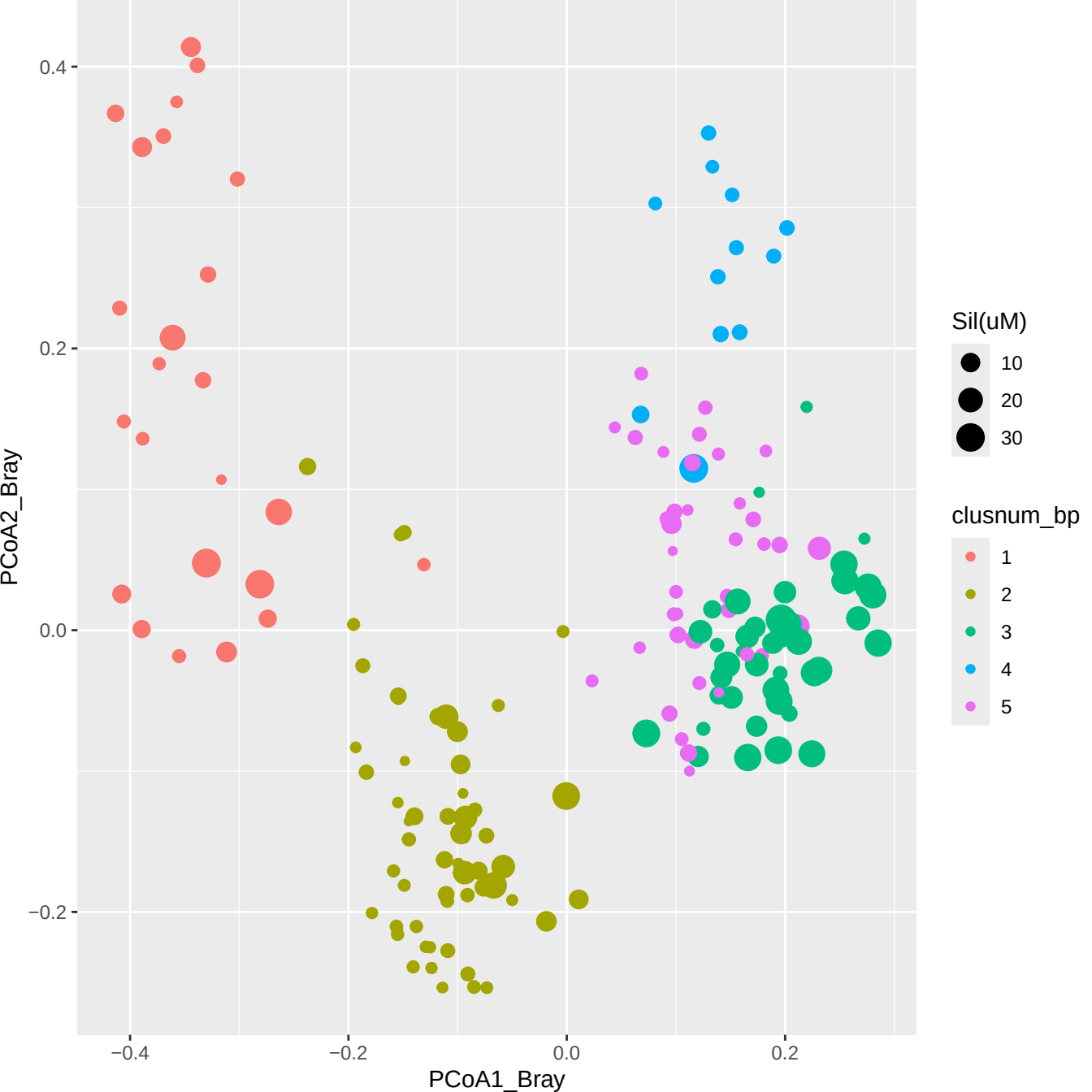

phaeo (ug/l)
8
6
4
2
0
1
2
3
4
5
clusnum_bp
clusnum_bp
1
2
3
4
5

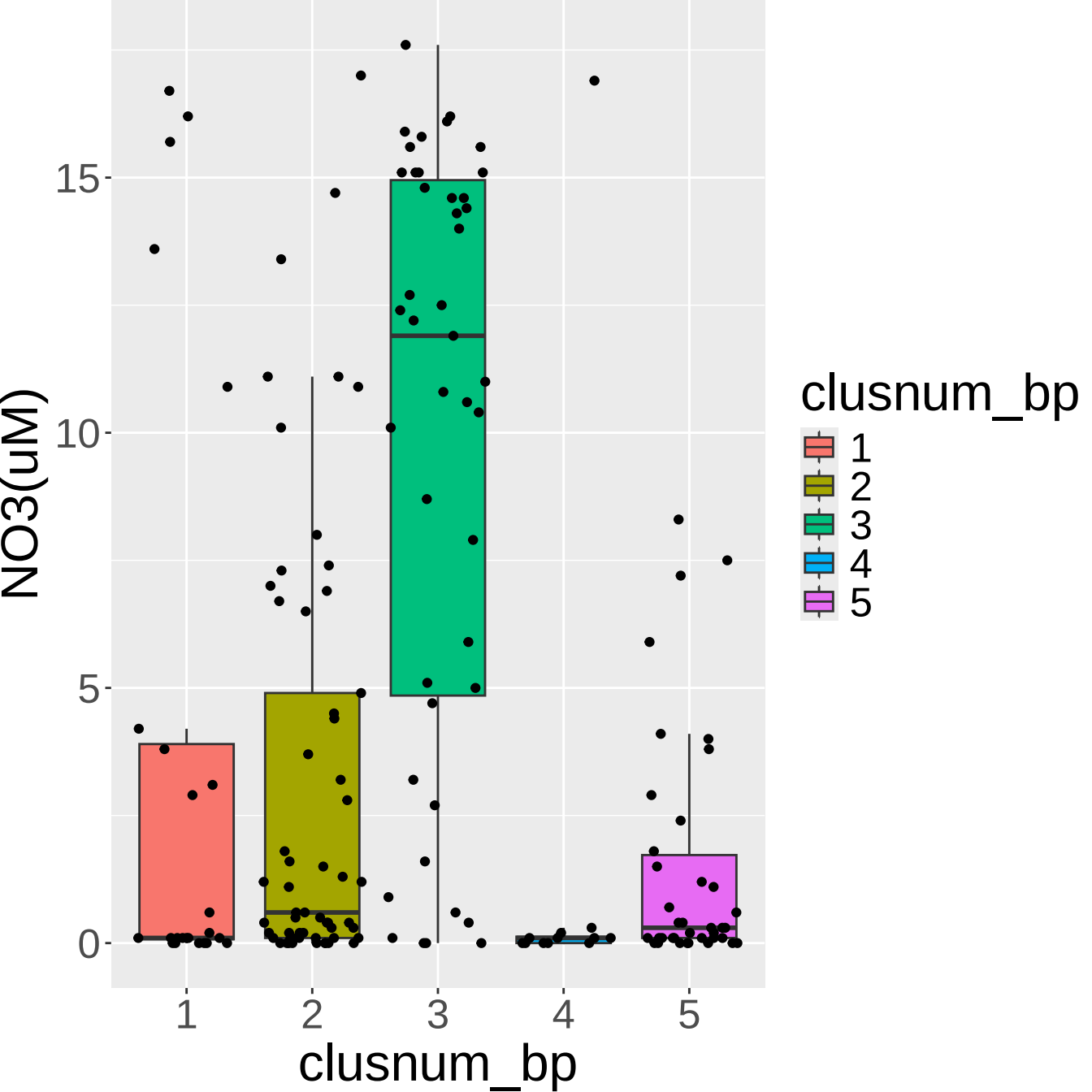

0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
phaeo (ug/l)
2
4
6
clusnum_bp
1
2
3
4
5

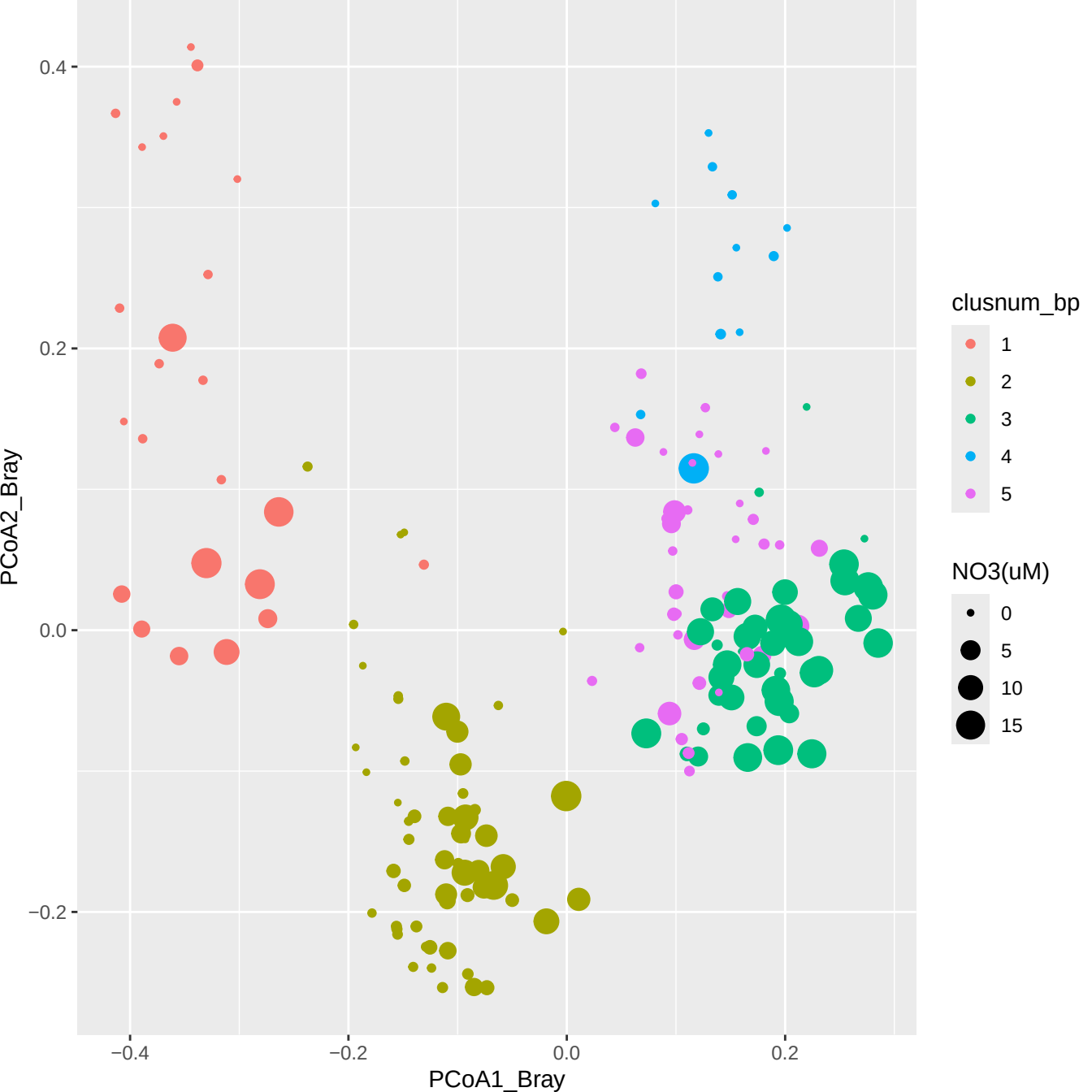

PO4(uM)
1.5
1.0
0.5
1
2
3
4
5
clusnum_bp
clusnum_bp
1
2
3
4
5

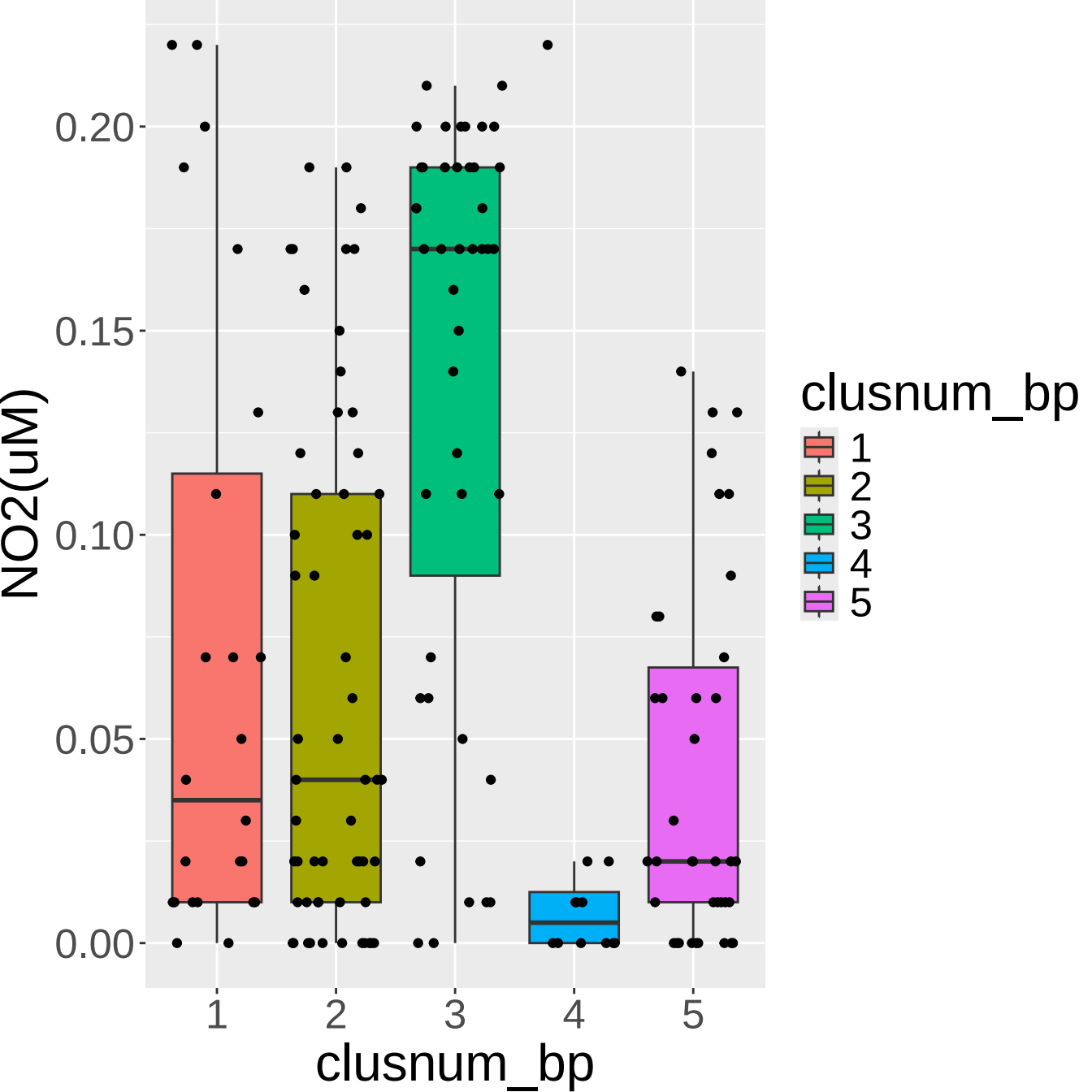

0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
clusnum_bp
1
2
3
4
5
PO4(uM)
0.5
1.0
1.5

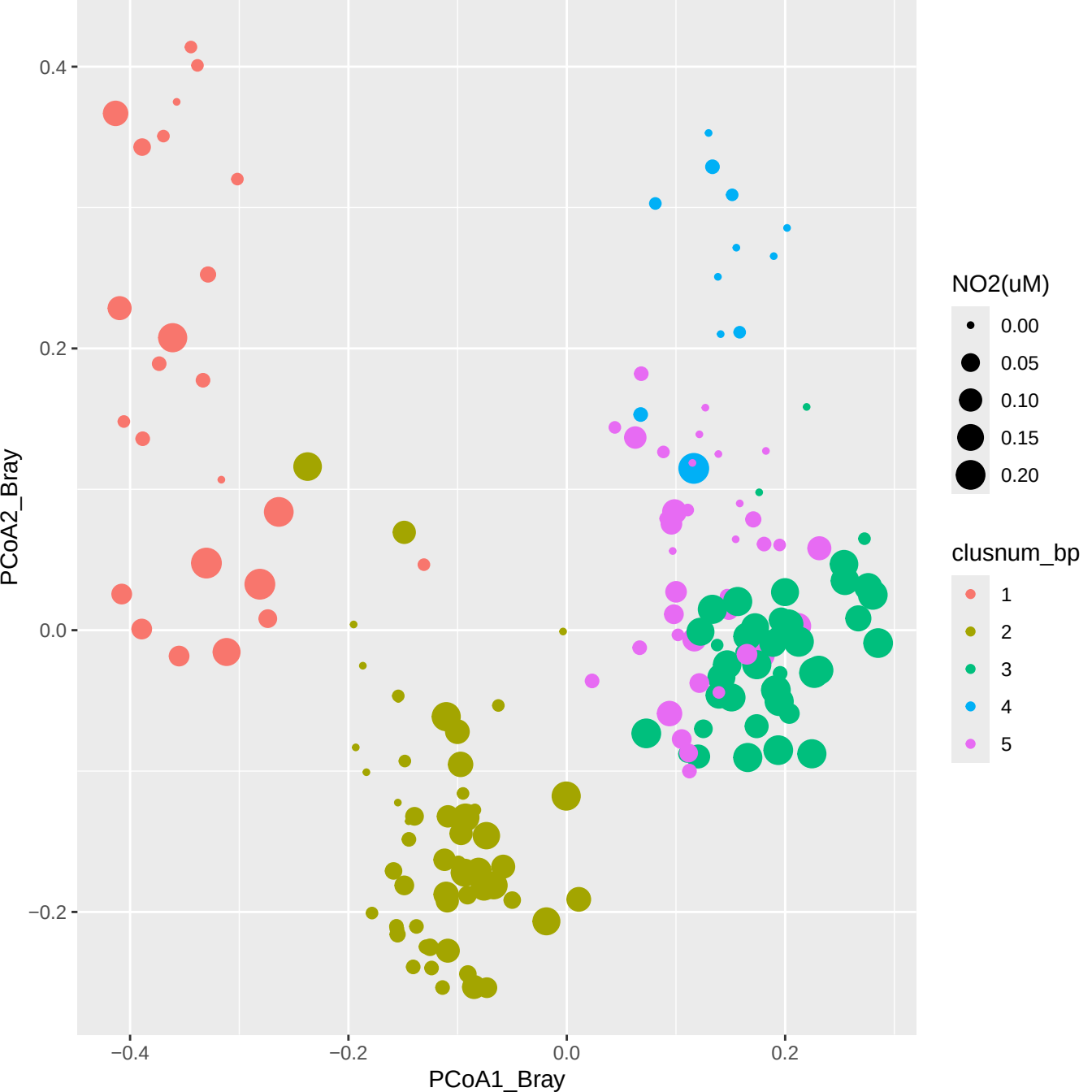

Sil(uM)
30
20
10
0
1
2
3
4
5
clusnum_bp
clusnum_bp
1
2
3
4
5

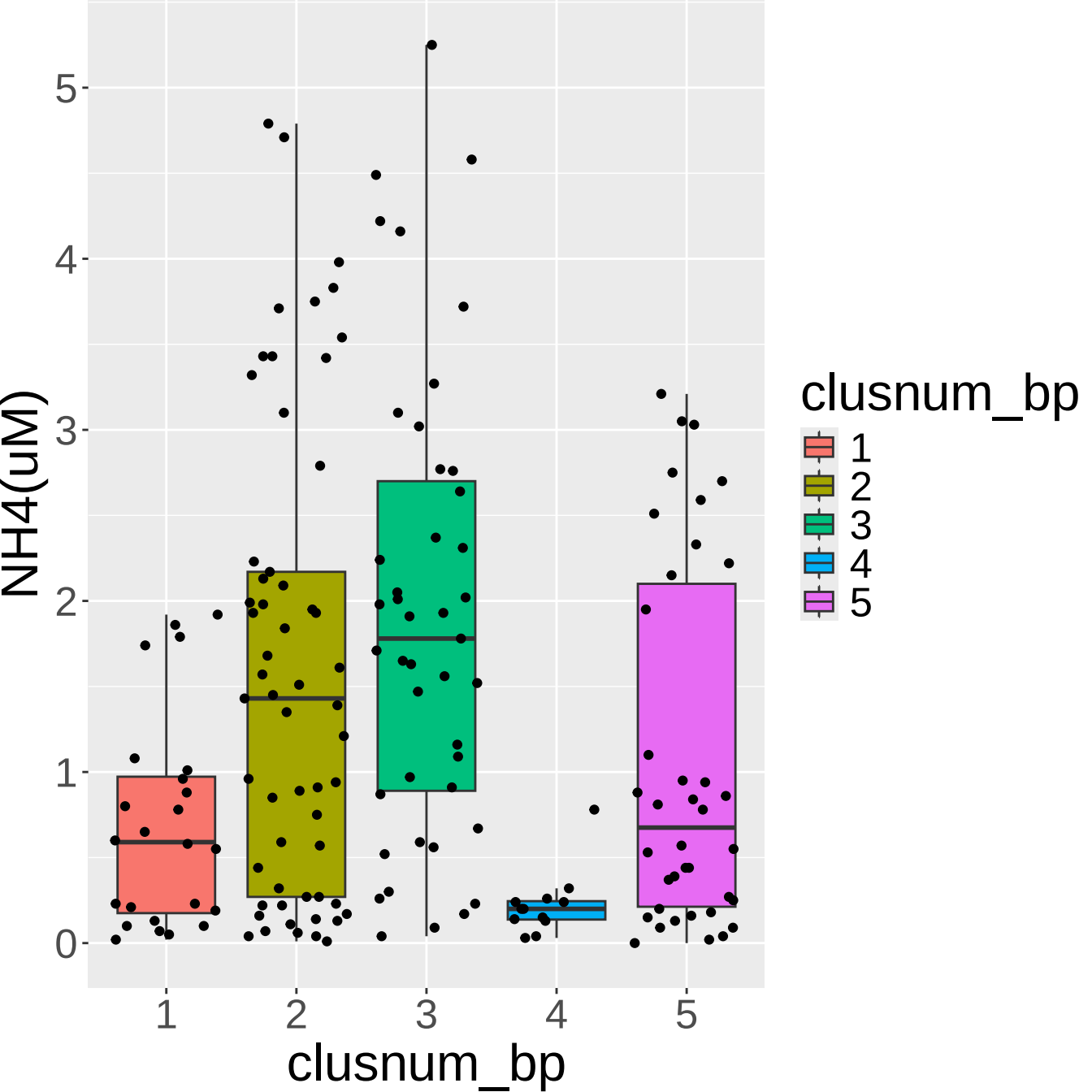

0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
Sil(uM)
10
20
30
clusnum_bp
1
2
3
4
5

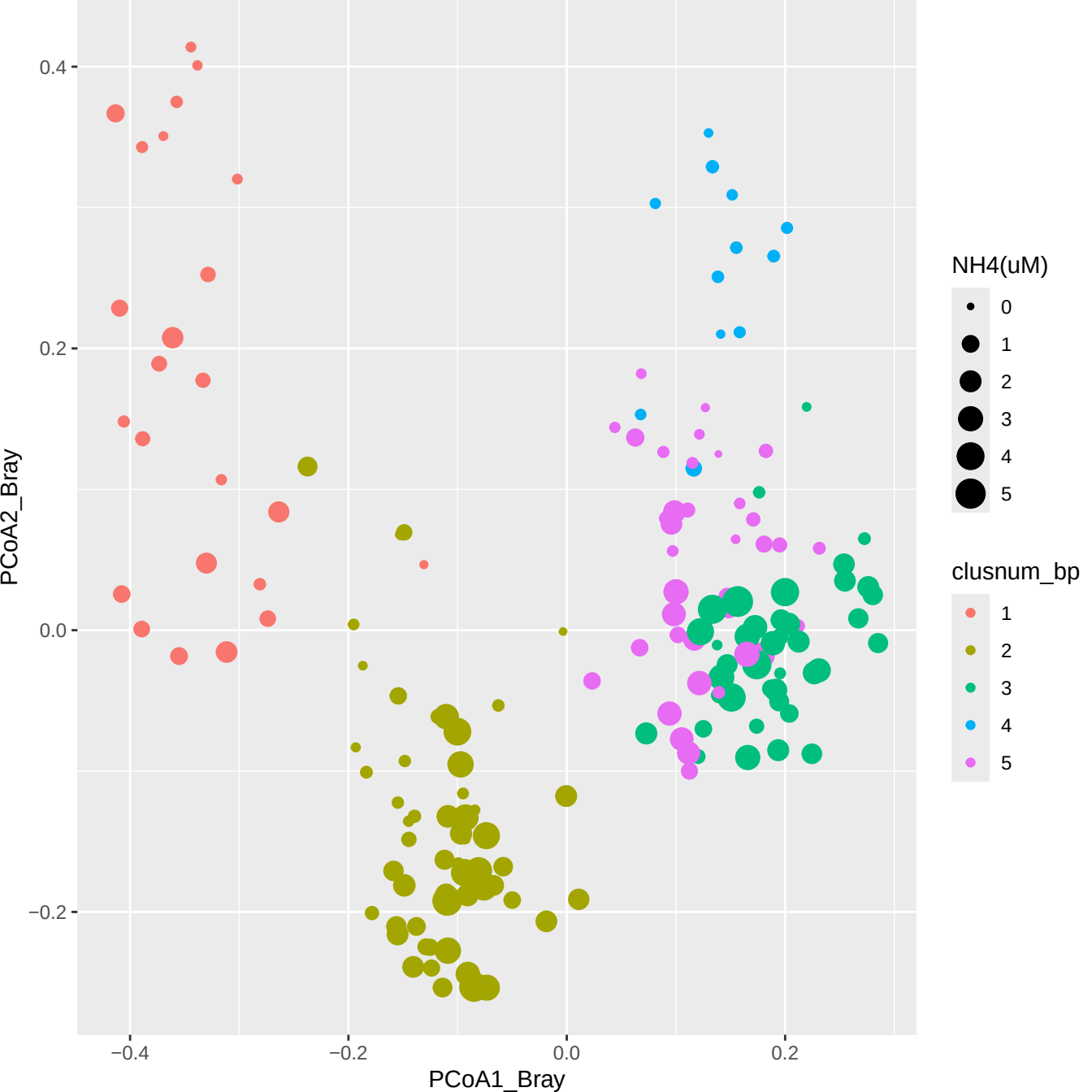

NO3(uM)
15
10
5
0
1
2
3
4
5
clusnum_bp
clusnum_bp
1
2
3
4
5

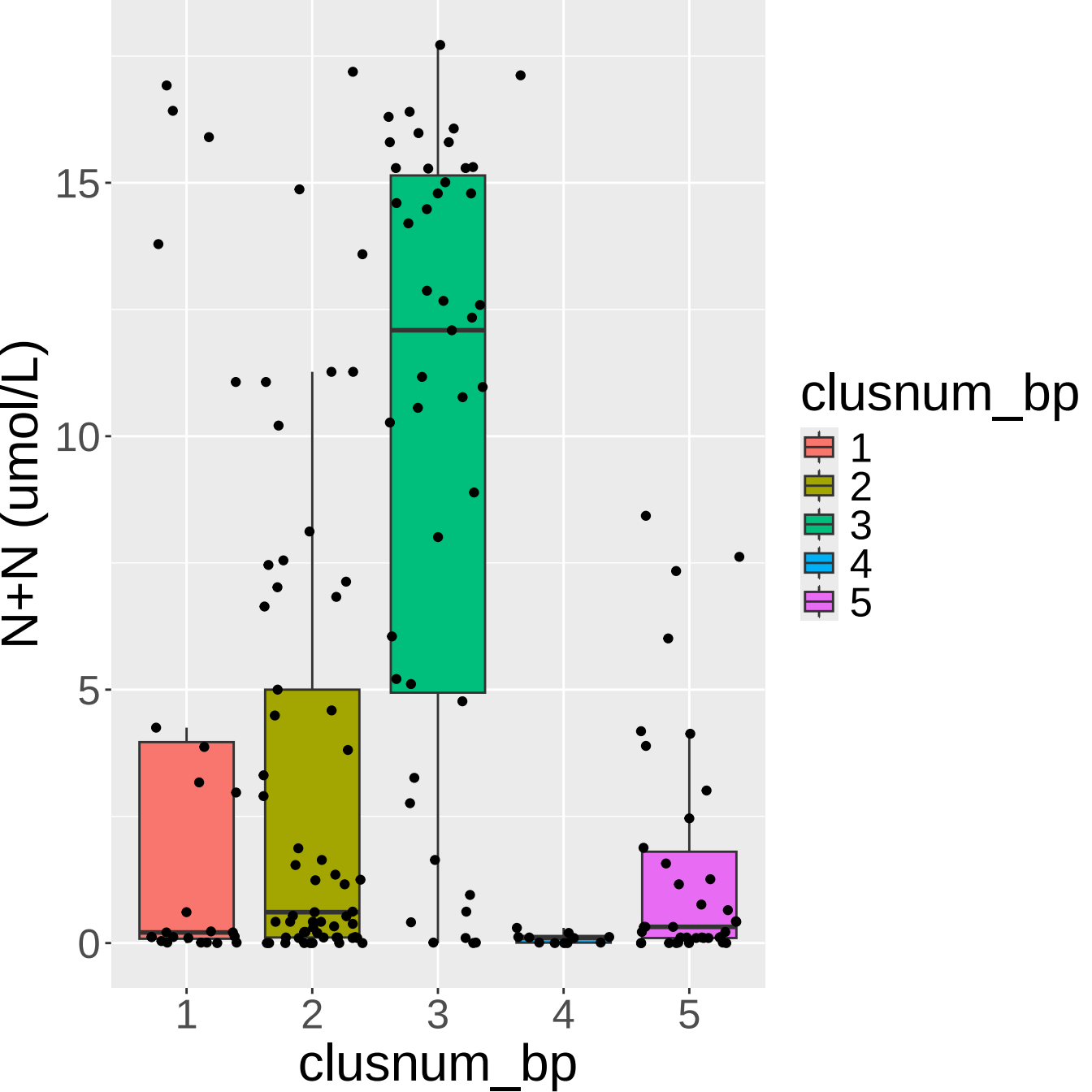

0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
clusnum_bp
1
2
3
4
5
NO3(uM)
0
5
10
15

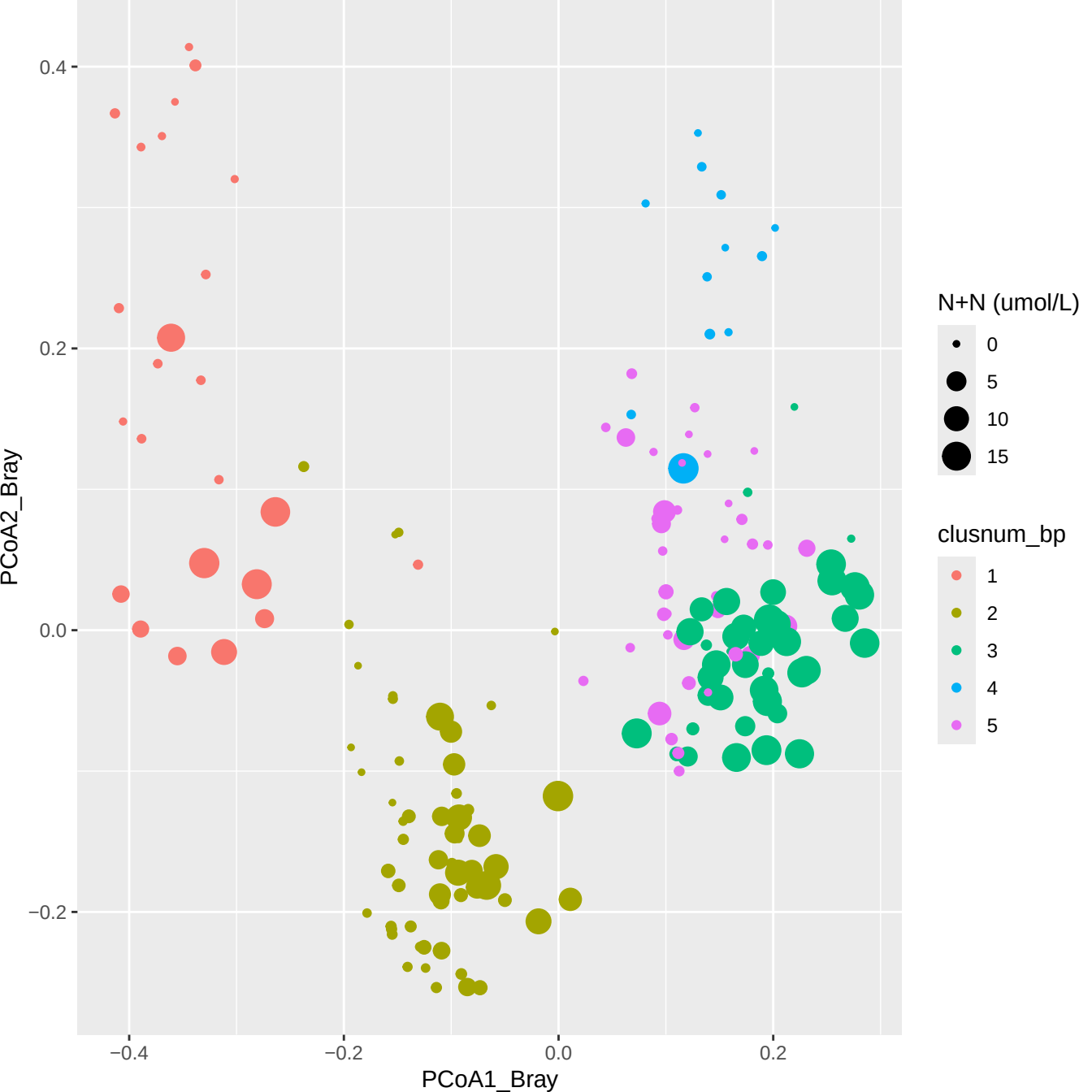

NO2(uM)
0.20
0.15
0.10
0.05
0.00
1
2
3
4
5
clusnum_bp
clusnum_bp
1
2
3
4
5

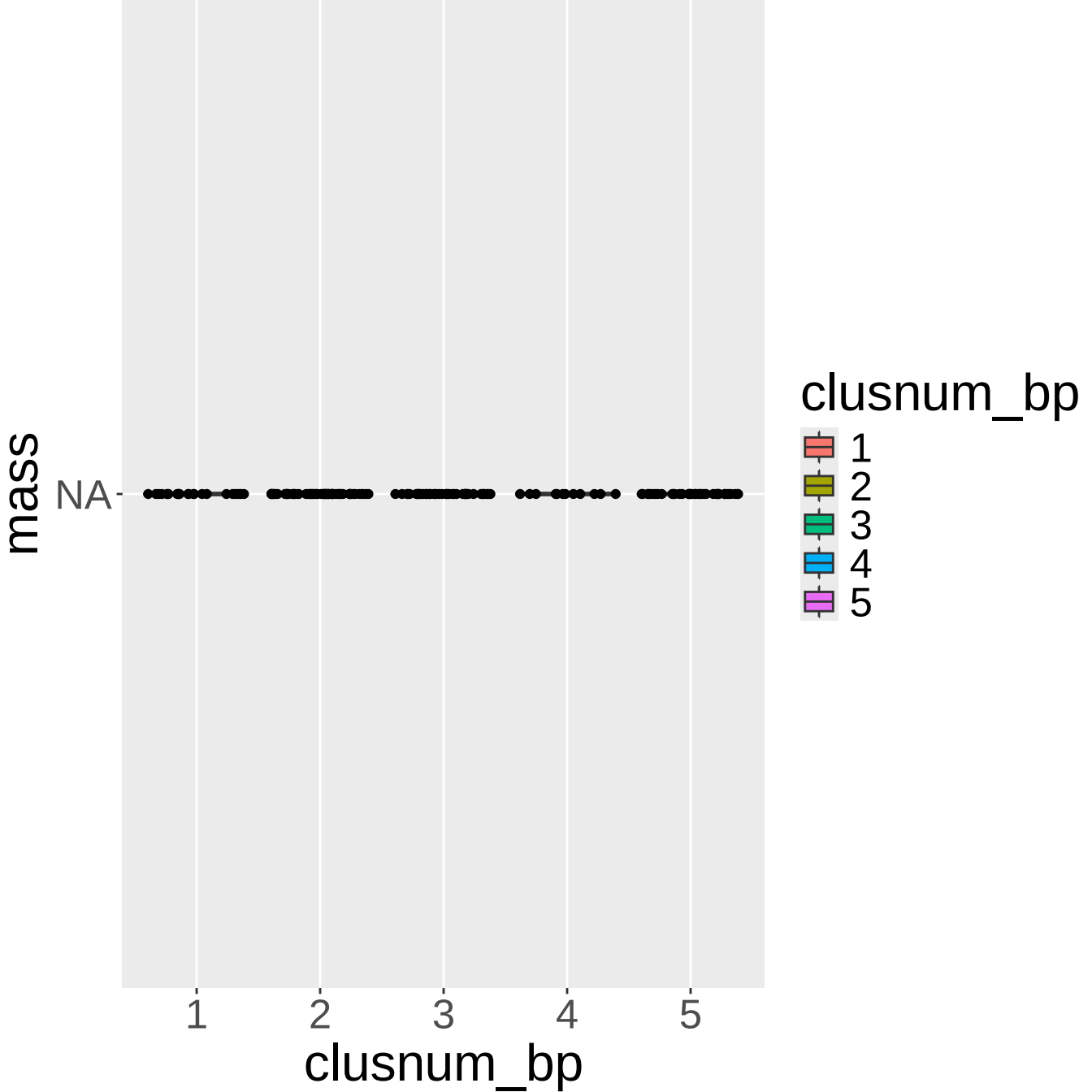

0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
NO2(uM)
0.00
0.05
0.10
0.15
0.20
clusnum_bp
1
2
3
4
5

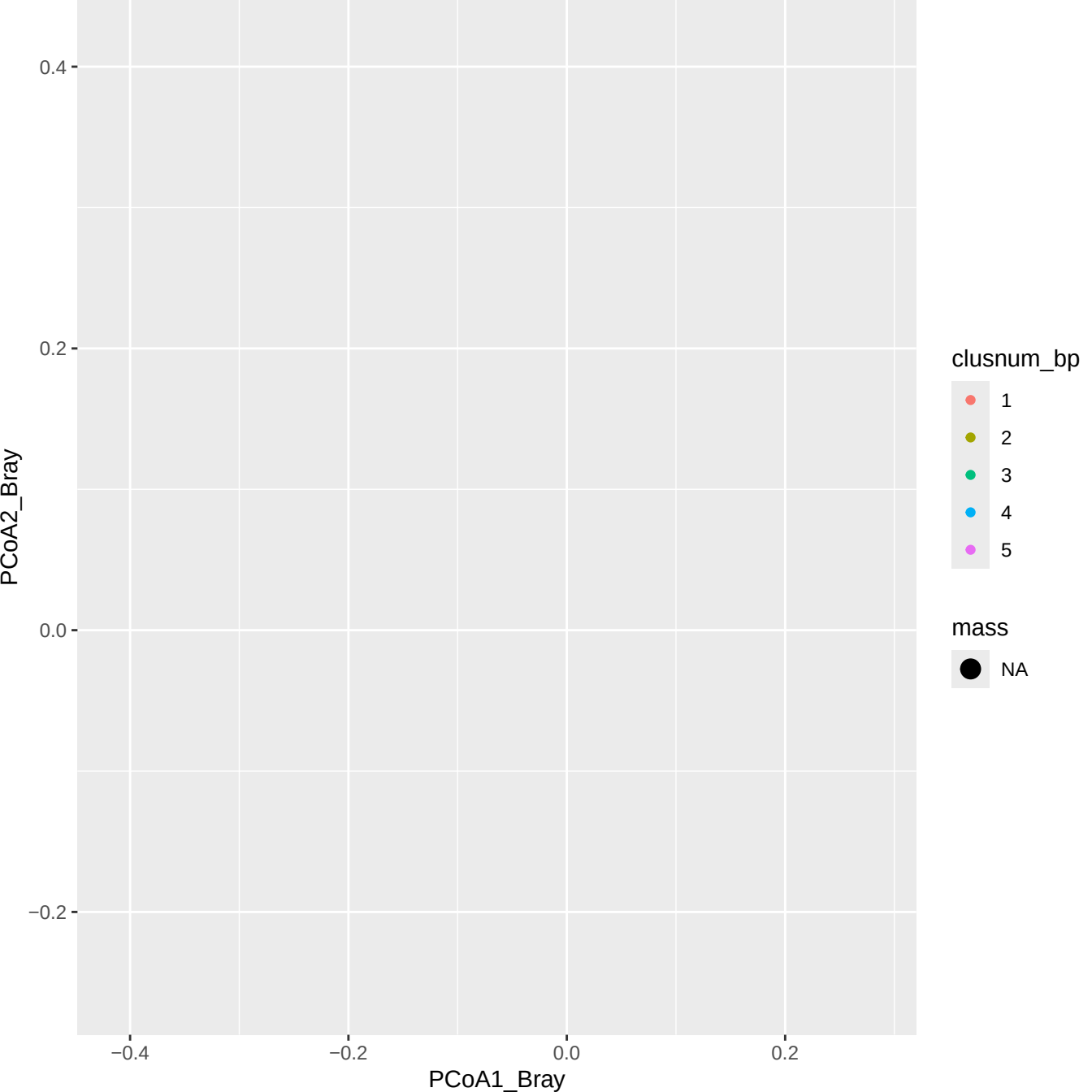

NH4(uM)
0
1
2
3
4
5
1
2
3
4
5
clusnum_bp
clusnum_bp
1
2
3
4
5

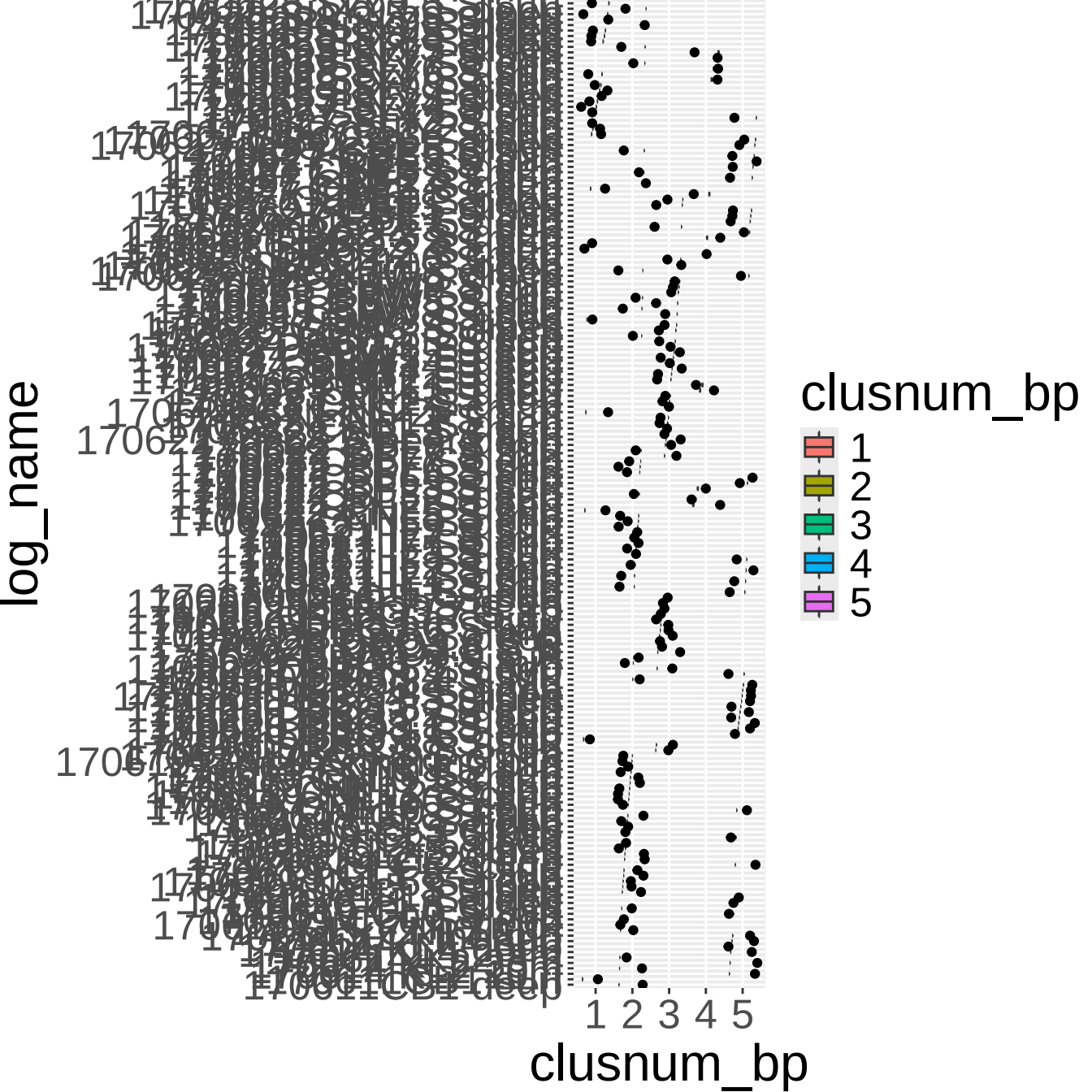

0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
NH4(uM)
0
1
2
3
4
5
clusnum_bp
1
2
3
4
5


N+N (umol/L)
15
10
5
0
1
2
3
4
5
clusnum_bp
clusnum_bp
1
2
3
4
5


0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
N+N (umol/L)
0
5
10
15
clusnum_bp
1
2
3
4
5

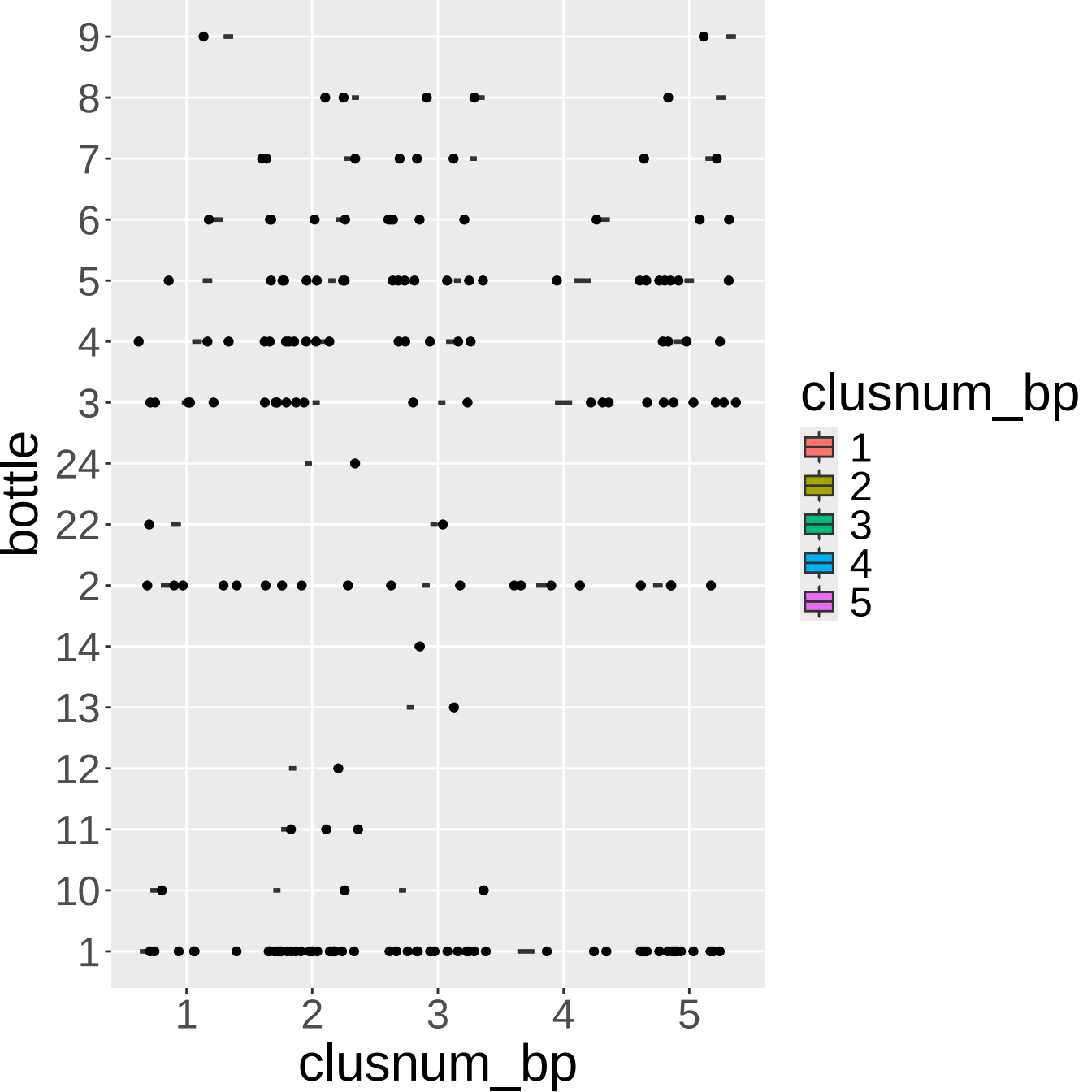

mass
NA
1
2
3
4
5
clusnum_bp
clusnum_bp
1
2
3
4
5

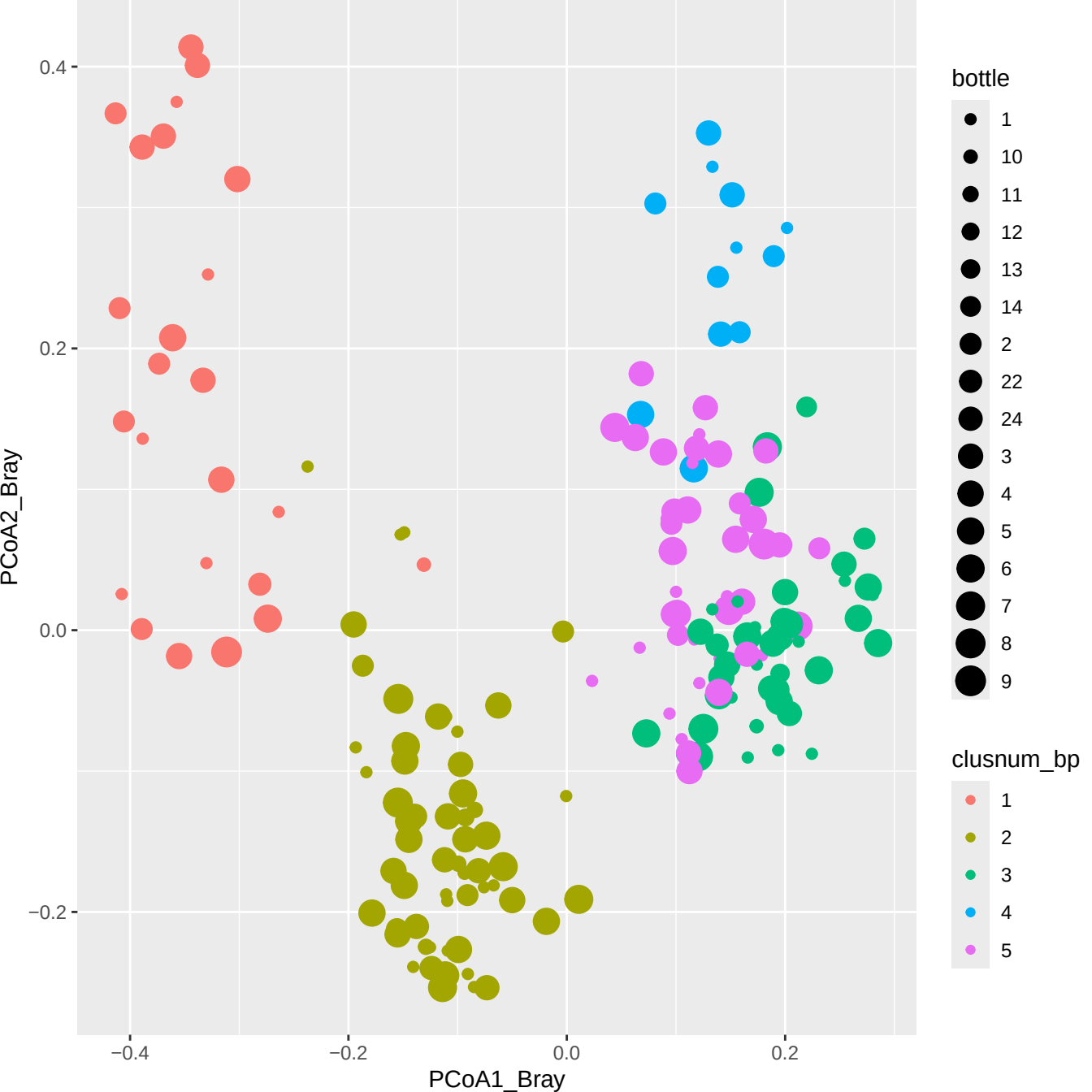

0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
clusnum_bp
1
2
3
4
5
mass
NA

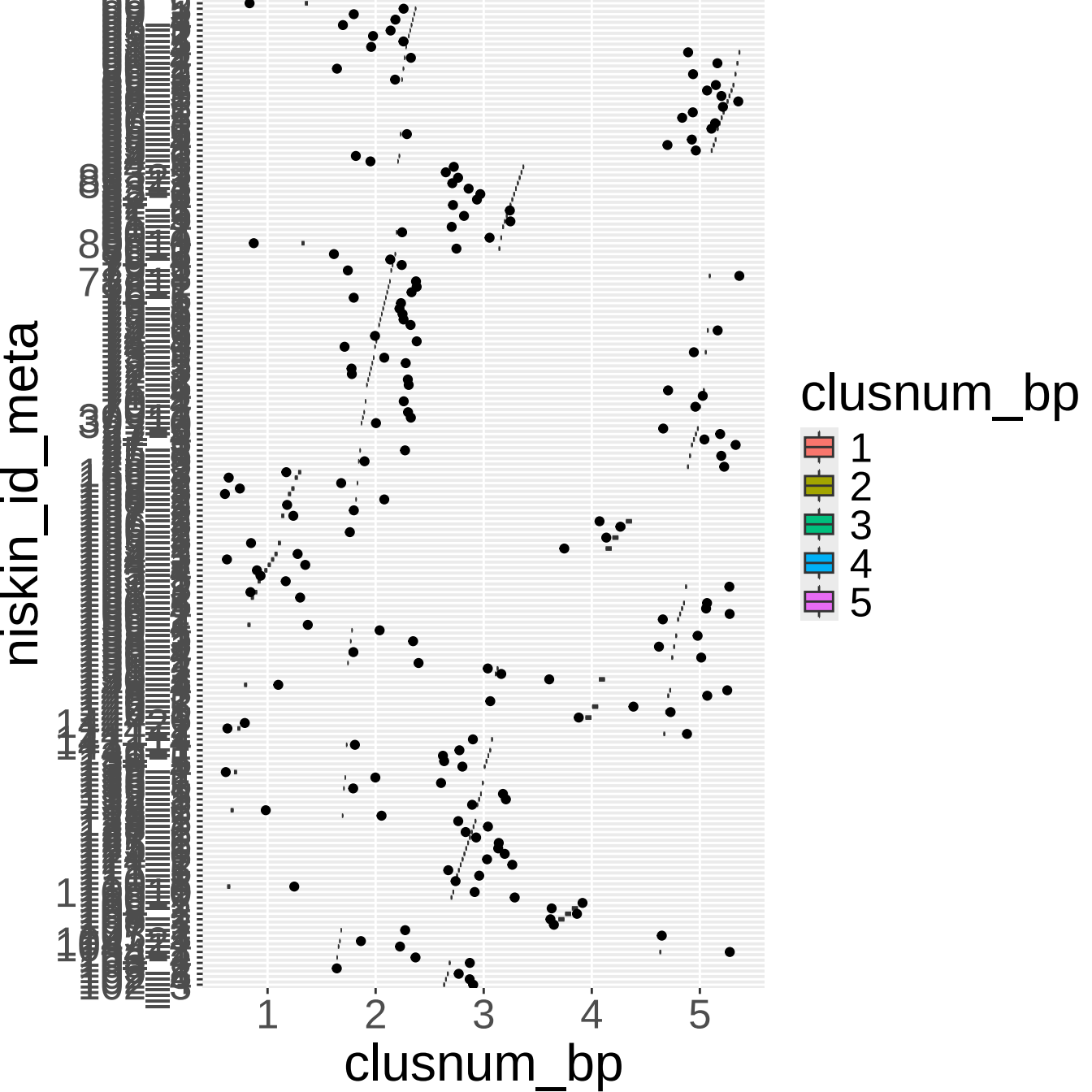

log_name
clusnum_bp
1
2
3
4
5
clusnum_bp
1
2
3
4
5

o blip
m
p
p
p
es
p

- 170620 DBO3.4 S surf
- 170620 DBO3.5 S deep
- 170620 DBO3.5 S mid
- 170620 DBO3.5 S surf
- 170620 DBO3.6 S A
- 170620 DBO3.6 S B
- 170620 DBO3.6 S C
- 170620 DBO3.6 S deep
- 170620 DBO3.6 S mid
- 170620 DBO3.6 S surf
- 170620 DBO3.6 S surf2
- 170620 DBO3.7 S 10
- 170620 DBO3.7 S 20
- 170620 DBO3.7 S deep
- 170620 DBO3.7 S surf
- 170621 IL1 S deep
- 170621 IL1 S mid
- 170621 IL1 S surf
- 170621 IL2 S deep
- 170621 IL2 S mid
- 170621 IL2 S surf
- 170621 IL3 S deep
- 170621 IL3 S surf
- 170621 IL5 S surf
- 170621 IL7 S deep
- 170621 IL7 S mid
- 170621 IL7 S surf
- 170621 IL9 S surf
- 170622 CNL5 S deep
- 170622 CNL5 S mid
- 170622 CNL5 S surf
- 170622 CPL2 S deep
- 170622 CPL2 S surf
- 170622 CPL3 S deep
- 170622 CPL3 S surf
- 170622 CPL5 S surf
- 170622 CPL6 S deep
- 170622 CPL6 S mid
- 170622 CPL6 S surf
- 170622 CPL7 S deep
- 170622 CPL7 S mid
- 170622 CPL7 S surf
- 170622 CPL8 S mid
- 170622 CPL8 S near deep
- 170622 CPL8 S surf
- 170623 CNL1 S chl m
- 170623 CNL1 S deep
- 170623 CNL1 S surf
- 170623 CNL2 S 2x chl m
- 170623 CNL2 S chl m
- 170623 CNL2 S deep
- 170623 CNL2 S surf
- 170623 CPL1 S deep
- 170623 CPL1 S surf
- 170624 CBW12 S deep
- 170624 CBW12 S surf
- 170624 CBW14 S deep
- 170624 CBW14 S su
- 170624 CBW15 S de
- 170624 CBW15 S su
- 170624 DBO2.5 S de
- 170624 DBO2.5 S su
- 170625 CBE5 S deep
- 170625 CBW3 S chl
- 170625 CBW3 S clin
- 170625 CBW3 S dee
- 170625 CBW3 S surf
- 170625 CBW5 S dee
- 170625 CBW5 S mid
- 170625 CBW5 S surf
- 170625 CBW9 S dee
- 170625 CBW9 S mid
- 170625 CBW9 S surf
- 170625 DBO1.10 S d
- 170625 DBO1.10 S d
- 170625 DBO1.10 S d


bottle
9
8
7
6
5
4
3
24
22
2
14
13
12
11
10
1
1
2
3
4
5
clusnum_bp
clusnum_bp
1
2
3
4
5


0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
bottle
1
10
11
12
13
14
2
22
24
3
4
5
6
7
8
9
clusnum_bp
1
2
3
4
5


niskin_id_meta
clusnum_bp
1
2
3
4
5
clusnum_bp
1
2
3
4
5

clusnum_bp

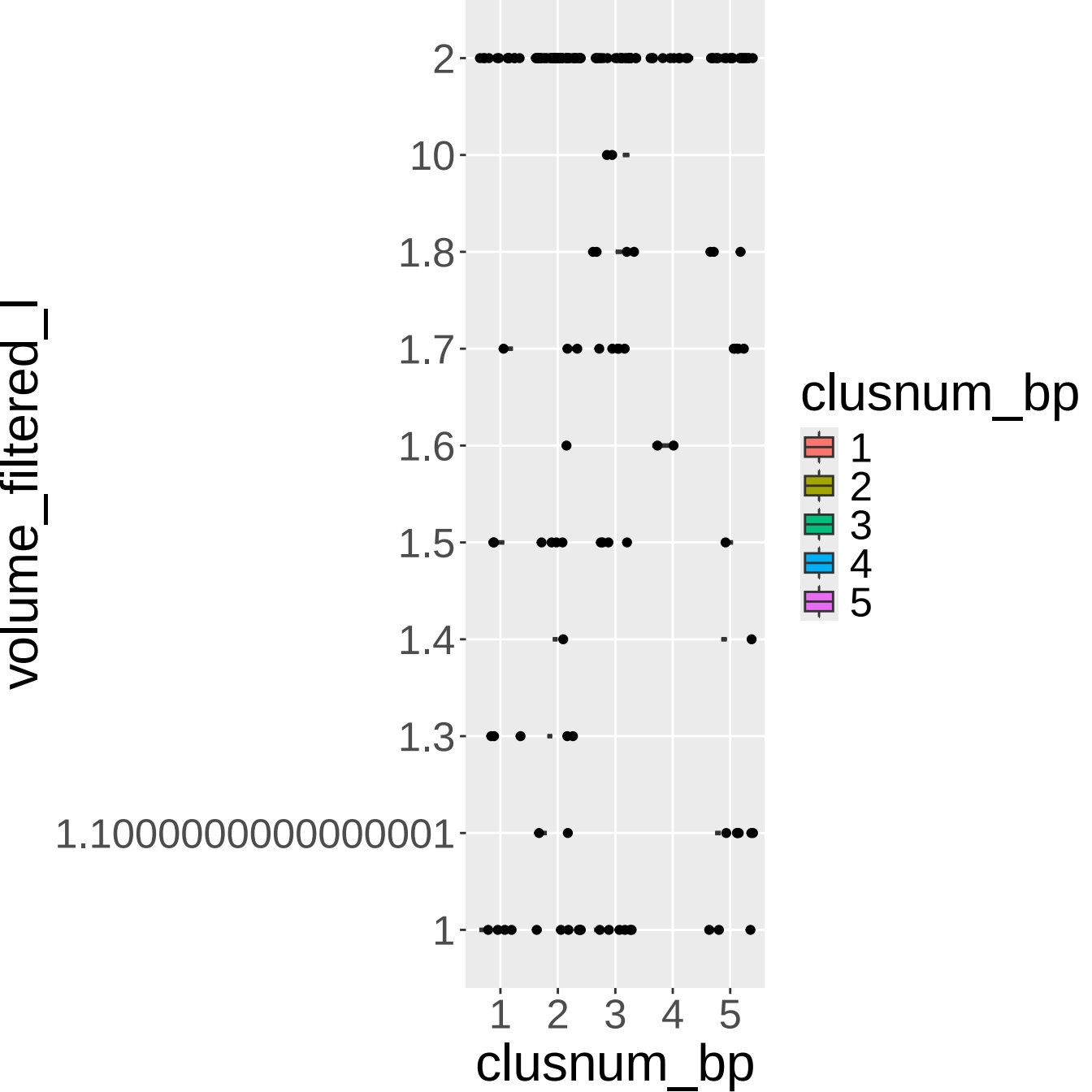
- 1
- 2
- 3
- 4
- 5

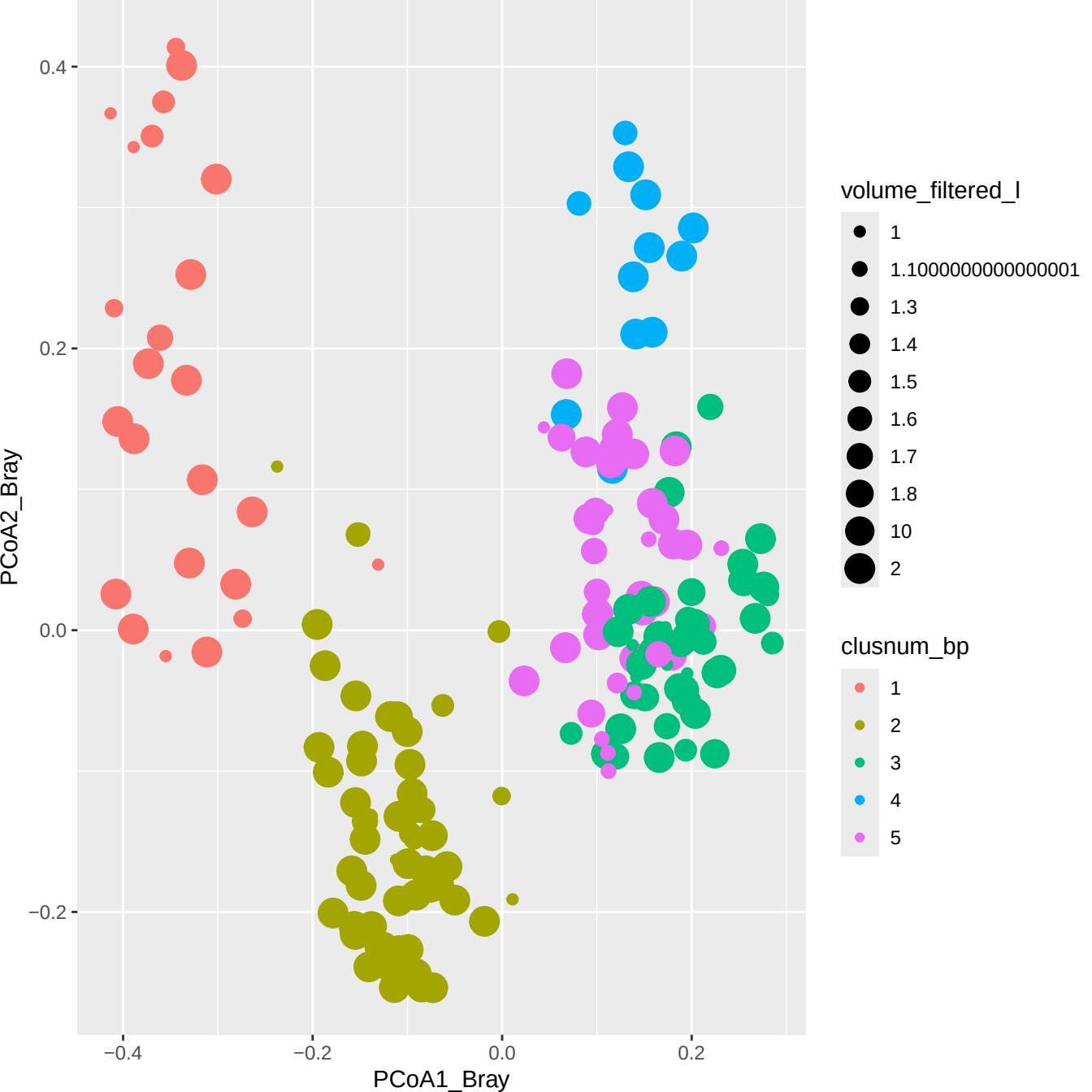
niskin_id_meta

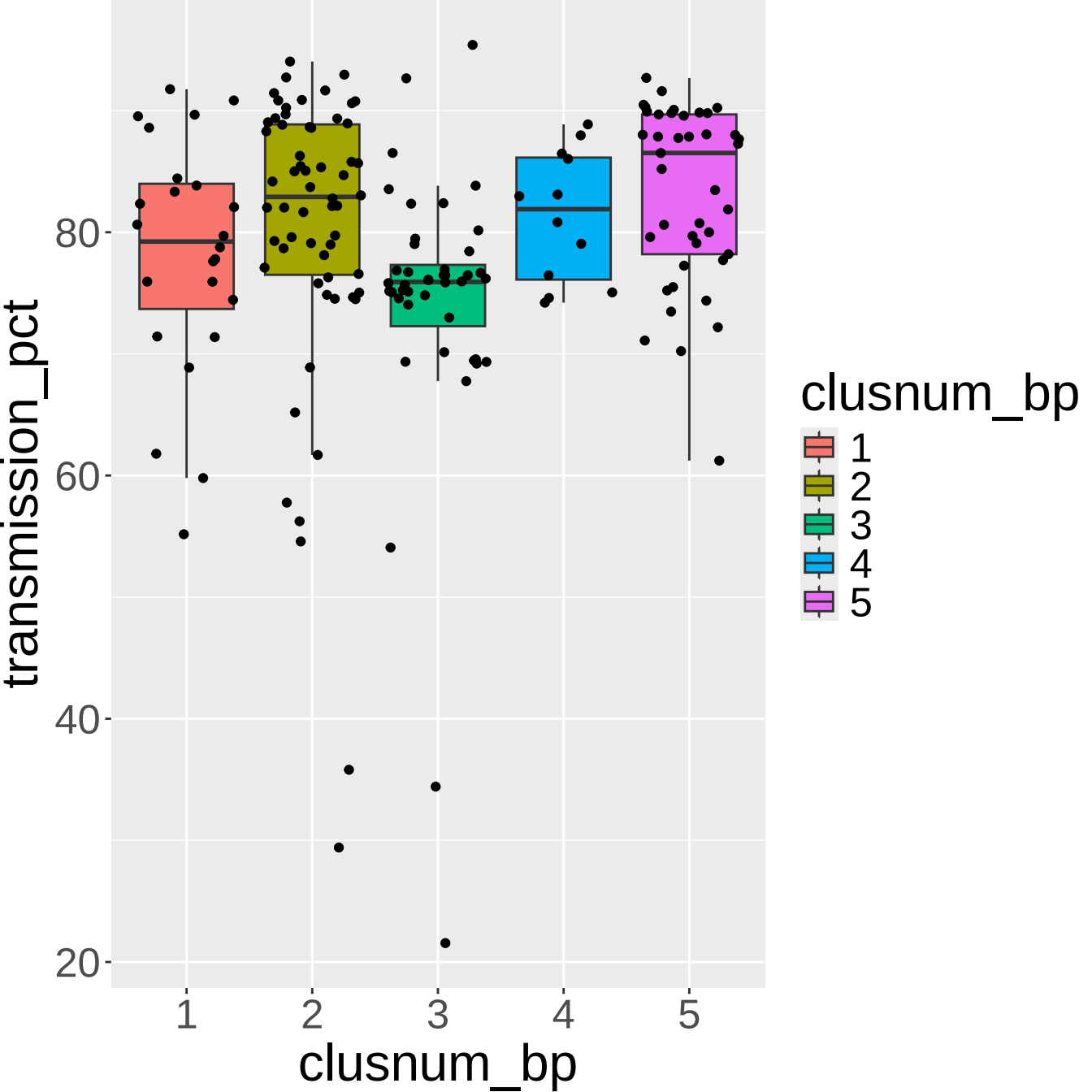
102_3	110_8	136_7	151_1	163_4	25_2	72_5	79_4	84_6	
102_4	111_1	139_1	151_2	164_1	25_3	73_1	79_6	85_1	
102_8	111_5	139_4	156_1	164_2	25_5	73_9	80_1	85_3	
103_1	111_7	139_5	156_2	164_3	25_8	74_1	80_10	85_4	
103_2	124_1	139_7	156_4	165_1	27_1	74_4	80_4	85_5	
103_4	124_6	141_1	158_1	165_2	27_3	74_5	80_7	85_6	
104_1	125_1	141_11	158_3	165_3	27_4	74_6	81_1	86_1	
104_11	125_6	141_14	158_5	166_1	27_5	75_1	81_4	86_5	
104_24	127_1	141_7	159_1	166_2	39_10	75_3	81_5	87_1	
105_3	127_6	144_1	159_4	166_3	39_12	75_6	81_6	87_5	
107_1	128_1	144_22	159_7	167_1	70_1	75_8	82_1	88_1	
107_2	128_5	144_6	160_1	167_2	70_2	76_1	82_4	88_3	
108_1	13_1	146_1	160_4	167_3	70_4	76_5	82_6	88_5	
108_2	13_6	146_5	160_5	168_1	71_1	76_7	83_1	89_1	
109_1	132_1	146_7	161_1	168_2	71_4	78_1	83_13	89_2	
109_2	132_3	148_1	161_2	168_3	71_6	78_11	83_22	89_3	
110_1	132_5	148_3	161_3	169_1	72_1	78_8	83_5	90_1	
110_10	136_1	148_4	163_1	169_2	72_2	79_1	84_1	90_2	
110_5	136_5	150_3	163_3	169_3	72_3	79_3	84_3	90_4	

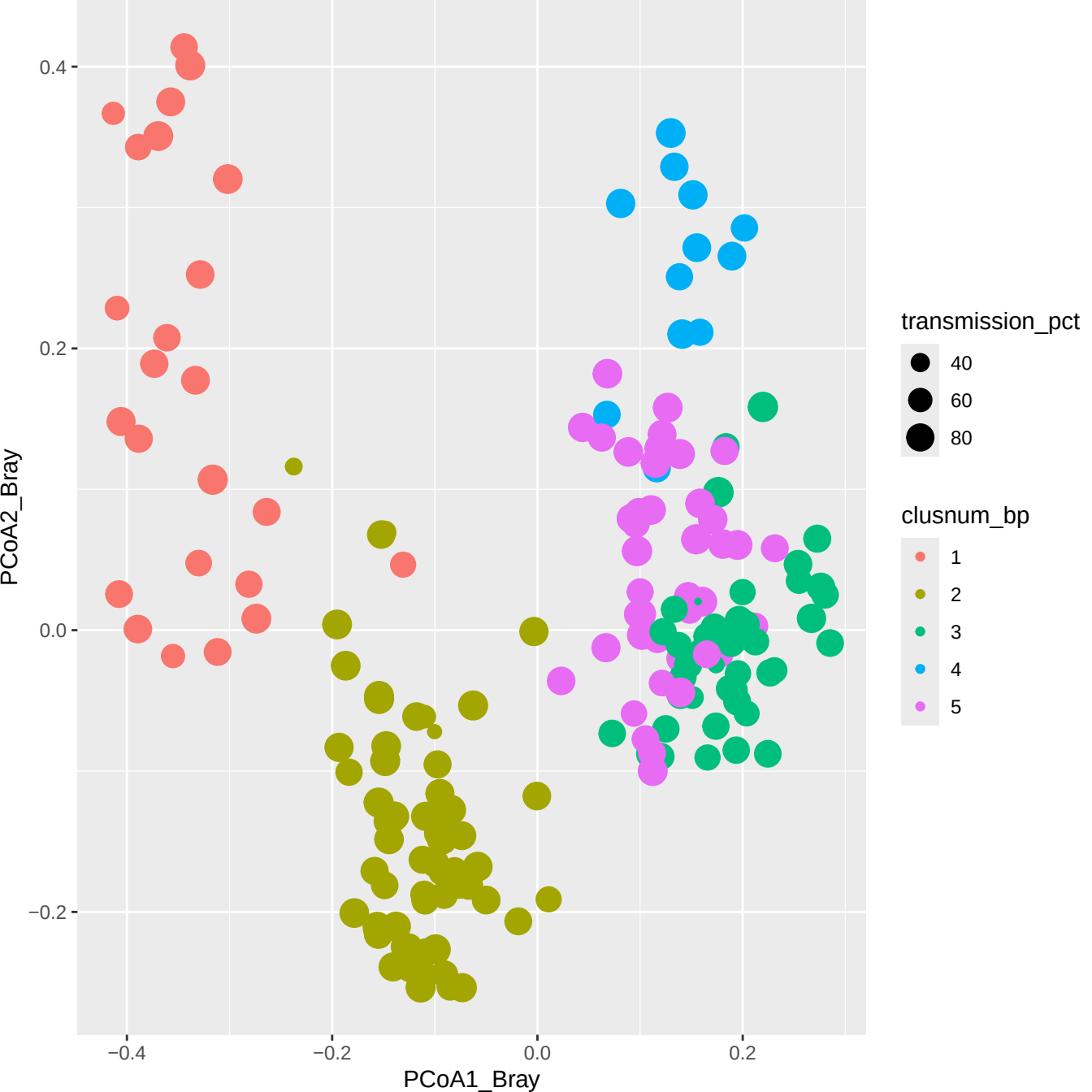
2

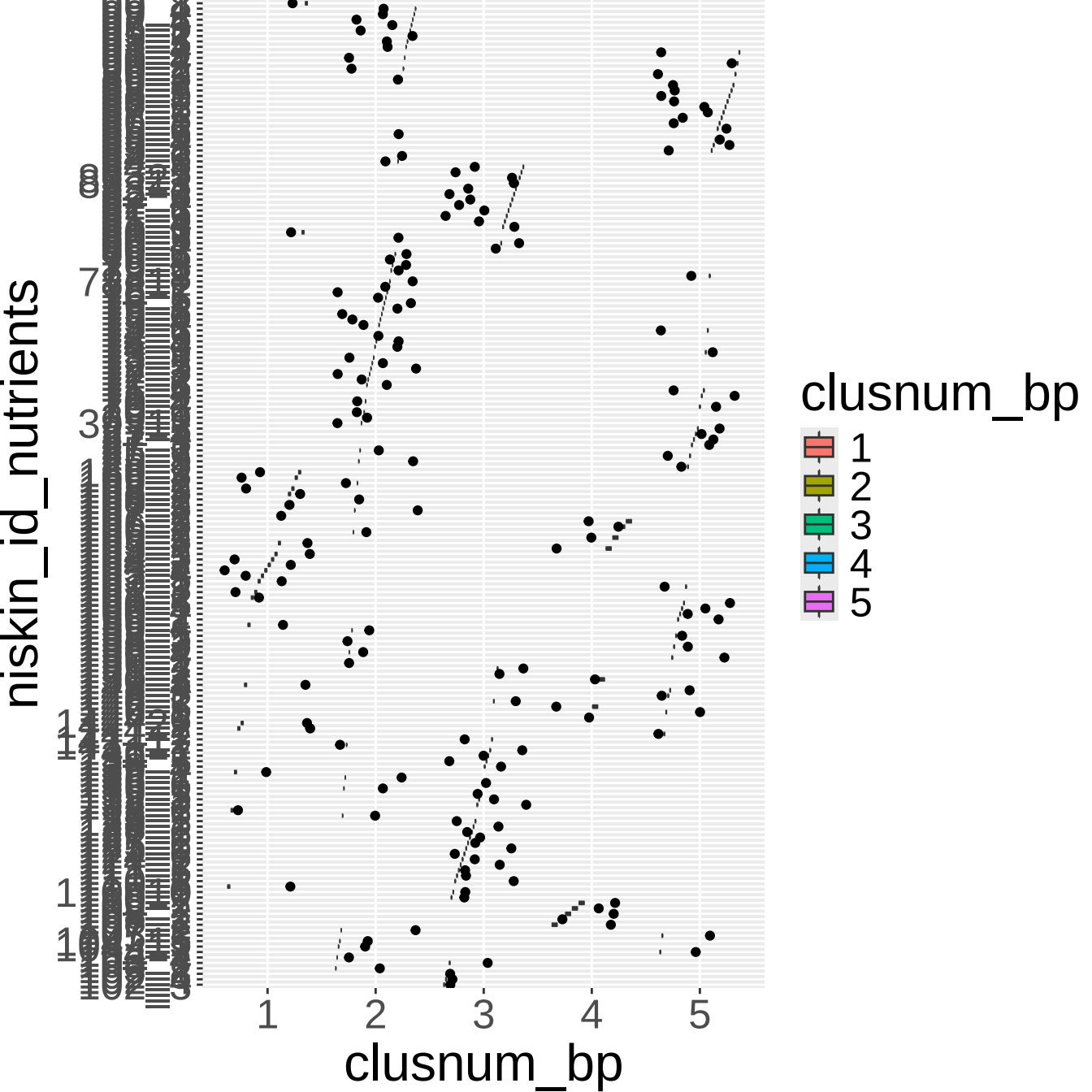
_Bray


volume_filtered_l
2
10
1.8
1.7
1.6
1.5
1.4
1.3
1.1000000000000001
1
1
2
3
4
5
clusnum_bp
clusnum_bp
1
2
3
4
5


0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
volume_filtered_l
1
1.1000000000000001
1.3
1.4
1.5
1.6
1.7
1.8
10
2
clusnum_bp
1
2
3
4
5


transmission_pct
80
60
40
20
1
2
3
4
5
clusnum_bp
clusnum_bp
1
2
3
4
5


0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
transmission_pct
40
60
80
clusnum_bp
1
2
3
4
5


niskin_id_nutrients
clusnum_bp
1
2
3
4
5
clusnum_bp
1
2
3
4
5

niskin_id_nutrients

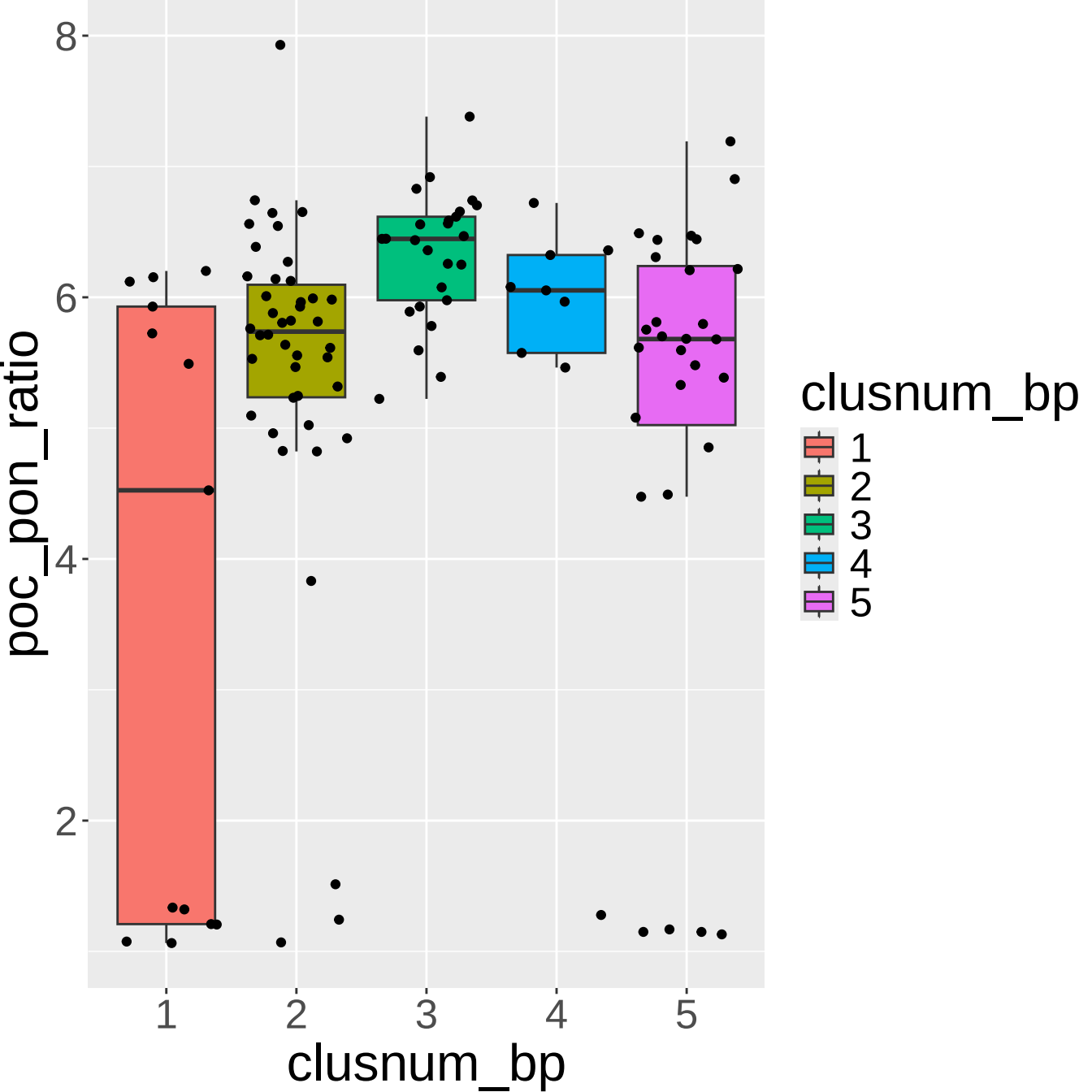
102_3	110_8	136_7	151_1	163_4	25_1	72_5	79_4	84_6	
102_4	111_1	139_1	151_2	164_1	25_3	73_1	79_6	85_1	
102_8	111_5	139_4	156_1	164_2	25_5	73_9	80_1	85_3	
103_1	111_7	139_5	156_2	164_3	25_7	74_1	80_4	85_4	
103_2	124_1	139_7	156_4	165_1	27_1	74_4	80_5	85_5	
103_4	124_6	141_1	158_1	165_2	27_3	74_5	80_9	85_6	
104_1	125_1	141_11	158_3	165_3	27_4	74_6	81_1	86_1	
104_11	125_6	141_12	158_5	166_1	27_5	75_1	81_4	86_5	
104_15	127_1	141_7	159_1	166_2	39_11	75_4	81_5	87_1	
105_3	127_6	144_1	159_4	166_3	39_9	75_6	81_6	87_5	
107_1	128_1	144_22	159_7	167_1	70_1	75_8	82_1	88_1	
107_2	128_5	144_6	160_1	167_2	70_2	76_1	82_4	88_3	
108_1	13_1	146_1	160_4	167_3	70_4	76_5	82_6	88_5	
108_2	13_6	146_5	160_5	168_1	71_1	76_7	83_1	89_1	
109_1	132_1	146_7	161_1	168_2	71_4	78_1	83_13	89_2	
109_2	132_3	148_1	161_2	168_3	71_6	78_11	83_22	89_3	
110_1	132_5	148_3	161_3	169_1	72_1	78_8	83_5	90_1	
110_10	136_1	148_4	163_1	169_2	72_2	79_1	84_1	90_2	
110_5	136_6	150_3	163_3	169_3	72_3	79_3	84_3	90_4	

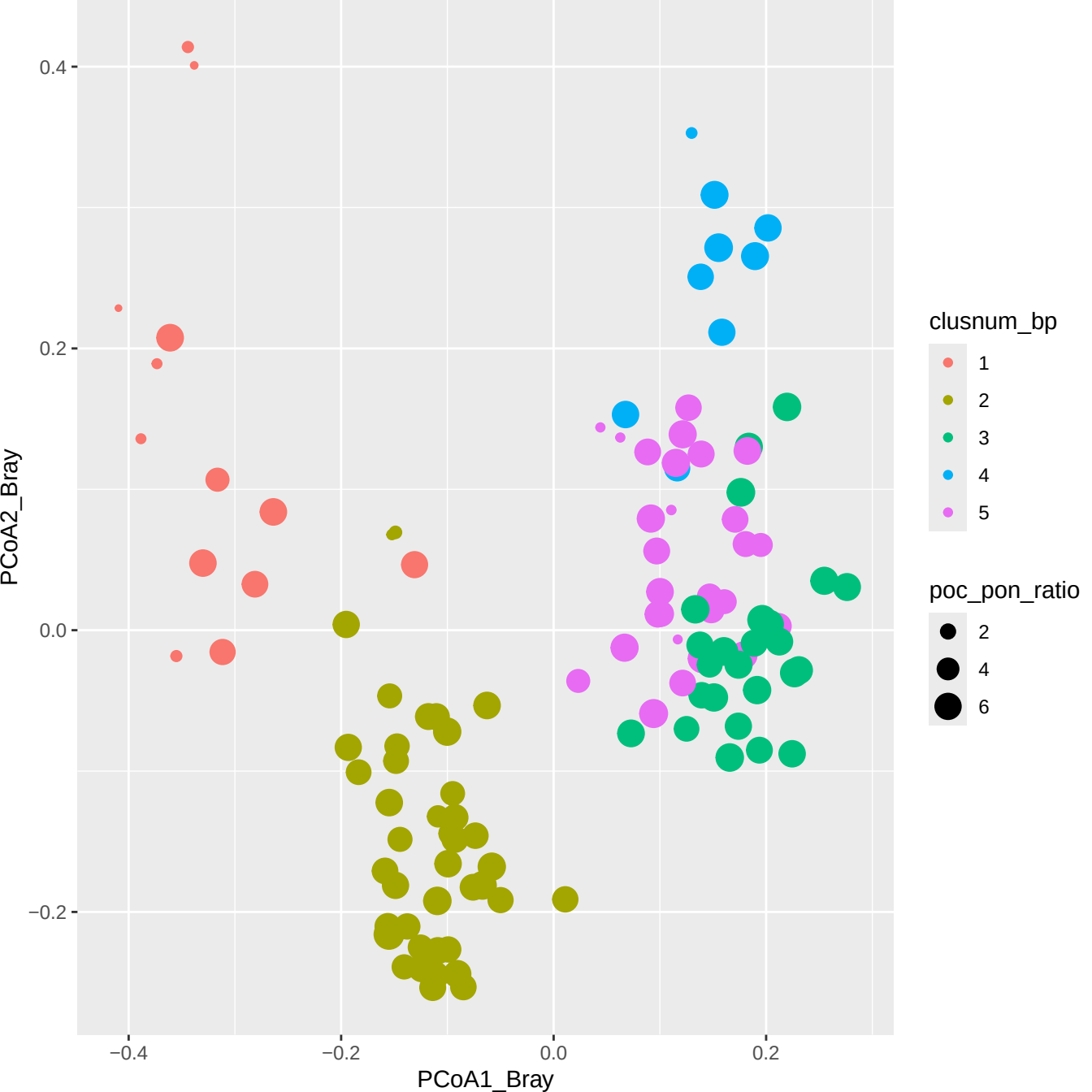
clusnum_bp

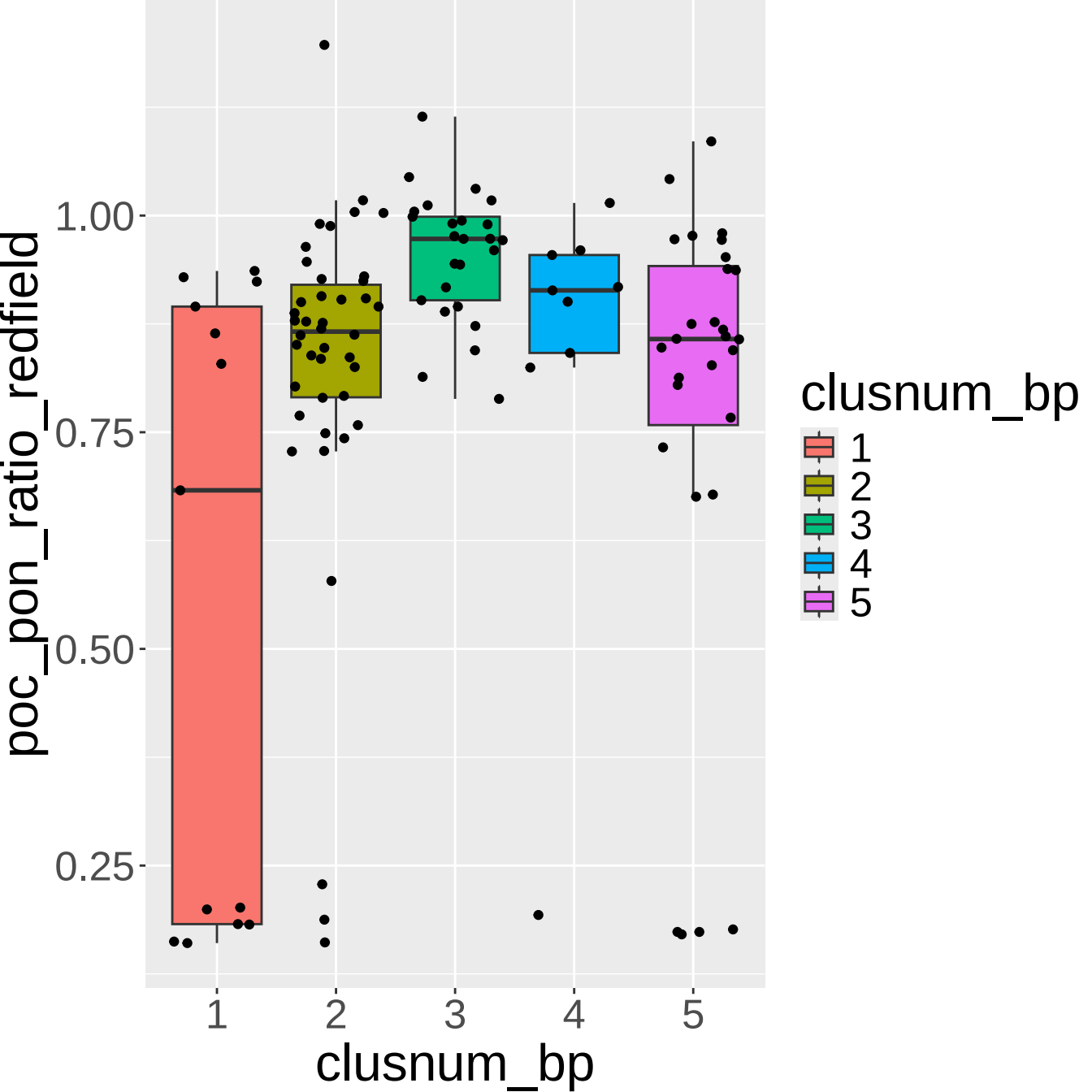
- 1
- 2
- 3
- 4
- 5

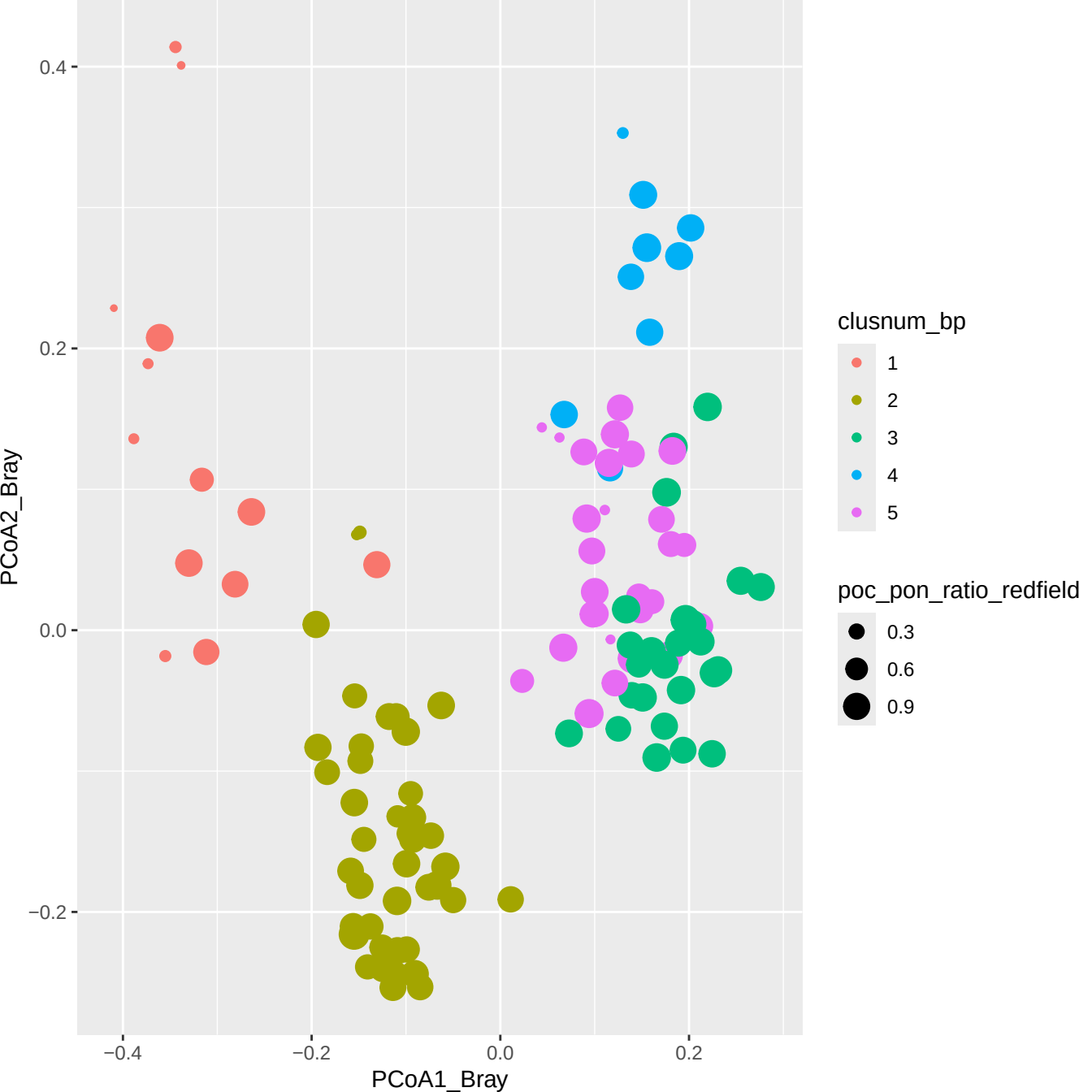
2

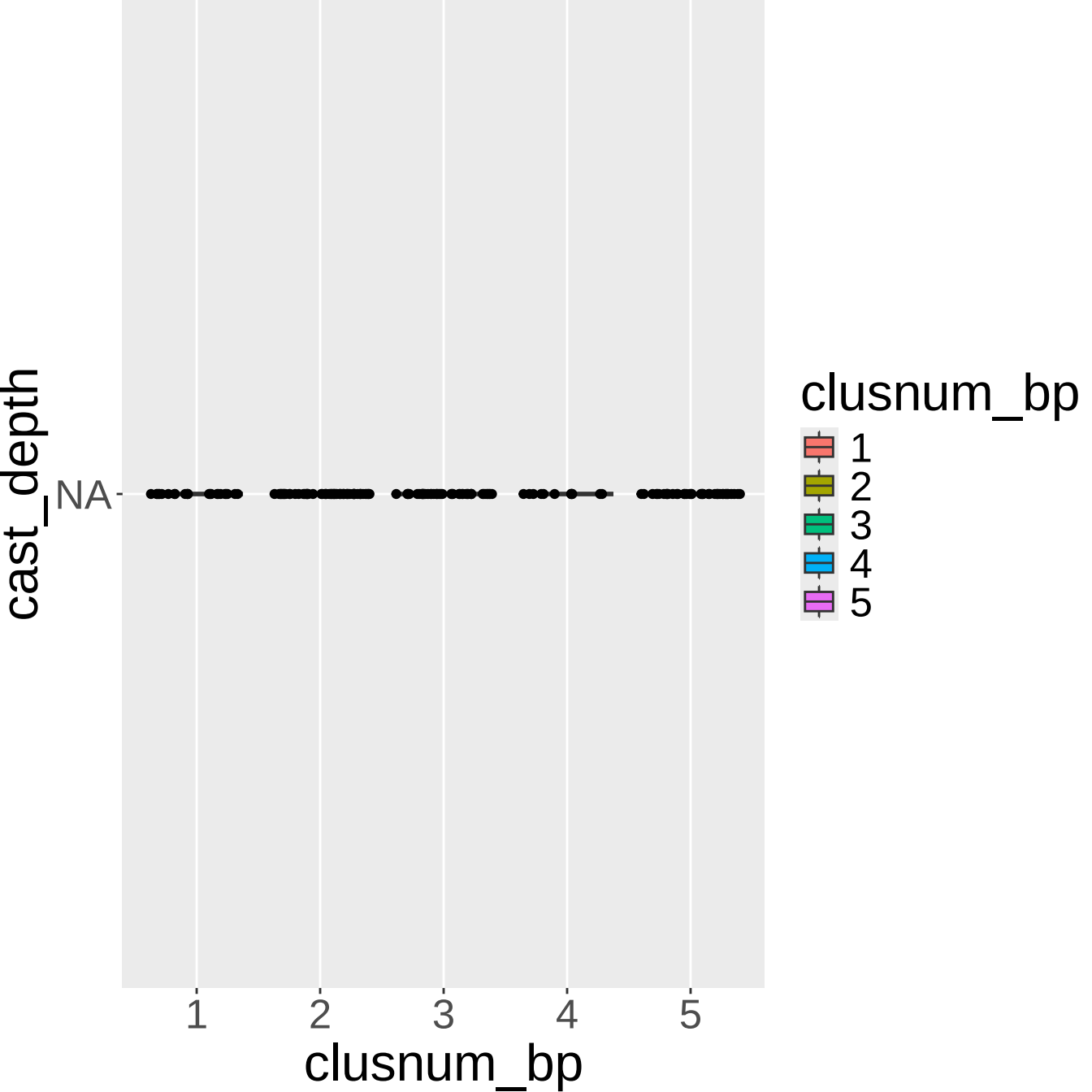
_Bray

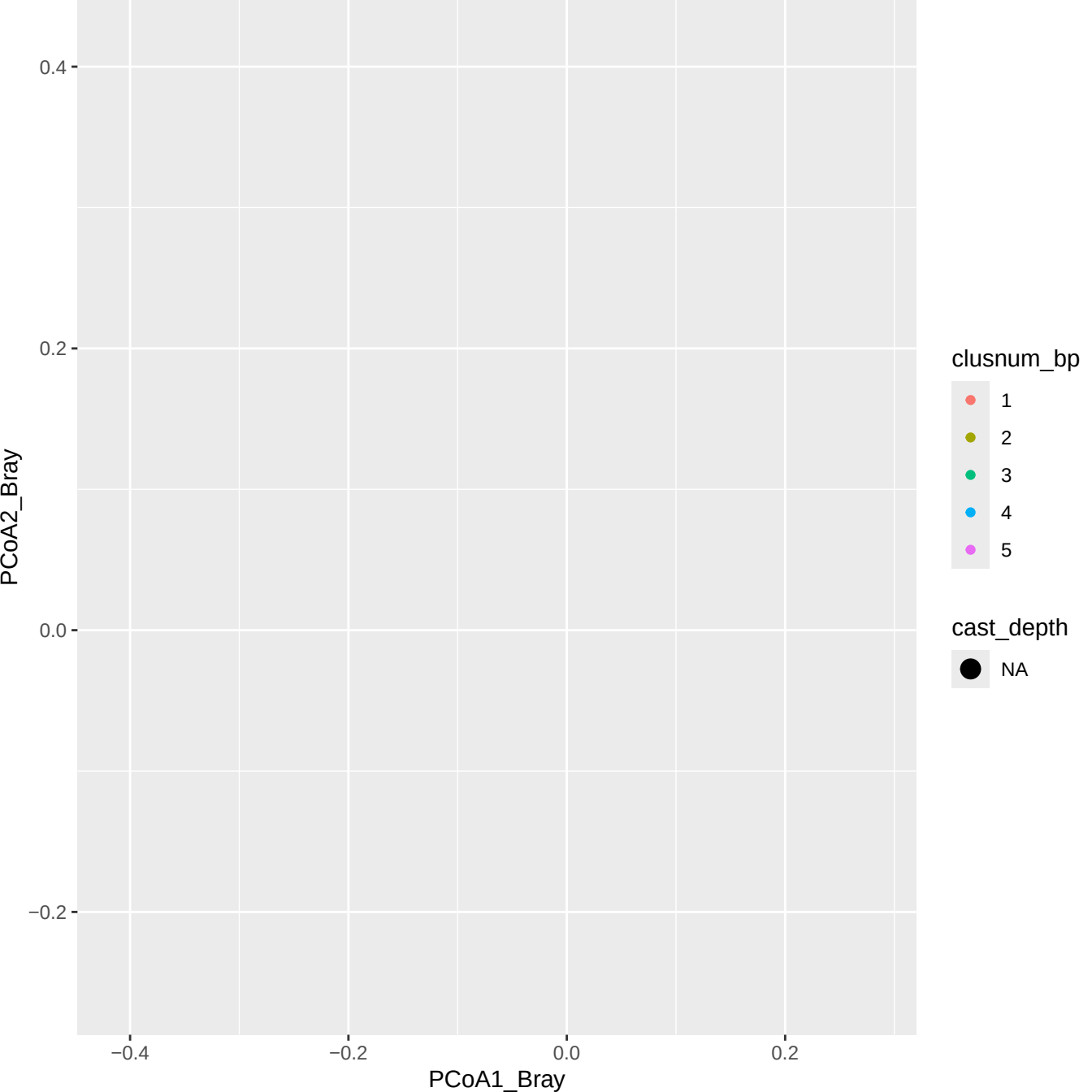

8
6
4
2
poc_pon_ratio
1
2
3
4
5
clusnum_bp
clusnum_bp
1
2
3
4
5

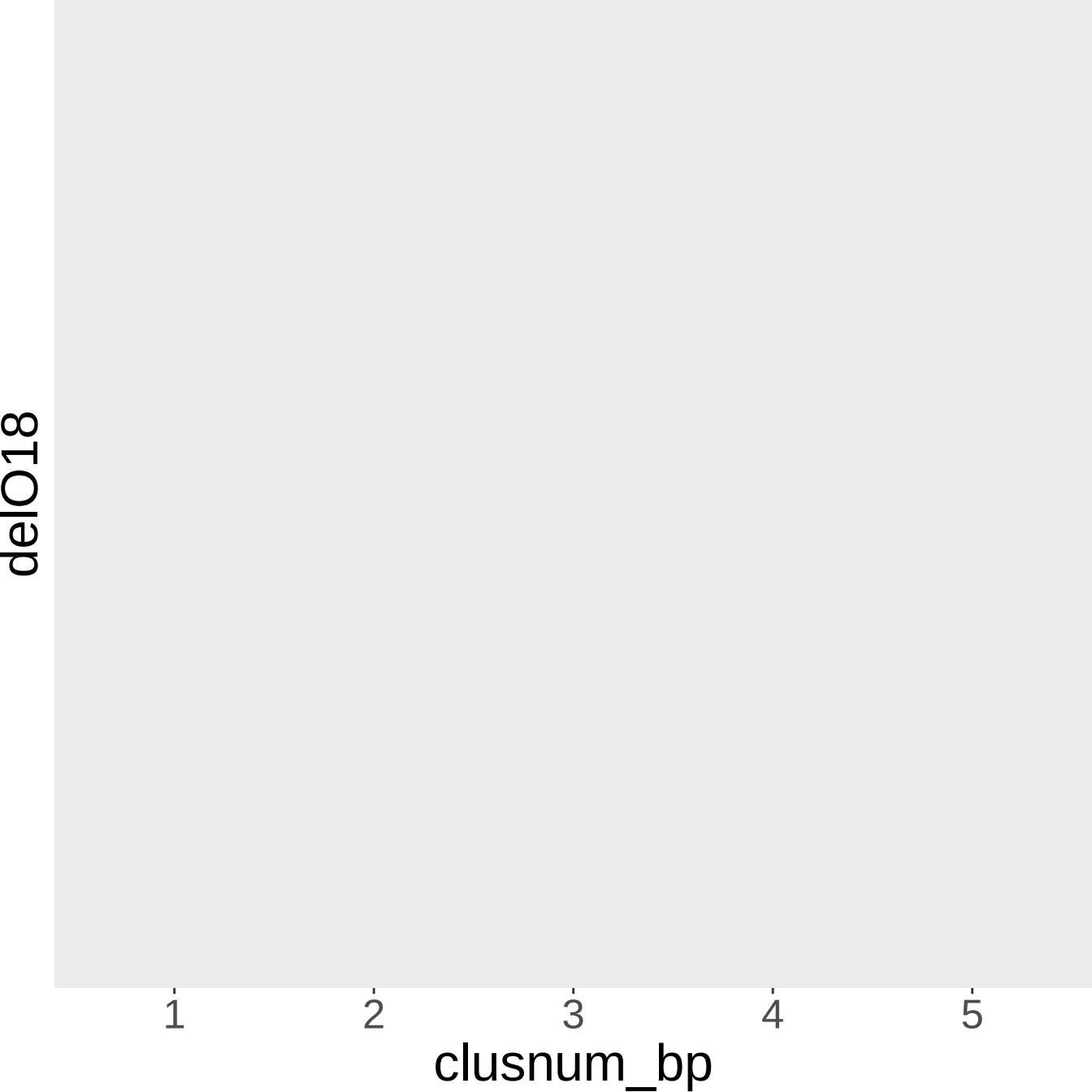

0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
clusnum_bp
1
2
3
4
5
poc_pon_ratio
2
4
6

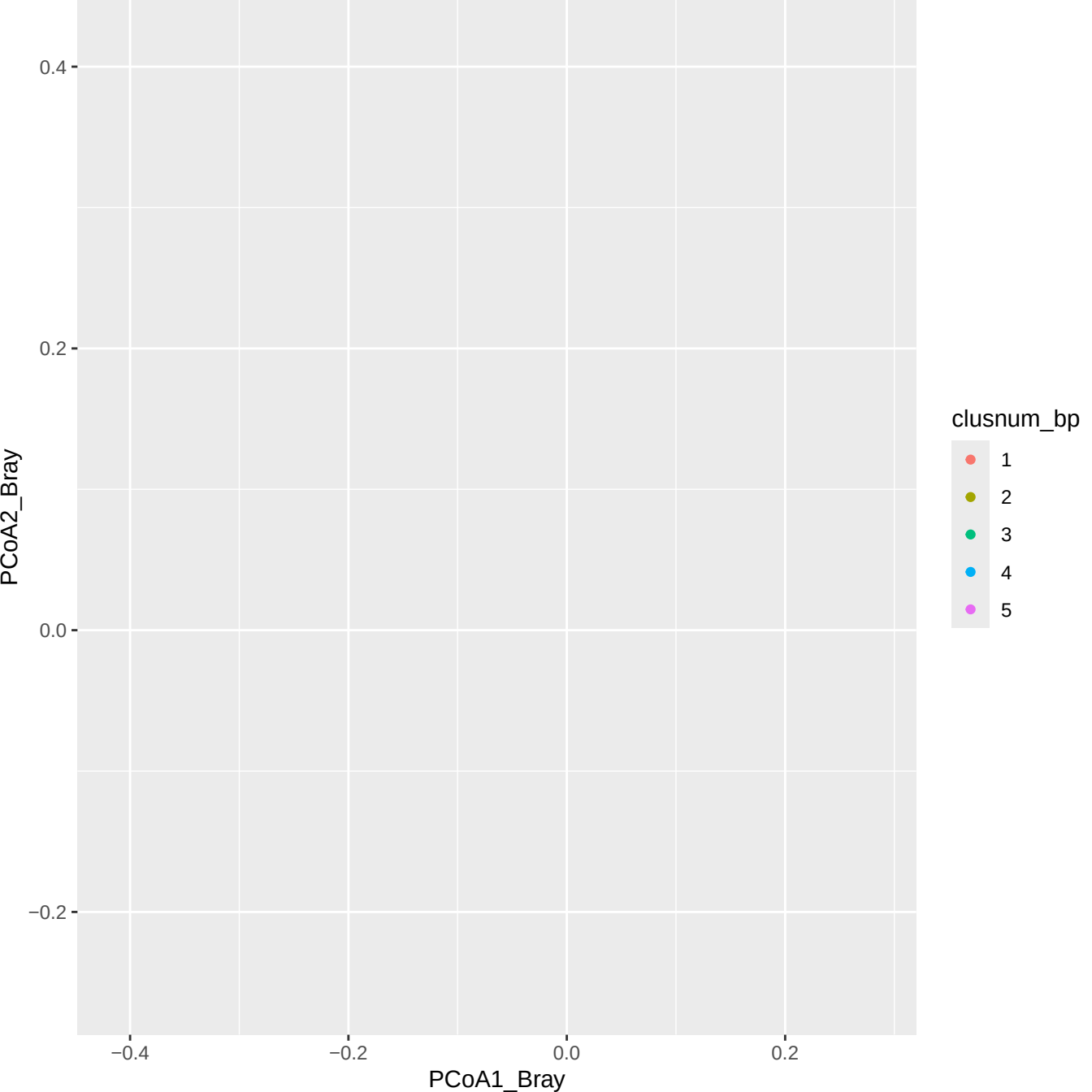

poc_pon_ratio_redfield
1.00
0.75
0.50
0.25
1
2
3
4
5
clusnum_bp
clusnum_bp
1
2
3
4
5

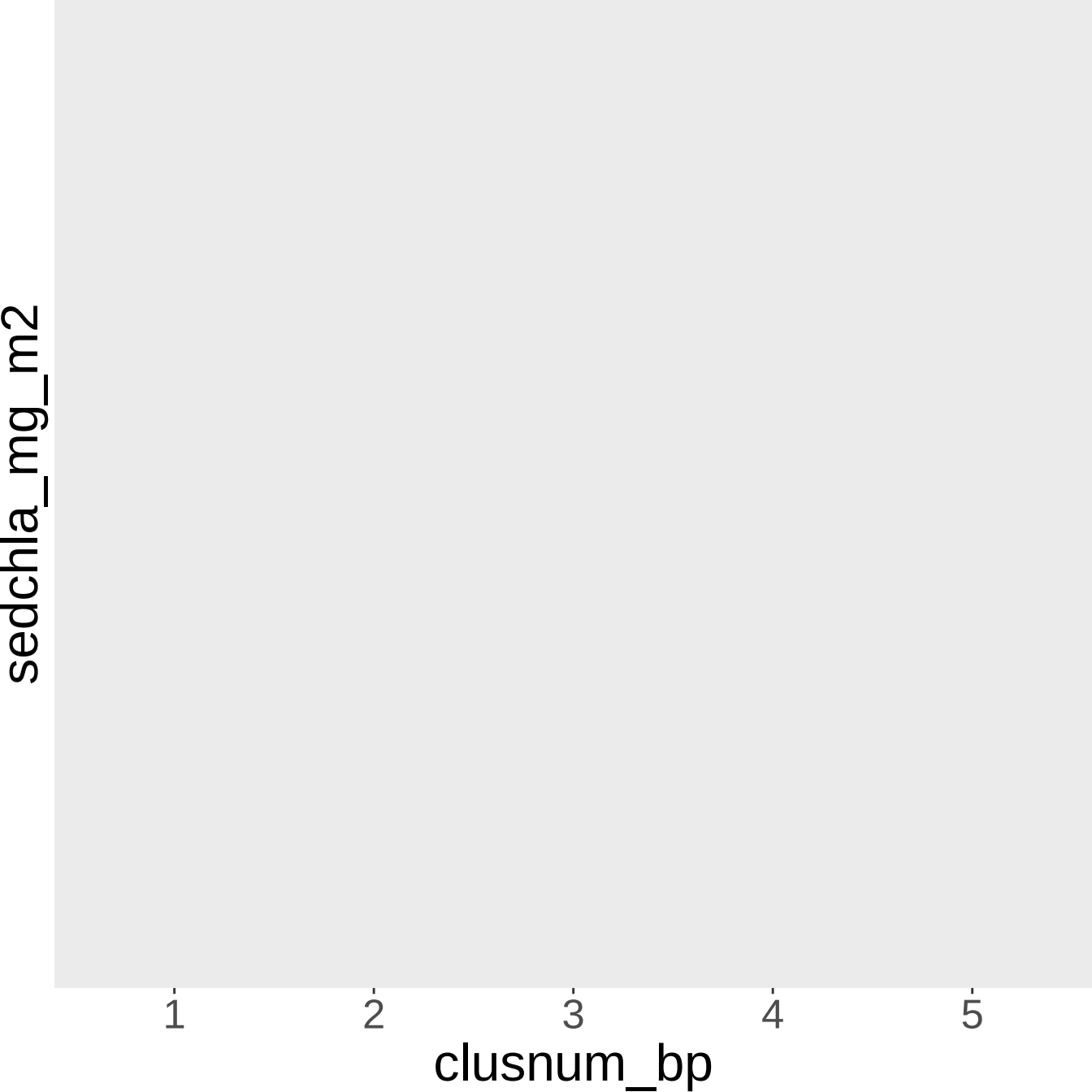

0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
clusnum_bp
1
2
3
4
5
poc_pon_ratio_redfield
0.3
0.6
0.9

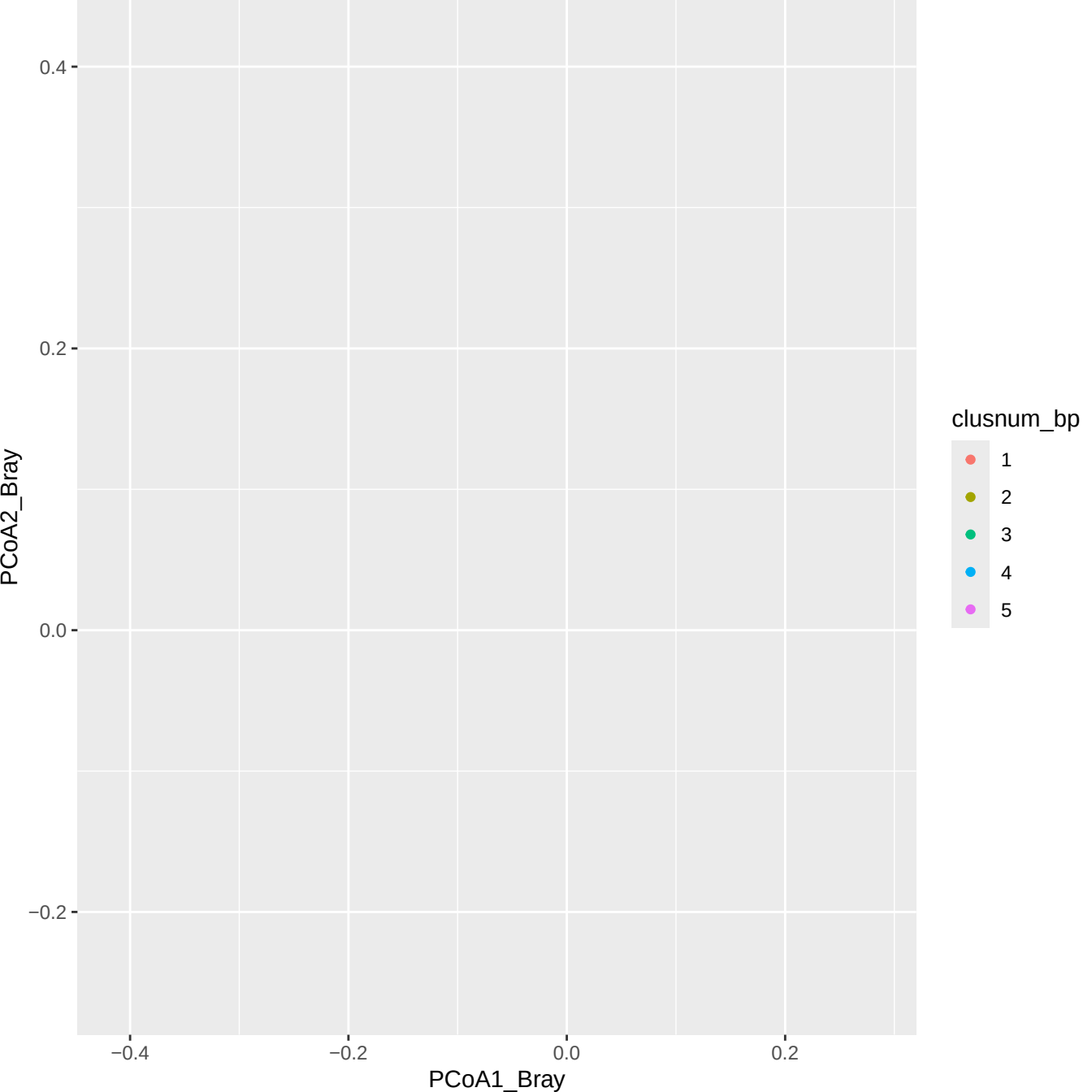

cast_depth
NA
1
2
3
4
5
clusnum_bp
clusnum_bp
1
2
3
4
5


0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
clusnum_bp
1
2
3
4
5
cast_depth
NA


delO18
1
2
3
4
5
clusnum_bp


0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
clusnum_bp
1
2
3
4
5


sedchla_mg_m2
1
2
3
4
5
clusnum_bp

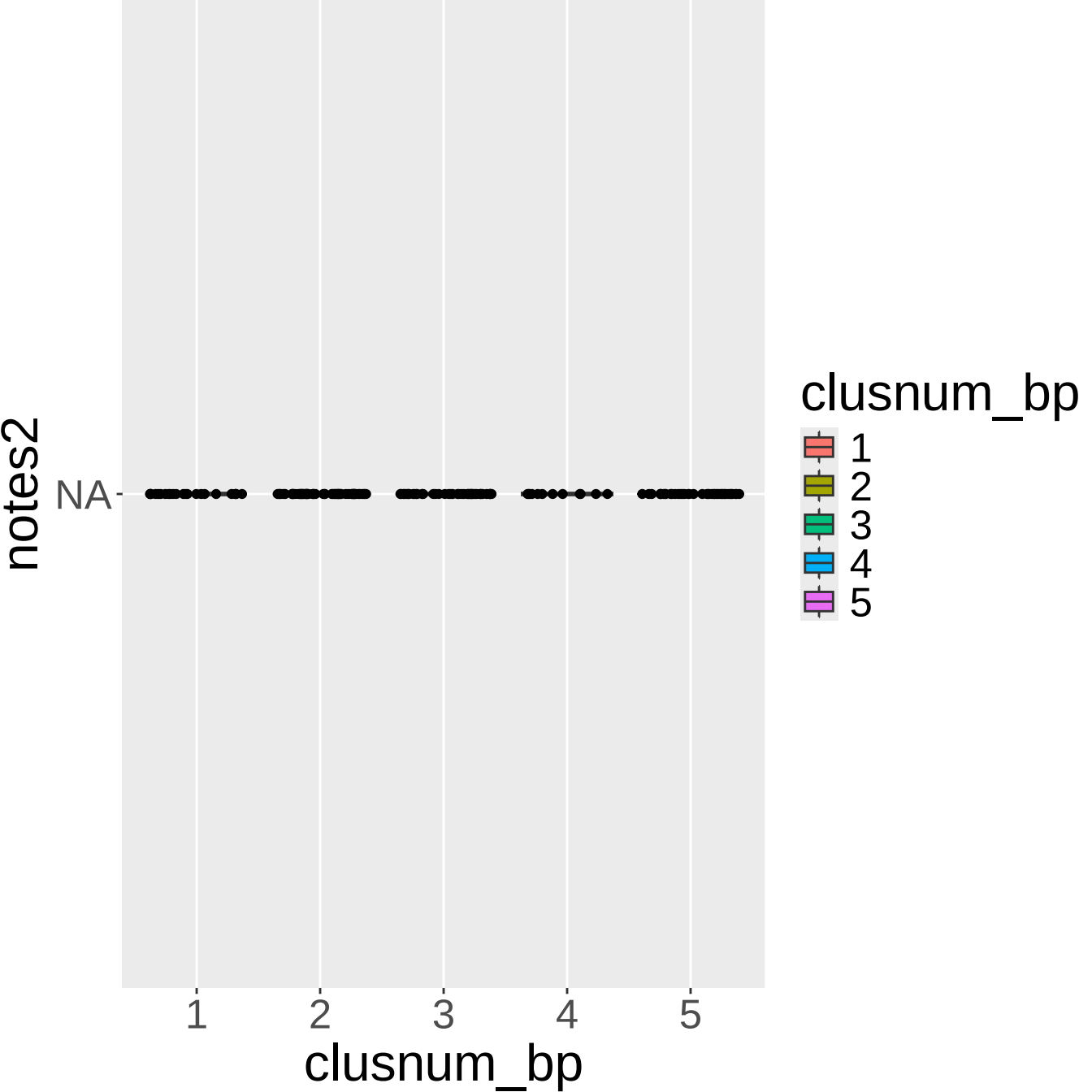

0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
clusnum_bp
1
2
3
4
5

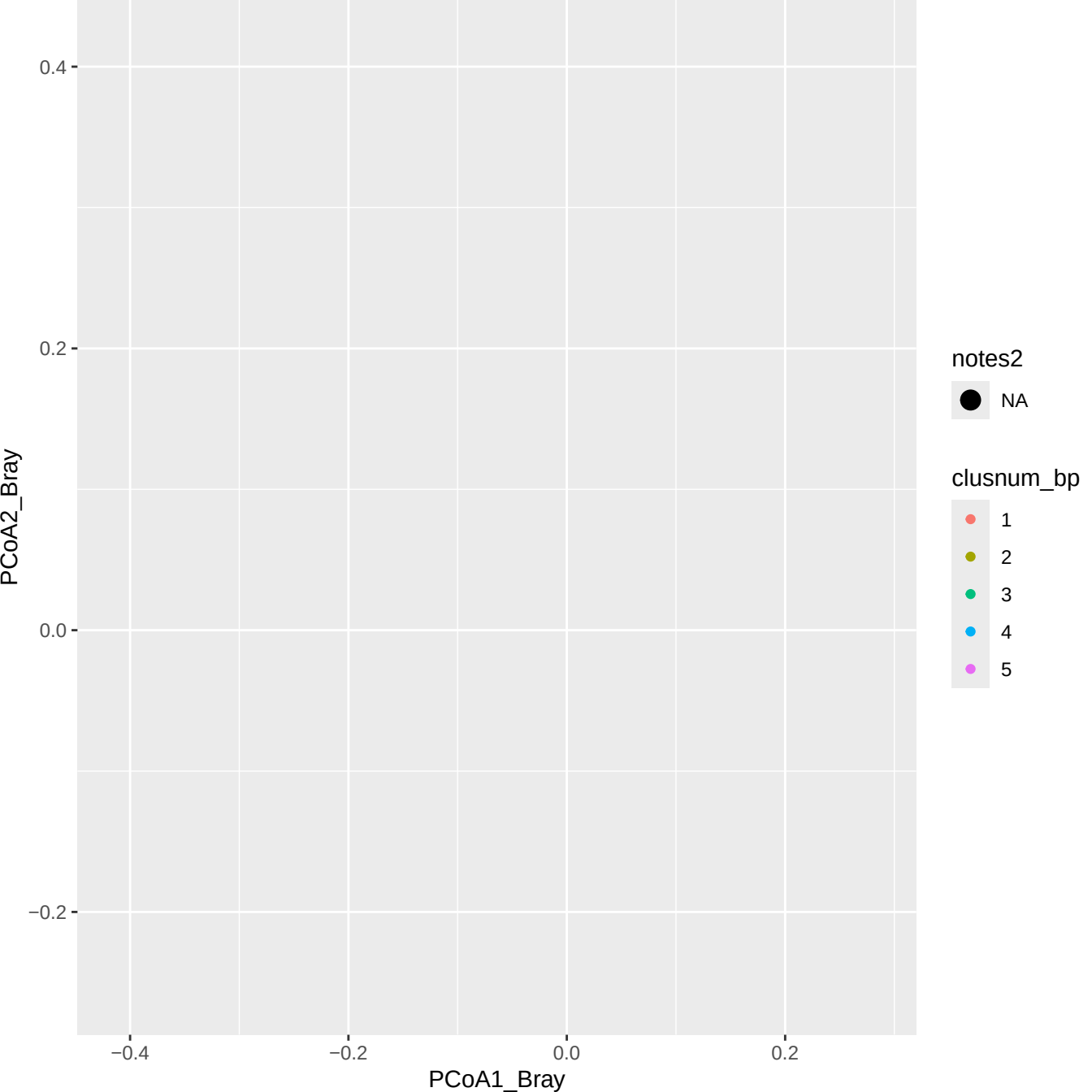
UVP was on for t
UVP not on for
thermocline! Halocline! Strat
n from ships underway seawater flow through positioned @ 6
surf and 10 m have same levels of c
SLY8 no
SLY6 no
SLY4 no
SLY2 no
ted DBO3.8 later in the morning after finding a strong chl ma
TD casts in between process stations due to being ahead of s
No UVP on DBO3.8 (was
no particle filters th
no particle filters fo
no particle filters
no particle filter f
lots of stuff in the water, reduced filtered volu
ect is an extra nutrient and profile line en route to next proces
first signs of returning stratification −− especially in the
may not actually be accurate, numbers in data sheet were cro
DBO3.6 − no partic
CTD non−cooperative−−bottles for depths above 13 di

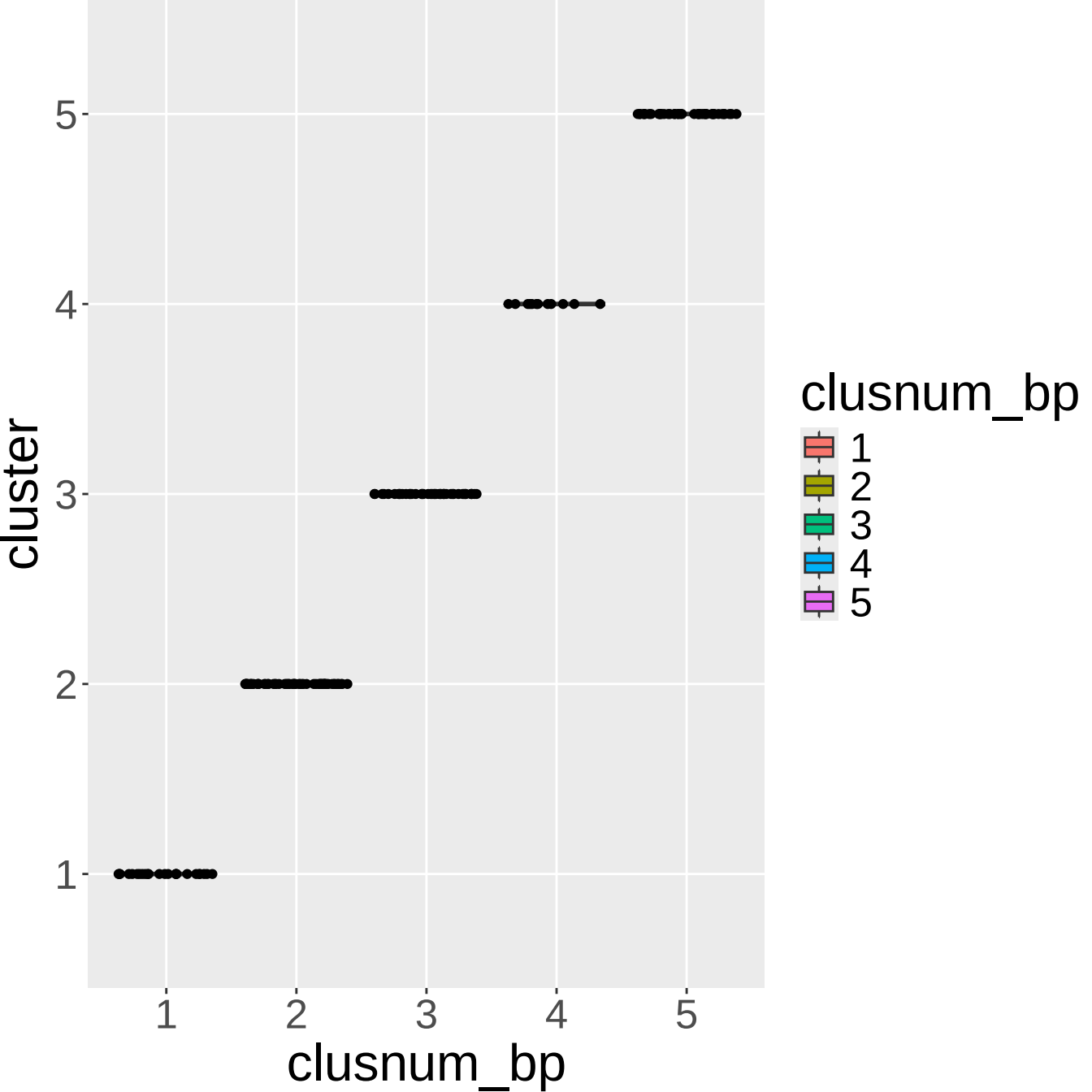
CBW15 no partic
nt starting to form around 15 meters, everything else still look
also revisiting process station as survey station (c
*very bottom of the
*clines = thermo and pyc
*cline = thermo and pycnoclines and increa
**oringally logged as DBO2.1, when it is actually static
**duplicate particle filters, same genetic

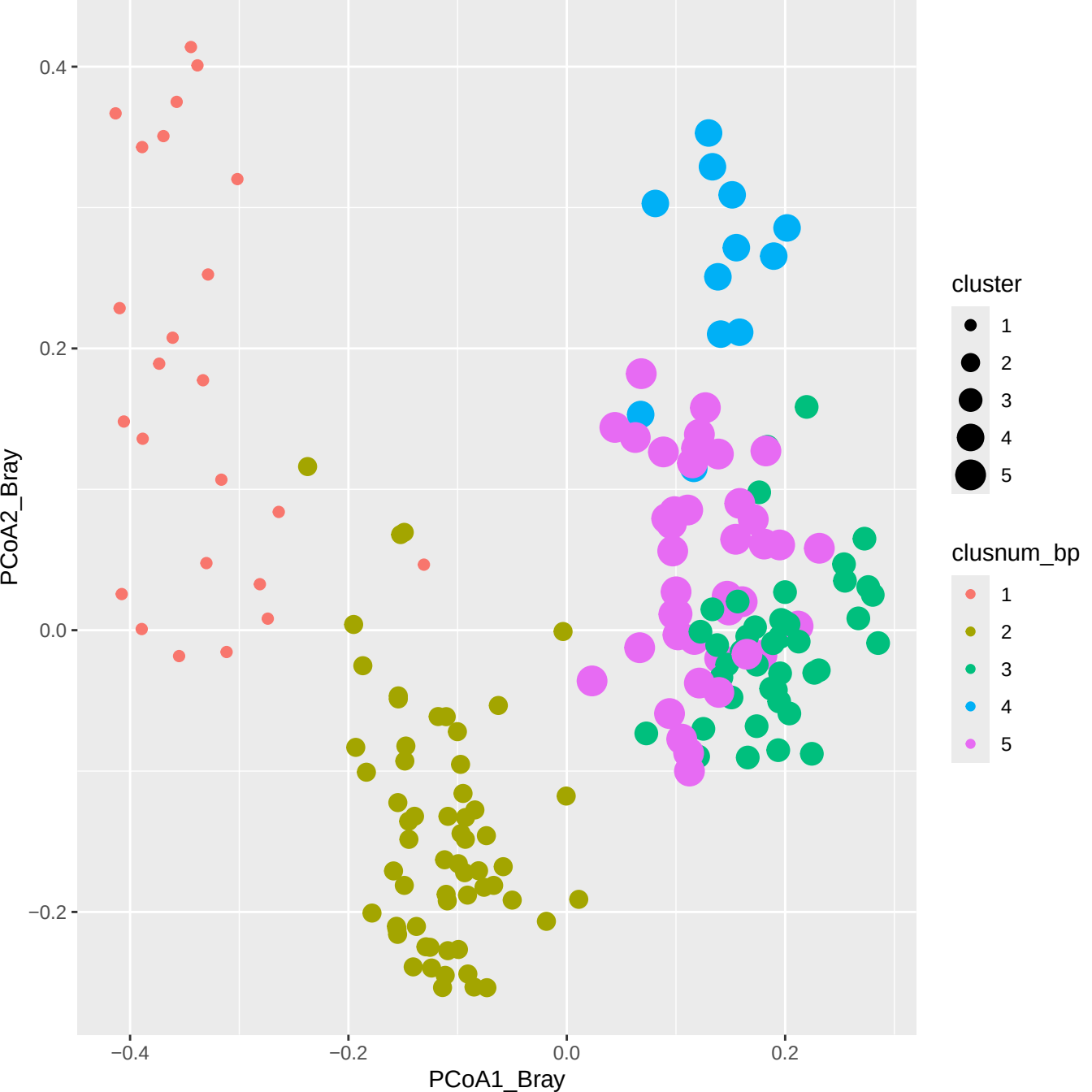
s, same genetic sample
BO2.1, when it is actually station DB2.2
noclines and increase in O2
cnoclines
line
tation as survey station (cast 012)
tarting to form around 15 meters, everything else still looks mixed
s
-bottles for depths above 13 did not fire
ers
nay not actually be accurate, numbers in data sheet were crossed off.
ratification −− especially in thermocline
utrient and profile line en route to next process station
reduced filtered volume to 1
s

- no particle filters for SLY8
- no particle filters for SLY9.5
- no particle filters this depth
- No UVP on DBO3.8 (was asleep)
- opportunistic CTD casts in between process station
- revisited DBO3.8 later in the morning after finding a
- SLY2 no particles
- SLY4 no particles
- SLY6 no particles
- SLY8 no particles
- surf and 10 m have same levels of chl about
- surface water was taken from ships underway seaw
- thermocline! Halocline! Stratification!
- UVP not on for this cast
- UVP was on for this cast!
- NA


notes2
NA
1
2
3
4
5
clusnum_bp
clusnum_bp
1
2
3
4
5


0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
notes2
NA
clusnum_bp
1
2
3
4
5


cluster
5
4
3
2
1
1
2
3
4
5
clusnum_bp
clusnum_bp
1
2
3
4
5


0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
cluster
1
2
3
4
5
clusnum_bp
1
2
3
4
5